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MEE 224 — Past Questions & Materials
by Renance

Course: MEE 224

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 5

Levels: 200

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Paper 1 of 5 — MEE 224 STRENGTH OF MATERIALS I 2023|2024

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF MECHANICAL ENGINEERING

Second Semester Examination 2023/2024 Session

COURSE CODE AND TITLE: MEE 224- STRENGTH OF MATERIALS I

TIME ALLOTTED: 2 HOURS

INSTRUCTION: Answer 4 questions in all with at least one question from section A

SECTION A

Q1) a) Define the following: i) True stress und strain (2 Marks) (ii) Ductility of a material (2 Marks) and (it) Engineering

stress and strain (2 Marks)

b) The ultimate stress for a hollow steel column which carries an axial load of 1.9MN is 480N/mm*. If the external

diameter of the column is 200mm, determine the internal diameter. Take the factor of safety as 4 (11.5 Marks)

Q2) a) Briefly discuss the deformation of a body due to its weight (8 Marks). (b) A steel wire of imm diamete= is freely

hanging under its own weight. If the extension of the wire should not exceed 2.5mm, what should be its maximum length?

Take E (modulus of elasticity) for the wire material as 200GPa and its specific weight as 78.5KN/m?

SECTION B

Q3) a) What is a (i) principal stress (11) two-dimensional stress system (4 marks)

b) At a point ina strained material, the principal stresses are 200 MPa and 80 MPa both tensile. Find the normal,

tangential and resultant stresses at a section inclined at 25° with the axis of the minor principal stress using both

analytical ane graphical (Mohr circle) methods. (13.5 marks)

Pie ed

- Q4a) List and sketch three og sles of beams classified by the type of support (4 marks)

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c) A cantilever beam 1.5m long is loaded as shown in figure. c Draw the free body diagram of any suitable scale. (11.5 marks)

3. KN

2 kN/m

. D 15

PO OPO SG a BS

Y t pas 10

pe PR ore !m —————+— 0.25 m— AYE af

Fig. 2c PP: FIG. 136

Q5 a) What is the difference between symmetrical and unsymmetrical bending. (3 marks)

b) Define the section modulus and moment of resistance of a section. (3 marks)

c) A beam with the cross section given in figure Sc is made of material with permissible stress in compression and tension

equal to 100 MPa and 140 MPa respectively. Calculate the section modulus and the moment of resistance of the cross

section when subjected to a moment causing compression at the top and tension at the bottom. (11,5 marks)

Paper 2 of 5 — MEE 224 STRENGTH OF MATERIAL (2022/2023)

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN
DEPARTMENT OF MECHANICAL ENGINEERING
Second Semester Examination 2022/2023 Session
MEE 224: STRENGTH OF MATERIALS I
TIME ALLOWED: 2 HOURS.
INSTRUCTION: ANSWER ANY FOUR QUESTIONS.

SECTION A

QUESTION 1

- Define normal stress and normal strain and show with a diagram of each (5.5marks).
- A hollow cylinder 2m long has an outside diameter of 50mm and an inside diameter of 30mm. (i) If the cylinder is carrying a load of 25kN, find the stress in the cylinder (6marks).
(ii) Find also the deformation of the cylinder, if the value of modulus of elasticity for the cylinder material is 100GPa (6marks).

QUESTION 2

- Briefly discuss the deformation of a body due to its weight (8marks).
- A steel wire of 1mm diameter is freely hanging under its own weight. If the extension of the wire should not exceed 2.5mm, what should be its maximum length? Take E for the wire material as 200GPa and its specific weight as 78.5 kN/m^3 (9.5marks).

SECTION B

QUESTION 3

- List and sketch four examples of beams classified by the type of support (4 marks)
- Differentiate between a shear force diagram and a bending moment diagram (2 marks)
- A cantilever beam 5m long is loaded as shown in figure 3c. Draw the shear force and bending moment diagrams using any suitable scale. (11.5 marks)

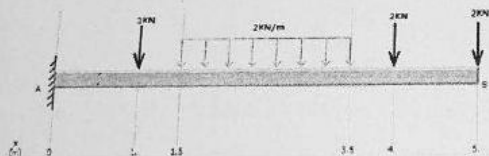


Fig. 3c

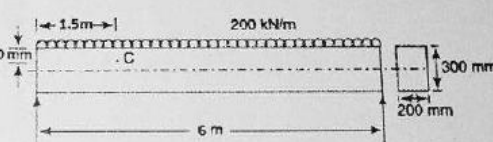


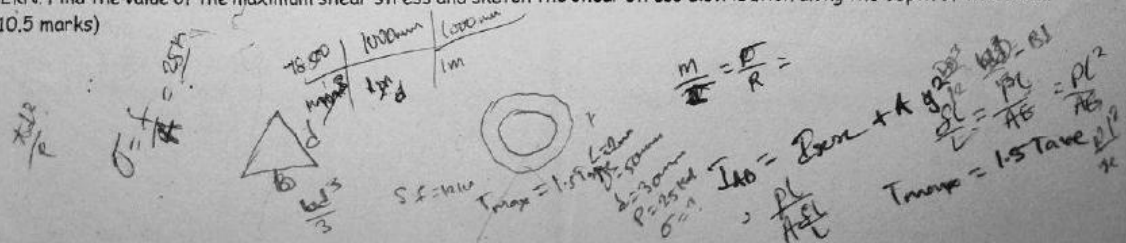
Fig. 4c

QUESTION 4

- Define i) The section modulus and moment of resistance of a section (3 marks)
- State the parallel axis theorem. (3 marks)
- Figure 4c shows a simply supported 200-mm wide, 300-mm deep and 6-m long beam. Determine the bending stress at the point C which is 50 mm below the top surface and 1.5 m from the left support. (11.5 marks)

QUESTION 5

- Given that $\tau = \frac{F}{3I}(hx - x^2)$ and $I = \frac{bh^3}{36}$ for a triangular section, show that $\tau_{max} = \tau_{ave}$ (7 marks)
- A beam of triangular cross section having base width of 50mm and height of 60mm is subjected to a shear force of 12kN. Find the value of the maximum shear stress and sketch the shear stress distribution along the depth of the beam. (10.5 marks)



Paper 3 of 5 — MEE 224 STRENGTH OF MATERIAL

Level 200 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN,
DEPARTMENT OF MECHANICAL ENGINEERING

Second Semester Examination 2022/2023 Session

MEE 224: STRENGTH OF MATERIALS I.

TIME ALLOWED: 2 HOURS.

INSTRUCTION: ANSWER ANY FOUR QUESTIONS,

SECTION A

QUESTION 1

- (a) Define normal stress and normal strain and show with a diagram of each (5.5marks).
 (b) A hollow cylinder 2m long has an outside diameter of 50mm and an inside diameter of 30mm.
 (i) If the cylinder is carrying a load of 25KN, find the stress in the cylinder (6marks).
 (ii) Find also the deformation of the cylinder, if the value of modulus of elasticity for the cylinder material is 100GPa (6marks).

QUESTION 2

- (a) Briefly discuss the deformation of a body due to its weight (8marks).
 (b) A steel wire of 1mm diameter is freely hanging under its own weight. If the extension of the wire should not exceed 2.5mm, what should be its maximum length? Take E for the wire material as 200GPa and its specific weight as 78.5 KN/m³ (9.5marks).

SECTION B

QUESTION 3,

- (a) List and sketch four examples of beams classified by the type of support (4 marks)
 (b) Differentiate between a shear force diagram and a bending moment diagram (2 marks)
 (c) A cantilever beam 5m long is loaded as shown in figure 3c. Draw the shear force and bending moment diagrams using any suitable scale. (11.5 marks)

aN

mo Las 344

Fig. 3c

QUESTION 4

- (a) Define i) The section modulus and moment of resistance of a section (3 marks)
 (b) State the parallel axis theorem. (3 marks)

(c) Figure 4c shows a simply supported 200-mm wide, 300-mm deep and 6-m long beam.

Determine the bending

stress at the point C which is 50 mm below the top surface and 1.5 m from the left support, (11.5 marks)

QUESTION 5

cy)

i

(a) Given that $c = E(hx - x^2)$ and $J = \frac{bh^3}{12}$ for a triangular section, show that $T_{max} = T_{ave}$ (7 marks)

b) A beam of triangular cross section having base width of 50mm and height of 60mm is subjected to a shear force of

12kN. Find the value of the maximum shear stress and erect the shear stress distribution along the depth of the beam,

(10.5 marks)

Paper 4 of 5 — MEE 224 PAST QUESTION

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROL: |
ENGINEERING .

DEPARTMENT OF MECHA. |

Second Semester Examauicn 22 Cdl De@ssron

COURSE CODE AND TITLE MEE ce* RENGTH OF MATERIALS I

TIME ALLOTTED. 2 HOURS

INSTRUCTION: ANSWER QUESTION ONE AND ANY OTHER THREE

Q1) c) Define (1) Stress (u) Strain (2 marks)

2m long and 25mm X 25mm in cross section 's subjected to a tensile force of

b) +) State Hooke's low (i) A steel rod

ulus of elastic:ty of steel 1s 200 GPa (8 marks)

60KN Determine the elongation of the rod if mod

1p hos three segments as shown in Figure 1c below

the middle seqment to have zero elongation of the bor

the bar (Toke E - 205 GPa) (14 marks)

c) 1) State the principle of superposition (1) A steel! circular

Dermune (I) the tote! elongation of the bar (II) the length of

(III) the cameter of the last segment to have zero elongation «|

Comey oS" ee

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& 0 %

A

J 150 men te— 200mm - ej S50 en —|

Fig ic

Q2) a) With the aid of simple sketches only, differentiate between a simply supported beam and a cantilever beam. (2

marks)

b) A simply supported beam of 10-m length carries three point loads of 8 KN, 4 KN and 2 KN at 2m, 3m and 6m

respectively from the ght end Draw the shear force and bending moment diagrams (10 marks)

Q3) a) State the perallel axis theorem. (2 marks)

b) The tension flange of a cast iron I-section beam is 280 mm wide and 60 mm deep. the compression flange is 120 mm

ond 30 mm deep whereas the web 1s 400 mm X 40 mm Find the load per m run which con be carried over a 4 m span by

a simply supported beam if the maximum permissible stresses are 100 MPa in compression and 30 MPa in tension. (10

marks)

Q4) a) What are principal stresses? (2 marks)

b) A rectangular block is subjected to two perpendicular tensile stresses of 120 MPa and 70 MPa. Determine normal,

tangential and resultant stresses on a plane inclined at 30° to the plane of 100 MPa stress using both the Mohr circle

and analytical method (10 marks)

Q5) c) What are composite beams? (2 marks)

b) Two rectangular bars, one of brass and the other of steel each 40mm by 10mm are placed together to form a beam

of 40mm by 20mm depth on two supports 800mm apart, the steel bar being at the bottom. Find the maximum central

load if the bars are firmly secured to each other throughout the length. Max permissible stress in steel is 102 MPa and

in brass is 72 MPa. (Take $E_s = 204 \text{ GPa}$ and $E_b = 85 \text{ GPa}$) (10 marks)

Q6) a) What is moment of resistance of a section? (2 marks)

b) A hollow circular bar used as a beam has its outside diameter twice the inside diameter. It is subjected to a maximum

bending moment of 60 kNm. Determine the inside diameter and the thickness of the beam if the permissible bending

stress is limited to 120 MPa (10 marks)

Paper 5 of 5 — MEE 224 handout

Level 200 • Second Semester • Handout

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Section 5

Strength of Materials

BY

JOHN SYMONDS Fellow Engineer (Retired), Oceanic Division, Westinghouse Electric Corporation.

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by John Symonds, Expanded by Staff

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5.3 PIPELINE FLEXURE STRESSES

5.2 MECHANICS OF MATERIALS

by Harold V. Hawkins

by J. P. Vidosic

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5.1 MECHANICAL PROPERTIES OF MATERIALS

by John Symonds, Expanded by Staff

REFERENCES: Davis et al., "Testing and Inspection of Engineering Materials," permanent strain. The permanent strain commonly used is 0.20 percent

McGraw-Hill, Timoshenko, "Strength of Materials," pt . II, Van Nostrand. of the original gage length. The intersection of this line with the curve

Richards, "Engineering Materials Science," Wadsworth. Nadai, "Plasticity," determines the stress value called the yield strength. In reporting the

McGraw-Hill. Tetelman and McEvily, "Fracture of Structural Materials," Wiley. yield strength, the amount of permanent set should be specified. The

"Fracture Mechanics," ASTM STP-833. McClintock and Argon (eds.), "Mechanical Behavior of Materials," Addison-Wesley. Dieter, "Mechanical Metal-

lurgy," McGraw-Hill. "Creep Data," ASME. ASTM Standards, ASTM. Blaz- hibiting a natural yield point such as nonferrous metals; but it is not

nynski (ed.), "Plasticity and Modern Metal Forming Technology," Elsevier limited to these. Plastic behavior is somewhat time-dependent, particu-

Science. larly at high temperatures. Also at high temperatures, a small amount of time-dependent reversible strain may be detectable, indicative of anelas- tic behavior.

STRESS-STRAIN DIAGRAMS

The Stress-Strain Curve The engineering tensile stress-strain curve is obtained by static loading of a standard specimen, that is, by applying the load slowly enough that all parts of the specimen are in equilibrium

at any instant. The curve is usually obtained by controlling the loading rate in the tensile machine. ASTM Standards require a loading rate not exceeding 100,000 lb/in² (70 kgf/mm²)/min. An alternate method of obtaining the curve is to specify the strain rate as the independent variable, in which case the loading rate is continuously adjusted to maintain the required strain rate. A strain rate of 0.05 in/in/(min) is commonly used. It is measured usually by an extensometer attached to the gage length of the specimen. Figure 5.1.1 shows several stress-strain curves. Fig. 5.1.2. General stress-strain diagram.

The ultimate tensile strength (UTS) is the maximum load sustained by the specimen divided by the original specimen cross-sectional area. The percent elongation at failure is the plastic extension of the specimen at failure expressed as (the change in original gage length $\div$ 100) divided by the original gage length. This extension is the sum of the uniform and nonuniform elongations. The uniform elongation is that which occurs prior to the UTS. It has an unequivocal significance, being associated with uniaxial stress, whereas the nonuniform elongation which occurs during localized extension (necking) is associated with triaxial stress. The nonuniform elongation will depend on geometry, particularly the ratio of specimen gage length L_0 to diameter D or square root of cross-sectional area A . ASTM Standards specify test-specimen geometry for a number of specimen sizes. The ratio $L_0 / \sqrt{A}$ is maintained at 4.5 for flat- and round-cross-section specimens. The original gage length should always be stated in reporting elongation values.

The specimen percent reduction in area (RA) is the contraction in cross-sectional area at the fracture expressed as a percentage of the original area. It is obtained by measurement of the cross section of the specimen at the fracture location. The RA along with the load at fracture can be used to obtain the fracture stress, that is, fracture load divided by cross-sectional area at the fracture. See Table 5.1.1.

For most engineering materials, the curve will have an initial linear elastic region (Fig. 5.1.2) in which deformation is reversible and time-independent. The slope in this region is Young's modulus E . The proportional elastic limit (PEL) is the point where the curve starts to deviate from a straight line. The elastic limit (frequently indistinguishable from the PEL in brittle materials. In general, fractures are either of the shear or of

PEL) is the point on the curve beyond which plastic deformation is the separation (loss of cohesion) type. Flat tensile specimens of ductile present after release of the load. If the stress is increased further, the metals often show shear failures if the ratio of width to thickness is stress-strain curve departs more and more from the straight line. Un- greater than 6 : 1. A completely shear-type failure may terminate in a loading the specimen at point X (Fig. 5.1.2), the portion XX $\diamond$ is linear chisel edge, for a flat specimen, or a point rupture, for a round specimen. and is essentially parallel to the original line OX $\diamond\diamond$. The horizontal dis- Separation failures occur in brittle materials, such as certain cast irons. tance OX $\diamond$ is called the permanent set corresponding to the stress at X. Combinations of both shear and separation failures are common on This is the basis for the construction of the arbitrary yield strength. To round specimens of ductile metal. Failure often starts at the axis in a determine the yield strength, a straight line XX $\diamond$ is drawn parallel to the necked region and produces a relatively flat area which grows until the initial elastic line OX $\diamond\diamond$ but displaced from it by an arbitrary value of material shears along a cone-shaped surface at the outside of the speci-

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STRESS-STRAIN DIAGRAMS 5-3

Table 5.1.1 Typical Mechanical Properties at Room Temperature
(Based on ordinary stress-strain values)

Tensile Yield Ultimate strength, strength, elongation, Reduction Brinell	Metal	1,000 lb/in 2	1,000 lb/in 2	% of area	% no.
Cast iron	18 – 60	8 – 40	0 0	100 – 300	
Wrought iron	45 – 55	25 – 35	35 – 25	55 – 30	100
Commercially pure iron, annealed	42	19	48	85	70
Hot-rolled	48	30	30	75	90
Cold-rolled	100	95	200		
Structural steel, ordinary	50 – 65	30 – 40	40 – 30	120	
Low-alloy, high-strength	65 – 90	40 – 80	30 – 15	70 – 40	150
Steel, SAE 1300, annealed	70	40	26	70	150
Quenched, drawn 1,300°F	100	80	24	65	200
Drawn 1,000°F	130	110	20	60	260
Drawn 700°F	200	180	14	45	400
Drawn 400°F	240	210	10	30	480
Steel, SAE 4340, annealed	80	45	25	70	170
Quenched, drawn 1,300°F	130	110	20	60	270
Drawn 1,000°F	190	170	14	50	395

Drawn 700°F 240 215 12 48 480
 Drawn 400°F 290 260 10 44 580
 Cold-rolled steel, SAE 1112 84 76 18 45 160
 Stainless steel, 18-S 85 – 95 30 – 35 60 – 55 75 – 65 145 – 160
 Steel castings, heat-treated 60 – 125 30 – 90 33 – 14 65 – 20 120 – 250
 Aluminum, pure, rolled 13 – 24 5 – 21 35 – 5 23 – 44
 Aluminum-copper alloys, cast 19 – 23 12 – 16 4-0 50 – 80
 Wrought , heat-treated 30 – 60 10 – 50 33 – 15 50 – 120
 Aluminum die castings 30 2
 Aluminum alloy 17ST 56 34 26 39 100
 Aluminum alloy 51ST 48 40 20 35 105
 Copper, annealed 32 5 58 73 45
 Copper, hard-drawn 68 60 4 55 100
 Brasses, various 40 – 120 8 – 80 60 – 3 50 – 170
 Phosphor bronze 40 – 130 55 – 5 50 – 200
 Tobin bronze, rolled 63 41 40 52 120
 Magnesium alloys, various 21 – 45 11 – 30 17 – 0.5 47 – 78
 Monel 400, Ni-Cu alloy 79 30 48 75 125
 Molybdenum, rolled 100 75 30 250
 Silver, cast , annealed 18 8 54 27
 Titanium 6 – 4 alloy, annealed 130 120 10 25 352
 Ductile iron, grade 80-55-06 80 55 6 225 – 255

NOTE: Compressive strength of cast iron, 80,000 to 150,000 lb/in 2.

Compressive yield strength of all metals, except those cold-worked □ tensile yield strength.

Stress 1,000 lb/in 2 □ 6.894 □ stress, MN/m 2.

men, resulting in what is known as the cup-and-cone fracture. Double to test temperature, test strain rate, and the characteristics of the tensile

cup-and-cone and rosette fractures sometimes occur. Several types of machine employed.

tensile fractures are shown in Fig. 5.1.3. The plastic behavior in a uniaxial tensile test can be represented as the

Annealed or hot-rolled mild steels generally exhibit a yield point (see true stress-strain curve. The true stress □ is based on the instantaneous

Fig. 5.1.4). Here, in a constant strain-rate test, a large increment of extension occurs under constant load at the elastic limit or at a stress just below the elastic limit. In the latter event the stress drops suddenly from the upper yield point to the lower yield point. Subsequent to the drop, the yield-point extension occurs at constant stress, followed by a rise to the UTS. Plastic flow during the yield-point extension is discontinuous;

Fig. 5.1.3. Typical metal fractures in tension.

successive zones of plastic deformation, known as Luder's bands or stretcher strains, appear until the entire specimen gage length has been

uniformly deformed at the end of the yield-point extension. This behavior causes a banded or stepped appearance on the metal surface. The exact form of the stress-strain curve for this class of material is sensitive Fig. 5.1.4. Yielding of annealed steel.

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5-4 MECHANICAL PROPERTIES OF MATERIALS

cross section A , so that $\sigma = \text{load}/A$. The instantaneous true strain increment $d\epsilon$ is dA/A , or dL/L prior to necking. Total true strain ϵ is scribed in the technical literature. In the range between initial

ϵ_0 and

ϵ_p a yielding and the neighborhood of the maximum load point the relationship $\sigma = k\epsilon^n$

holds. In ship between plastic strain ϵ_p and true stress often approximates $\sigma = k\epsilon^n$

$\sigma = k\epsilon^n$

or $\ln(\sigma/\sigma_0) = n \ln(L/L_0)$ prior to necking. The true stress-strain curve or flow curve

obtained has the typical form shown in Fig. 5.1.5. In the part of the test where k is the strength coefficient and n is the work-hardening exponent.

subsequent to the maximum load point (UTS), when necking occurs, For a material which shows a yield point the relationship applies only to

the true strain of interest is that which occurs in an infinitesimal length the rising part of the curve beyond the lower yield. It can be shown that

at the region of minimum cross section. True strain for this element can at the maximum load point the slope of the true stress-strain curve

still be expressed as $\ln(A_0/A)$, where A refers to the minimum cross equals the true stress, from which it can be deduced that for a material

obeying the above exponential relationship between ϵ_p and n , $\epsilon_p = n$ at the maximum load point. The exponent strongly influences the spread

between Y_S and UTS on the engineering stress-strain curve. Values of n and k for some materials are shown in Table 5.1.2. A point on the flow curve identifies the flow stress corresponding to a certain strain, that is, the stress required to bring about this amount of plastic deformation.

The concept of true strain is useful for accurately describing large amounts of plastic deformation. The linear strain definition $(L - L_0)/L_0$ fails to correct for the continuously changing gage length, which leads to an increasing error as deformation proceeds.

During extension of a specimen under tension, the change in the specimen cross-sectional area is related to the elongation by Poisson's ratio ν , which is the ratio of strain in a transverse direction to that in the longitudinal direction. Values of ν for the elastic region are shown in

Table 5.1.3. For plastic strain it is approximately 0.5.

Table 5.1.2 Room-Temperature Plastic-Flow Constants for a
Number of Metals

k, 1,000 in 2

Material Condition (MN/m²) n

0.40% C steel Quenched and tempered at 416 (2,860) 0.088

400°F (478K)

0.05% C steel Annealed and temper-rolled 72 (49.6) 0.235

2024 aluminum Precipitation-hardened 100 (689) 0.16

2024 aluminum Annealed 49 (338) 0.21

Copper Annealed 46.4 (319) 0.54

70 – 30 brass Annealed 130 (895) 0.49

SOURCE: Reproduced by permission from “Properties of Metals in Materials Engineering,”

Fig. 5.1.5. True stress-strain curve for 20°C annealed mild steel. ASM, 1949.

Table 5.1.3 Elastic Constants of Metals

(Mostly from tests of R. W. Vose)

E G K ν

Modulus of Modulus of

elasticity rigidity

(Young’s (shearing Bulk

modulus). modulus). modulus.

1,000,000 1,000,000 1,000,000 Poisson’s

Metal lb/in² lb/in² lb/in² ratio

Cast steel 28.5 11.3 20.2 0.265

Cold-rolled steel 29.5 11.5 23.1 0.287

Stainless steel 18 – 8 27.6 10.6 23.6 0.305

All other steels, including high-carbon, heat-treated 28.6 – 30.0 11.0 – 11.9 22.6 – 24.0 0.283 – 0.292

Cast iron 13.5 – 21.0 5.2 – 8.2 8.4 – 15.5 0.211 – 0.299

Malleable iron 23.6 9.3 17.2 0.271

Copper 15.6 5.8 17.9 0.355

Brass, 70 – 30 15.9 6.0 15.7 0.331

Cast brass 14.5 5.3 16.8 0.357

Tobin bronze 13.8 5.1 16.3 0.359

Phosphor bronze 15.9 5.9 17.8 0.350

Aluminum alloys, various 9.9 – 10.3 3.7 – 3.9 9.9 – 10.2 0.330 – 0.334

Monel metal 25.0 9.5 22.5 0.315

Inconel 31 11 0.27 – 0.38

Z-nickel 30 11 ν 0.36

Beryllium copper 17 7 ν 0.21

Elektron (magnesium alloy) 6.3 2.5 4.8 0.281

Titanium (99.0 Ti), annealed bar 15 – 16 6.5 0.34

Zirconium, crystal bar 11 – 14

Molybdenum, arc-cast 48 – 52

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STRESS-STRAIN DIAGRAMS 5-5

The general effect of increased strain rate is to increase the resistance Tension or compression Bending

to plastic deformation and thus to raise the flow curve. Decreasing test temperature also raises the flow curve. The effect of strain rate is expressed as $\frac{1}{2} \log \frac{\dot{\epsilon}_2}{\dot{\epsilon}_1}$

pressed as strain-rate sensitivity m . Its value can be measured in the double punch tension test if the strain rate is suddenly increased by a small increment $\frac{1}{2} \log \frac{\dot{\epsilon}_2}{\dot{\epsilon}_1}$

during the plastic extension. The flow stress will then jump to a higher value. The strain-rate sensitivity is the ratio of incremental changes of $\log \sigma$ and $\log \dot{\epsilon}$ $m = \frac{\Delta \log \sigma}{\Delta \log \dot{\epsilon}}$

$\frac{\Delta \sigma}{\sigma}$

$\frac{\Delta \dot{\epsilon}}{\dot{\epsilon}}$

$\frac{\Delta \sigma}{\sigma}$

$\frac{\Delta \log \sigma}{\Delta \log \dot{\epsilon}}$

$m = \frac{\Delta \log \sigma}{\Delta \log \dot{\epsilon}}$

$\frac{\Delta \log \sigma}{\Delta \log \dot{\epsilon}}$

$\frac{\Delta \sigma}{\sigma}$

$\frac{\Delta \dot{\epsilon}}{\dot{\epsilon}}$

$\frac{\Delta \sigma}{\sigma}$

For most engineering materials at room temperature the strain rate sensitivity is of the order of 0.01. The effect becomes more significant at elevated temperatures, with values ranging to 0.2 and sometimes higher.

Compression Testing The compressive stress-strain curve is similar to the tensile stress-strain curve up to the yield strength. Thereafter, the progressively increasing specimen cross section causes the compressive stress-strain curve to diverge from the tensile curve. Some ductile metals will not fail in the compression test. Complex behavior occurs when the direction of stressing is changed, because of the Bauschinger effect, which can be described as follows: If a specimen is first plastically strained in tension, its yield stress in compression is reduced

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3.4

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Stress concentration factor, K

and vice versa. 3.0

Combined Stresses This refers to the situation in which stresses are present on each of the faces of a cubic element of the material. For a given cube orientation the applied stresses may include shear stresses over the cube faces as well as stresses normal to them. By a suitable rotation of axes the problem can be simplified: applied stresses on the

new cubic element are equivalent to three mutually orthogonal principal stresses σ_1 , σ_2 , σ_3 alone, each acting normal to a cube face. Combined stress behavior in the elastic range is described in Sec. 5.2, Mechanics of Materials. V

Prediction of the conditions under which plastic yielding will occur under combined stresses can be made with the help of several empirical Note; in all cases $D \leq 2r$

theories. In the maximum-shear-stress theory the criterion for yielding is that yielding will occur when

$$\sigma_1 - \sigma_3 \geq \sigma_{ys}$$

in which σ_1 and σ_3 are the largest and smallest principal stresses, respectively, and σ_{ys} is the uniaxial tensile yield strength. This is the simplest theory for predicting yielding under combined stresses. A more accurate prediction can be made by the distortion-energy theory, according to which the criterion is

$$(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \leq 2(\sigma_{ys})^2$$

Stress-strain curves in the plastic region for combined stress loading can be constructed. However, a particular stress state does not determine a unique strain value. The latter will depend on the stress-state path which is followed.

02

Plane strain is a condition where strain is confined to two dimensions. There is generally stress in the third direction, but because of mechanical constraints, strain in this dimension is prevented. Plane strain occurs in certain metalworking operations. It can also occur in the neighborhood of stress concentration factors, K

cal constraints, strain in this dimension is prevented. Plane strain occurs in certain metalworking operations. It can also occur in the neighborhood of stress concentration factors, K

hood of a crack tip in a tensile loaded member if the member is sufficiently thick. The material at the crack tip is then in triaxial tension, which condition promotes brittle fracture. On the other hand, ductility is

5

enhanced and fracture is suppressed by triaxial compression. 2.2 Blunt Sharp Stress Concentration In a structure or machine part having a notch grooves

1

or any abrupt change in cross section, the maximum stress will occur at this location and will be greater than the stress calculated by elementary

2

formulas based upon simplified assumptions as to the stress distribution. The ratio of this maximum stress to the nominal stress (calculated by the elementary formulas) is the stress-concentration factor K_t . This is a constant for the particular geometry and is independent of the material, provided it is isotropic. The stress-concentration factor may be

0.4 1.0 1.5 2 3 4 5 6

determined experimentally or, in some cases, theoretically from the mathematical theory of elasticity. The factors shown in Figs. 5.1.6 to h Sharpness of groove,

5.1.13 were determined from both photoelastic tests and the theory of elasticity. Stress concentration will cause failure of brittle materials if Fig. 5.1.7. Flat plate with grooves, in tension.

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5-6 MECHANICAL PROPERTIES OF MATERIALS

h

the concentrated stress is larger than the ultimate strength of the material. In ductile materials, concentrated stresses higher than the yield strength will generally cause local plastic deformation and redistribution

h h

of stresses (rendering them more uniform). On the other hand, even with ductile materials areas of stress concentration are possible sites for fatigue if the component is cyclically loaded.

3.4

02

0.

5

d □

0 .0

Stress concentration factor, K

h

3.0

 $h \leq D \leq 2h$

1

0.

r h 2.6 Full fillets

 $D \leq (h/r) \leq 2$

0.

 $h/r \leq h$

2.2

Blunt Sharp

fillets fillets

1.8 0.5

3.4

1

02

05 1.4 2

0.

Stress concentration factor, K

0.

 $3.0 \leq D \leq 2h$ $d \leq$ $\leq h$

Full fillets 1

0.

 $(h/r) \leq 1.0$

et

2.6 0.4 1.0 1.5 2 3 4 5 6

fill

0.2

of

h

th

Sharpness of fillet,

p

r

De

2.2 Blunt Sharp 0.5 Fig. 5.1.10. Flat plate with fillets, in bending.

fillets fillets

1.8 1.0

2.0

1.4

1.0

0.4 1.0 1.5 2 3 4 5 6

h

Sharpness of fillet,

r

Fig. 5.1.8. Flat plate with fillets, in tension.

Fig. 5.1.11. Flat plate with angular notch, in tension or bending.

h

r h

h d D

r h

D d

h r

h

3.4

1

$d \geq 0$.

3.4

5

0.4

0.0

0.1

h

3.0

$d \geq$

Stress concentration factor, K

$D \geq d \leq 2h$

Stress concentration factor, K

h

3.0 Semi circ.

Semi-circle grooves $h \geq r$

0.2

grooves

1

2.6

2.6 ($h \geq r$)

2.2 Blunt Sharp

5

2.2 grooves grooves

0.

Sharp

grooves h

1.8 $\square$ 0.04

1.8 1 d

4

2

1.4 1.4

10

1.0 1.0

0.4 1.0 1.5 2 3 4 5 6 0.5 1.0 1.5 2 3 4 5 6 8 10 15 20

h h

Sharpness of groove, Sharpness of groove,

r r

Fig. 5.1.9. Flat plate with grooves, in bending. Fig. 5.1.12. Grooved shaft in torsion.

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FRACTURE AT LOW STRESSES 5-7

h

r h transition temperature of a material selected for a particular application

D d is suitably matched to its intended use temperature. The DBT can be

detected by plotting certain measurements from tensile or impact tests

against temperature. Usually the transition to brittle behavior is com-

plex, being neither fully ductile nor fully brittle. The range may extend

over 200°F (110 K) interval. The nil-ductility temperature (NDT), deter-

3.4 h mined by the drop weight test (see ASTM Standards), is an important

 $\square$ 0.05

d reference point in the transition range. When NDT for a particular steel

Stress concentration factor, K

3.0 is known, temperature-stress combinations can be specified which de-

D $\square$ d $\leq$ 2h fine the limiting conditions under which catastrophic fracture can occur.

0.1

2

In the Charpy V-notch (CVN) impact test, a notched-bar specimen

0.

2.6 Full fillets (h $\square$ r)

(Fig. 5.1.26) is used which is loaded in bending (see ASTM Standards).

Sharp fillets The energy absorbed from a swinging pendulum in fracturing the speci-

2.2 5

0. men is measured. The pendulum strikes the specimen at 16 to 19 ft

Blunt

(4.88 to 5.80 m)/s so that the specimen deformation associated with

1.8 1 fracture occurs at a rapid strain rate. This ensures a conservative mea-

fillets

sure of toughness, since in some materials, toughness is reduced by high

1.4 strain rates. A CVN impact energy vs. temperature curve is shown in

Fig. 5.1.14, which also shows the transitions as given by percent brittle

1.0 fracture and by percent lateral expansion. The CVN energy has no

0.5 1.0 2 3 4 5 7 10 20 40 analytical significance. The test is useful mainly as a guide to the frac-

h

ture behavior of a material for which an empirical correlation has been

Sharpness of fillet, established between impact energy and some rigorous fracture criterion.

r

For a particular grade of steel the CVN curve can be correlated with

Fig. 5.1.13. Filleted shaft in torsion.

NDT. (See ASME Boiler and Pressure Vessel Code.)

Fracture Mechanics This analytical method is used for ultra-high-

FRACTURE AT LOW STRESSES

strength alloys, transition-temperature materials below the DBT tem-

Materials under tension sometimes fail by rapid fracture at stresses perature, and some low-strength materials in heavy section thickness.

much below their strength level as determined in tests on carefully Fracture mechanics theory deals with crack extension where plastic

prepared specimens. These brittle, unstable, or catastrophic failures origi- effects are negligible or confined to a small region around the crack tip.

nate at preexisting stress-concentrating flaws which may be inherent in The present discussion is concerned with a through-thickness crack in a

a material. tension-loaded plate (Fig. 5.1.15) which is large enough so that the

The transition-temperature approach is often used to ensure fracture- crack-tip stress field is not affected by the plate edges. Fracture me-

safe design in structural-grade steels. These materials exhibit a charac- chanics theory states that unstable crack extension occurs when the

teristic temperature, known as the ductile brittle transition (DBT) tem- work required for an increment of crack extension, namely, surface

perature, below which they are susceptible to brittle fracture. The tran- energy and energy consumed in local plastic deformation, is exceeded

sition-temperature approach to fracture-safe design ensures that the by the elastic-strain energy released at the crack tip. The elastic-stress

Fig. 5.1.14. CVN transition curves. (Data from Westinghouse R & D Lab.)

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5-8 MECHANICAL PROPERTIES OF MATERIALS

field surrounding one of the crack tips in Fig. 5.1.15 is characterized by Table 5.1.4
Room-Temperature K_{Ic} Values on High-Strength

the stress intensity K_I , which has units of $(\text{lb } \sqrt{\text{in}}) / \text{in}^2$ or $(\text{N} \sqrt{\text{m}}) / \text{m}^2$. It is Materials*

a function of applied nominal stress σ , crack half-length a , and a geom- 0.2% YS, 1,000 in 2 K_{Ic} ,
1,000 in 2

etry factor Q : Material $(\text{MN}/\text{m}^2) \sqrt{\text{in}}$ $(\text{MN m}^{1/2} / \text{m}^2)$

$K_{Ic} \geq Q \sqrt{a}$ (5.1.1) 18% Ni maraging steel 300 (2,060) 46 (50.7)

for the situation of Fig. 5.1.15. For a particular material it is found that 18% Ni maraging steel 270
(1,850) 71 (78)

18% Ni maraging steel 198 (1,360) 87 (96)

as K_I is increased, a value K_{Ic} is reached at which unstable crack propa- Titanium 6-4 alloy 152
(1,022) 39 (43)

Titanium 6-4 alloy 140 (960) 75 (82.5)

Aluminum alloy 7075-T6 75 (516) 26 (28.6)

Aluminum alloy 7075-T6 64 (440) 30 (33)

* Determined at Westinghouse Research Laboratories.

crack, and loadings (Paris and Sih, "Stress Analysis of Cracks," STP-
381, ASTM, 1965). Failure occurs in all cases when K_t reaches K_{Ic} .

Fracture mechanics also provides a framework for predicting the occur-
rence of stress-corrosion cracking by using Eq. (5.1.2) with K_{Ic} replaced
by K_{Isc} , which is the material parameter denoting resistance to stress-
corrosion-crack propagation in a particular medium.

Two standard test specimens for K_{Ic} determination are specified in
ASTM standards, which also detail specimen preparation and test pro-
cedure. Recent developments in fracture mechanics permit treatment of
crack propagation in the ductile regime. (See "Elastic-Plastic Frac-
ture," ASTM.)

Fig. 5.1.15. Through-thickness crack geometry.

gation occurs. K_{Ic} depends on plate thickness B , as shown in Fig. 5.1.16. FATIGUE

It attains a constant value when B is great enough to provide plane-strain

Fatigue is generally understood as the gradual deterioration of a mate-

conditions at the crack tip. The low plateau value of K_{Ic} is an important

rial which is subjected to repeated loads. In fatigue testing, a specimen

material property known as the plane-strain critical stress intensity or

is subjected to periodically varying constant-amplitude stresses by

fracture toughness K_{Ic} . Values for a number of materials are shown in

means of mechanical or magnetic devices. The applied stresses may

Table 5.1.4. They are influenced strongly by processing and small

alternate between equal positive and negative values, from zero to max-

changes in composition, so that the values shown are not necessarily minimum positive or negative values, or between unequal positive and negative values. K_{Ic} can be used in the critical form of Eq. (5.1.1):

$$K_{Ic} \geq Q \sqrt{\sigma} \quad (5.1.2)$$

compression of equal numerical values obtained by rotating a smooth cylindrical specimen while under a bending load. A series of fatigue tests are made on a number of specimens of the material at different stress levels. The stress endured is then plotted against the number of cycles sustained. The predictions will be accurate so long as plate thickness B satisfies the plane-strain criterion: $B \geq 2.5(K_{Ic}/\sigma_{ys})^2$. They will be conservative if a plane-strain condition does not exist. A big advantage of the fracture mechanics approach is that stress intensity can be calculated by the stress-cycle diagram or S-N diagram. Instead of recording the equations analogous to (5.1.1) for a wide variety of geometries, types of data on cartesian coordinates, either stress is plotted vs. the logarithm of the number of cycles (Fig. 5.1.17) or both stress and cycles are plotted to logarithmic scales. Both diagrams show a relatively sharp bend in the curve near the fatigue limit for ferrous metals. The fatigue limit may be established for most steels between 2 and 10 million cycles. Nonferrous metals usually show no clearly defined fatigue limit. The S-N curves in these cases indicate a continuous decrease in stress values to several hundred million cycles, and both the stress value and the number of cycles sustained should be reported. See Table 5.1.5.

The mean stress (the average of the maximum and minimum stress values for a cycle) has a pronounced influence on the stress range (the algebraic difference between the maximum and minimum stress values). Several empirical formulas and graphical methods such as the "modified Goodman diagram" have been developed to show the influence of the mean stress on the stress range for failure. A simple but conservative approach (see Soderberg, Working Stresses, Jour. Appl. Mech., 2, Sept. 1935) is to plot the variable stress S_v (one-half the stress range) as ordinate vs. the mean stress S_m as abscissa (Fig. 5.1.18). At zero mean stress, the ordinate is the fatigue limit under completely reversed stress. Yielding will occur if the mean stress exceeds the yield stress S_o , and this establishes the extreme right-hand point of the diagram. A straight line is drawn between these two points. The coordi-

Fig. 5.1.16. Dependence of K_c and fracture appearance (in terms of percentage nates of any other point along this line are values of S_m and S_v which of square fracture) on thickness of plate specimens. Based on data for aluminum may produce failure.

7075-T6. (From Scrawly and Brown, STP-381, ASTM.) Surface defects, such as roughness or scratches, and notches or

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FATIGUE 5-9

Accordingly, the pragmatic approach to arrive at a solution to a design problem often takes a conservative route and sets $q \approx 1$. The exact material properties at play which are responsible for notch sensitivity are not clear.

Further, notch sensitivity seems to be higher, and ordinary fatigue strength lower in large specimens, necessitating full-scale tests in many cases (see Peterson, Stress Concentration Phenomena in Fatigue of

Fig. 5.1.18. Effect of mean stress on the variable stress for failure.

Metals, Trans. ASME, 55, 1933, p. 157, and Buckwalter and Horger, Investigation of Fatigue Strength of Axles, Press Fits, Surface Rolling and Effect of Size, Trans. ASM, 25, Mar. 1937, p. 229). Corrosion and

Fig. 5.1.17. The S-N diagrams from fatigue tests. (1) 1.20% C steel, quenched galling (due to rubbing of mating surfaces) cause great reduction of

and drawn at 860°F (460°C); (2) alloy structural steel; (3) SAE 1050, quenched fatigue strengths, sometimes amounting to as much as 90 percent of the

and drawn at 1,200°F (649°C); (4) SAE 4130, normalized and annealed; (5) ordi- original endurance limit. Although any corroding agent will promote

nary structural steel; (6) Duralumin; (7) copper, annealed; (8) cast iron (reversed bending).

severe corrosion fatigue, there is so much difference between the effects of "sea water" or "tap water" from different localities that numerical values are not quoted here.

shoulders all reduce the fatigue strength of a part. With a notch of Overstressing specimens above the fatigue limit for periods shorter

prescribed geometric form and known concentration factor, the reduc- than necessary to produce failure at that stress reduces the fatigue limit

tion in strength is appreciably less than would be called for by the in a subsequent test. Similarly, understressing below the fatigue limit

concentration factor itself, but the various metals differ widely in their may increase it. Shot peening, nitriding, and cold work usually improve

susceptibility to the effect of roughness and concentrations, or notch fatigue properties.

sensitivity. No very good overall correlation exists between fatigue properties

For a given material subjected to a prescribed state of stress and type and any other mechanical property of a material. The best correlation is of loading, notch sensitivity can be viewed as the ability of that material between the fatigue limit under completely reversed bending stress and to resist the concentration of stress incidental to the presence of a notch. the ordinary tensile strength. For many ferrous metals, the fatigue limit Alternately, notch sensitivity can be taken as a measure of the degree to is approximately 0.40 to 0.60 times the tensile strength if the latter is which the geometric stress concentration factor is reduced. An attempt below 200,000 lb/in². Low-alloy high-yield-strength steels often show is made to rationalize notch sensitivity through the equation $q = (K_f - 1) / (K_f + 1)$ ($K_f \geq 1$), where q is the notch sensitivity, K is the geometric stress concentration factor (from data similar to those in Figs. 5.1.5 to 5.1.13 in reversed shear is approximately 0.57 times that in reversed bending. and the like), and K_f is defined as the ratio of the strength of unnotched In some very important engineering situations components are cyclically stressed into the plastic range. Examples are thermal strains result- laboratory tests, and K is deduced either theoretically or from laboratory ing from temperature oscillations and notched regions subjected to secondary stresses. Fatigue life in the plastic or "low-cycle" fatigue range The value of q lies between 0 and 1, so that (1) if $q = 0$, $K_f = 1$ and the has been found to be a function of plastic strain, and low-cycle fatigue material is not notch-sensitive (soft metals such as copper, aluminum, testing is done with strain as the controlled variable rather than stress. and annealed low-strength steel); (2) if $q = 1$, $K_f = K$, the material is Fatigue life N and cyclic plastic strain ϵ_p tend to follow the relationship fully notch-sensitive and the full value of the geometric stress concentration factor is not diminished (hard, high-strength steel). In practice, q will lie somewhere between 0 and 1, but it may be hard to quantify. where C is a constant for a material when $N \geq 10^5$. (See Coffin, A Study

Table 5.1.5 Typical Approximate Fatigue Limits for Reversed Bending

Tensile strength, limit	Tensile strength, limit	Tensile strength, limit	Tensile strength, limit
Metal 1,000 lb/in ²	1,000 lb/in ²	Metal 1,000 lb/in ²	1,000 lb/in ²
Cast iron 20 - 50	6 - 18	Copper 32 - 50	12 - 17
Malleable iron 50	24	Monel 70 - 120	20 - 50
Cast steel 60 - 80	24 - 32	Phosphor bronze 55	12
Armco iron 44	24	Tobin bronze, hard 65	21
Plain carbon steels 60 - 150	25 - 75	Cast aluminum alloys 18 - 40	6 - 11

SAE 6150, heat-treated 200 80 Wrought aluminum alloys 25 – 70 8 – 18

Nitralloy 125 80 Magnesium alloys 20 – 45 7 – 17

Brasses, various 25 – 75 7 – 20 Molybdenum, as cast 98 45

Zirconium crystal bar 52 16 – 18 Titanium (Ti-75A) 91 45

NOTE: Stress, 1,000 lb/in² $\square$ 6.894 $\square$ stress, MN/m².

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of Cyclic-Thermal Stresses in a Ductile Material, Trans. ASME, 76, curve OA in Fig. 5.1.19 is the region of primary creep, AB the region

1954, p. 947.) of secondary creep, and BC that of tertiary creep. The strain rates, or

The type of physical change occurring inside a material as it is re- the slopes of the curve, are decreasing, constant, and increasing,

peatedly loaded to failure varies as the life is consumed, and a number respectively, in these three regions. Since the period of the creep test

of stages in fatigue can be distinguished on this basis. The early stages is usually much shorter than the duration of the part in service,

comprise the events causing nucleation of a crack or flaw. This is most various extrapolation procedures are followed (see Gittus, "Creep,

likely to appear on the surface of the material; fatigue failures generally Viscoelasticity and Creep Fracture in Solids," Wiley, 1975). See

originate at a surface. Following nucleation of the crack, it grows during Table 5.1.6.

the crack-propagation stage. Eventually the crack becomes large In practical applications the region of constant-strain rate (secondary

enough for some rapid terminal mode of failure to take over such as creep) is often used to estimate the probable deformation throughout the

ductile rupture or brittle fracture. The rate of crack growth in the crack- life of the part. It is thus assumed that this rate will remain constant

propagation stage can be accurately quantified by fracture mechanics during periods beyond the range of the test-data. The working stress is

methods. Assuming an initial flaw and a loading situation as shown in chosen so that this total deformation will not be excessive. An arbitrary

Fig. 5.1.15, the rate of crack growth per cycle can generally be ex- creep strength, which is defined as the stress which at a given tempera-

pressed as ture will result in 1 percent deformation in 100,000 h, has received a certain amount of recognition, but it is advisable to determine the proper

$da/dN \square C_0(\square KI)^n$ (5.1.3)

stress for each individual case from diagrams of stress vs. creep rate

where C_0 and n are constants for a particular material and $\square KI$ is the (Fig. 5.1.20) (see "Creep Data," ASTM and ASME).

range of stress intensity per cycle. KI is given by (5.1.1). Using (5.1.3),

it is possible to predict the number of cycles for the crack to grow to a size at which some other mode of failure can take over. Values of the

constants C_0 and n are determined from specimens of the same type as those used for determination of K_{Ic} but are instrumented for accurate measurement of slow crack growth.

Constant-amplitude fatigue-test data are relevant to many rotary-machinery situations where constant cyclic loads are encountered.

There are important situations where the component undergoes variable loads and where it may be advisable to use random-load testing.

In this method, the load spectrum which the component will experience in service is determined and is applied to the test specimen artificially.

CREEP

Experience has shown that, for the design of equipment subjected to sustained loading at elevated temperatures, little reliance can be placed on the usual short-time tensile properties of metals at those temperatures. Under the application of a constant load it has been found that materials, both metallic and nonmetallic, show a gradual flow or creep even for stresses below the proportional limit at elevated temperatures.

Similar effects are present in low-melting metals such as lead at room temperature. Fig. 5.1.20. Creep rates for 0.35% C steel.

The deformation which can be permitted in the satisfactory operation of most high-temperature equipment is limited.

In metals, creep is a plastic deformation caused by slip occurring at elevated temperatures ($^{\circ}\text{F}$) and stresses (in 1,000 lb/in²) for stated

directions in the individual crystals, together creep rates (percent per 1,000 h) for wrought nonferrous metals are as

with some flow of the grain-boundary material. After complete release follows:

of load, a small fraction of this plastic deformation is recovered with 60-40 Brass: Rate 0.1, temp. 350 (400), stress 8 (2); rate 0.01, temp

time. Most of the flow is nonrecoverable for metals. 300 (350) [400], stress 10 (3) [1].

Since the early creep experiments, many different types of tests have been used. Phosphor bronze: Rate 0.1, temp 400 (550) [700] [800], stress 15 (6)

come into use. The most common are the long-time creep test under [4] [4]; rate 0.01, temp 400 (550) [700], stress 8 (4) [2].

constant tensile load and the stress-rupture test. Other special forms are Nickel: Rate 0.1, temp 800 (1000), stress 20 (10).

the stress-relaxation test and the constant-strain-rate test. 70 CU, 30 NI. Rate 0.1, temp 600 (750), stress 28 (13 - 18); rate 0.01,

The long-time creep test is conducted by applying a dead weight to one end of a lever system, the other end being attached to the specimen Aluminum alloy 17 S (Duralumin): Rate 0.1, temp 300 (500) [600],

surrounded by a furnace and held at constant temperature. The axial stress 22 (5) [1.5].

deformation is read periodically throughout the test and a curve is plotted. Lead pure (commercial) (0.03 percent Ca): At 110°F, for rate 0.1

ted of the strain ϵ_0 as a function of time t (Fig. 5.1.19). This is repeated percent the stress range, lb/in², is 150 – 180 (60 – 140) [200 – 220]; for

for various loads at the same testing temperature. The portion of the rate of 0.01 percent, 50 – 90 (10 – 50) [110 – 150].

Stress, 1,000 lb/in² $\approx$ 6.894 $\times$ stress, MN/m², $t_k \approx 5/9(t_F \leq 459.67)$.

Structural changes may occur during a creep test, thus altering the metallurgical condition of the metal. In some cases, premature rupture appears at a low fracture strain in a normally ductile metal, indicating that the material has become embrittled. This is a very insidious condition and difficult to predict. The stress-rupture test is well adapted to study this effect. It is conducted by applying a constant load to the specimen in the same manner as for the long-time creep test. The nominal stress is then plotted vs. the time for fracture at constant temperature Fig. 5.1.19. Typical creep curve. on a log-log scale (Fig. 5.1.21).

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CREEP 5-11

Table 5.1.6 Stresses for Given Creep Rates and Temperatures*

Creep rate 0.1% per 1,000 h Creep rate 0.01% per 1,000 h

Material Temp, °F 800 900 1,000 1,100 1,200 800 900 1,000 1,100 1,200

Wrought steels:

SAE 1015 17 – 27 11 – 18 3 – 12 2-7 1 10 – 18 6 – 14 3-8 1

0.20 C, 0.50 Mo 26 – 33 18 – 25 9 – 16 2-6 1-2 16 – 24 11 – 22 4 – 12 2 1

0.10 – 0.25 C, 4 – 6 Cr $\leq$ Mo 22 15 – 18 9 – 11 3-6 2-3 14 – 17 11 – 15 4-7 2-3 1-2

SAE 4140 27 – 33 20 – 25 7 – 15 4-7 1-2 19 – 28 12 – 19 3-8 2-4 1

SAE 1030 – 1045 8 – 25 5 – 15 5 2 1 5 – 15 3-7 2-4 1

Commercially pure iron 7 4 3 5 2

0.15 C, 1 – 2.5 Cr, 0.50 Mo 25 – 35 18 – 28 8 – 20 6-8 3-4 20 – 30 12 – 18 3 – 12 2-5 1-2

SAE 4340 20 – 40 15 – 30 2 – 12 1-3 8 – 20 1-6

SAE X3140 7 – 10 5-4 3-8 1-2

0.20 C, 4 – 6 Cr 30 10 – 20 7 – 10 1 3-5

0.25 C, 4 – 6 Cr $\leq$ W 30 10 – 15 4 – 10 2-8 6 – 11 2-7

0.16 C, 1.2 Cu 18 10 – 15 3 1 10 – 18 7 – 12

0.20 C, 1 Mo 35 27 12 25 12 6

0.10 – 0.40 C, 0.2 – 0.5 Mo,

1 – 2 Mn 30 – 40 12 – 20 4 – 14 25 – 28 8 – 15 2-8 0.5

SAE 2340 7 – 12 5 2

SAE 6140 30 12 4 7 6 1

SAE 7240 30 21 6 – 15 2 30 11 3-9 1

$Cr \leq Va \leq W$, various 20 - 70 14 - 30 5 - 15 18 - 50 8 - 18 2 - 13

Temp, °F 1,100 1,200 1,300 1,400 1,500 1,000 1,100 1,200 1,300 1,400

Wrought chrome-nickel steels:

18-8† 10 - 18 5 - 11 3 - 10 2-5 2.5 11 - 16 5 - 12 2 - 10 1-2

10 - 25 Cr, 10 - 30 Ni‡ 10 - 20 5 - 15 3 - 10 2-5 6 - 15 3 - 10 2-8 1-3

Temp, °F 800 900 1,000 1,100 1,200 800 900 1,000 1,100 1,200

Cast steels:

0.20 - 0.40 C 10 - 20 5 - 10 3 8 - 15 1

0.10 - 0.30 C, 0.5 - 1 Mo 28 20 - 30 6 - 12 2 20 10 - 15 2-5

0.15 - 0.30 C, 4 - 6 Cr $\leq$ Mo 25 - 30 15 - 25 8 - 15 8 20 - 25 9 - 15 2-7 2

18 - 8§ 20 - 25 15 10 20 15 8

Cast iron 20 8 4 10 2

Cr Ni cast iron 9 3

* Based on 1,000-h tests. Stresses in 1,000 lb/in².

† Additional data. At creep rate 0.1 percent and 1,000 (1,600)°F the stress is 18 - 25 (1); at creep rate 0.01 percent at 1,500°F, the stress is 0.5.

‡ Additional data. At creep rate 0.1 percent and 1,000 (1,600)°F the stress is 10 - 30 (1).

§ Additional data. At creep rate 0.1 percent and 1,600°F the stress is 3; at creep rate 0.01 and 1,500°F, the stress is 2 - 3.

The stress reaction is measured in the constant-strain-rate test while of a velocity-modified temperature (see MacGregor and Fisher, A Ve-

the specimen is deformed at a constant strain rate. In the relaxation test, locity-modified Temperature for the Plastic Flow of Metals, Jour. Appl.

the decrease of stress with time is measured while the total strain (elastic Mech., Mar. 1945) simplifies the creep problem in reducing the number

$\leq$ plastic) is maintained constant. The latter test has direct application to of variables.

the loosening of turbine bolts and to similar problems. Although some Superplasticity Superplasticity is the property of some metals

correlation has been indicated between the results of these various types and alloys which permits extremely large, uniform deformation at

of tests, no general correlation is yet available, and it has been found elevated temperature, in contrast to conventional metals which neck

necessary to make tests under each of these special conditions to obtain down and subsequently fracture after relatively small amounts of

satisfactory results. plastic deformation. Superplastic behavior requires a metal with

The interrelationship between strain rate and temperature in the form small equiaxed grains, a slow and steady rate of deformation (strain

Fig. 5.1.21 Relation between time to failure and stress for a 3% chromium steel. (1) Heat treated 2 h at 1,740°F

(950°C) and furnace cooled; (2) hot rolled and annealed 1,580°F (860°C).

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5-12 MECHANICAL PROPERTIES OF MATERIALS

rate), and a temperature elevated to somewhat more than half the known load into the surface of a material and measuring the diameter of melting point. With such metals, large plastic deformation can be the indentation left after the test. The Brinell hardness number, or simply brought about with lower external loads; ultimately, that allows the the Brinell number, is obtained by dividing the load used, in kilograms, use of lighter fabricating equipment and facilitates production of by the actual surface area of the indentation, in square millimeters. The finished parts to near-net shape. result is a pressure, but the units are rarely stated.

$$H_{\text{B}} = \frac{P}{A} \quad \text{m 1.0 BHN } P \text{ mm } D$$

2

□

$$(D \geq \sqrt{D^2 \geq d^2})$$

where BHN is the Brinell hardness number; P the imposed load, kg; D the diameter of the spherical indenter, mm; and d the diameter of the d m

□ □ □ □ 0.5 resulting impression, mm.

m

In □

⦿ □ □ Hardened-steel bearing balls may be used for hardness up to 450, but beyond this hardness specially treated steel balls or jewels should be ⦿ □ □ • used to avoid flattening the indenter. The standard-size ball is 10 mm m □ tan □ □ ⦿ □ □ □ □ □ •

and the standard loads 3,000, 1,500, and 500 kg, with 100, 125, and 0 250 kg sometimes used for softer materials. If for special reasons any In □ • In □ •

other size of ball is used, the load should be adjusted approximately as (a) (b)

follows: for iron and steel, $P \geq 30D^2$; for brass, bronze, and other soft

Fig. 5.1.22. Stress and strain rate relations for superplastic alloys. (a) Log-log metals, $P \geq 5D^2$; for extremely soft metals, $P \geq D^2$ (see “Methods of plot of □ □ $K \dot{\epsilon}^m$; (b) m as a function of strain rate. Brinell Hardness Testing,” ASTM). Readings obtained with other than

the standard ball and loadings should have the load and ball size ap-

Stress and strain rates are related for a metal exhibiting superplas- pended, as such readings are only approximately equal to those obtained

ticity. A factor in this behavior stems from the relationship between under standard conditions.

the applied stress and strain rates. This factor m — the strain rate The size of the specimen should be sufficient to ensure that no part of

sensitivity index — is evaluated from the equation □ □ $K \dot{\epsilon}^m$, where □ the plastic flow around the impression reaches a free surface, and in no

is the applied stress, K is a constant, and $\dot{\epsilon}$ is the strain rate. Figure 5.1.22a shows the thickness of the impression should be less than 10 times the depth of the impression.

5.1.22a plots a stress/strain rate curve for a superplastic alloy. The load should be applied steadily and should remain on for at

log-log coordinates. The slope of the curve defines m , which is maximum at least 15 s in the case of ferrous materials and 30 s in the case of most

metals at the point of inflection. Figure 5.1.22b shows the variation of m for nonferrous materials. Longer periods may be necessary on certain soft

metals of m versus $\ln \dot{\epsilon}$. Ordinary metals exhibit low values of m — 0.2 or materials that exhibit creep at room temperature. In testing thin materi-

als; for those behaving superplastically, $m \approx 0.6$ to 0.8 . As m approaches 1, it is not permissible to pile up several thicknesses of material under

the indenter, the behavior of the metal will be quite similar to that of a Newtonian viscous solid, which elongates plastically without the true readings. With such materials, smaller indenters and loads, or

necking down. different methods of hardness testing, are necessary.

In Fig. 5.1.22a, in region I, the stress and strain rates are low and in the standard Brinell test, the

diameter of the impression is measured. In region III, the stress is measured with a low-power hand microscope, but for production work sev-

eral testing machines are available which automatically measure the

depth of the impression and from this give readings of hardness. Such

machines should be calibrated frequently on test blocks of known hard-

ness.

ness.

HARDNESS In the Rockwell method of hardness testing, the depth of penetration of an indenter under certain arbitrary conditions of test is determined. The

Hardness has been variously defined as resistance to local penetration, indenter may be either a steel ball of some specified diameter or a

to scratching, to machining, to wear or abrasion, and to yielding. The spherical-tipped conical diamond of 120° angle and 0.2-mm tip radius,

multiplicity of definitions, and corresponding multiplicity of hardness — called a "Brale." A minor load of 10 kg is first applied which causes an

measuring instruments, together with the lack of a fundamental definition, initial penetration and holds the indenter in place. Under this condition,

tion, indicates that hardness may not be a fundamental property of a material but rather a composite one including yield strength, work hard-

ness, true tensile strength, modulus of elasticity, and others. taken while the minor load is still on. The hardness number may then be

are 60, 100, or 150 kg. Upon removal of the major load, the reading is

are 60, 100, or 150 kg.

Scratch hardness is measured by Mohs scale of minerals (Sec. 1.2) read directly from the scale which measures penetration, and this scale which is so arranged that each mineral will scratch the mineral of the is so arranged that soft materials with deep penetration give low hardness numbers. In recent mineralogical work and in certain microscopic metallurgical work, jeweled scratching points either with a set A variety of combinations of indenter and major load are possible; load or else loaded to give a set width of scratch have been used. Hardness - the most commonly used are RB using as indenter a 1/16-in ball and a hardness in its relation to machinability and to wear and abrasion is generally major load of 100 kg and RC using a Brale as indenter and a major load ally dealt with in direct machining or wear tests, and little attempt is of 150 kg (see "Rockwell Hardness and Rockwell Superficial Hardness made to separate hardness itself, as a numerically expressed quantity, of Metallic Materials," ASTM).

from the results of such tests. Compared with the Brinell test, the Rockwell method makes a The resistance to localized penetration, or indentation hardness, is smaller indentation, may be used on thinner material, and is more rapid, widely used industrially as a measure of hardness, and indirectly as an since hardness numbers are read directly and need not be calculated. indicator of other desired properties in a manufactured product. The However, the Brinell test may be made without special apparatus and is indentation tests described below are essentially nondestructive, and in somewhat more widely recognized for laboratory use. There is also a most applications may be considered nonmarring, so that they may be Rockwell superficial hardness test similar to the regular Rockwell, except applied to each piece produced; and through the empirical relationships that the indentation is much shallower. of hardness to such properties as tensile strength, fatigue strength, and The Vickers method of hardness testing is similar in principle to the impact strength, pieces likely to be deficient in the latter properties may Brinell in that it expresses the result in terms of the pressure under be detected and rejected. the indenter and uses the same units, kilograms per square millimeter. Brinell hardness is determined by forcing a hardened sphere under a The indenter is a diamond in the form of a square pyramid with an apical

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TESTING OF MATERIALS 5-13

angle of 136°, the loads are much lighter, varying between 1 and of the work in hardness see Williams, "Hardness and Hardness Measurements," ASM. 120 kg, and the impression is measured by means of a medium-power microscope.

TESTING OF MATERIALS

V $\square$ P/(0.5393d²)

Testing Machines Machines for the mechanical testing of materials

where V is the Vickers hardness number, sometimes called the diamond- usually contain elements (1) for gripping the specimen, (2) for deform- pyramid hardness (DPH); P the imposed load, kg; and d the diagonal of ing it, and (3) for measuring the load required in performing the defor- indentation, mm. The Vickers method is more flexible and is considered mation. Some machines (ductility testers) omit the measurement of load to be more accurate than either the Brinell or the Rockwell, but the and substitute a measurement of deformation, whereas other machines equipment is more expensive than either of the others and the Rockwell include the measurement of both load and deformation through appa- is somewhat faster in production work.

ratus either integral with the testing machine (stress-strain recorders) or

Among the other hardness methods may be mentioned the Sclero- auxiliary to it (strain gages). In most general-purpose testing machines, scope, in which a diamond-tipped “hammer” is dropped on the surface the deformation is controlled as the independent variable and the result- and the rebound taken as an index of hardness. This type of apparatus is ing load measured, and in many special-purpose machines, particularly seriously affected by the resilience as well as the hardness of the mate- those for light loads, the load is controlled and the resulting deformation rial and has largely been superseded by other methods. In the Monotron is measured. Special features may include those for constant rate of method, a penetrator is forced into the material to a predetermined depth loading (pacing disks), for constant rate of straining, for constant load and the load required is taken as the indirect measure of the hardness. maintenance, and for cyclical load variation (fatigue).

This is the reverse of the Rockwell method in principle, but the loads

In modern testing systems, the load and deformation measurements are and indentations are smaller than those of the latter. In the Herbert made with load-and-deformation-sensitive transducers which generate pendulum, a 1-mm steel or jewel ball resting on the surface to be tested electrical outputs. These outputs are converted to load and deformation acts as the fulcrum for a 4-kg compound pendulum of 10-s period. The readings by means of appropriate electronic circuitry. The readings are swinging of the pendulum causes a rolling indentation in the material, commonly displayed automatically on a recorder chart or digital meter, or and from the behavior of the pendulum several factors in hardness, such they are read into a computer. The transducer outputs are typically used as work hardenability, may be determined which are not revealed by also as feedback signals to control the test mode (constant loading, con-

other methods. Although the Herbert results are of considerable significance, the instrument is suitable for laboratory use only (see Herbert, load cell attached to the test machine frame, with electrical output to a bridge circuit and amplifier. The load cell operation depends on change of electrical resistivity with deformation (and load) in the transducer element. The Pendulum Hardness Tester, and Some Recent Developments in Hardness Testing, Engineer, 135, 1923, pp. 390, 686). In the Herbert cloudburst test, a shower of steel balls, dropped from a predetermined height, dulls the surface of a hardened part in proportion to its softness to the test specimen gage length, and operates on the same principle as the load cell transducer: the change in electrical resistance in the specimen and thus reveals defective areas. A variety of mutual indentation methods, in which crossed cylinders or prisms of the material to be tested are forced together, give results comparable with the Brinell test. These are also available which do not make physical contact with the specimen. particularly useful on wires and on materials at high temperatures. Verification and classification of extensometers is controlled by ASTM Standards. The relation among the scales of the various hardness methods is not exact, since no two measure exactly the same sort of hardness, and a usually by means of a screw-driven mechanism, but it may also be applied relationship determined on steels of different hardnesses will be found by means of hydraulic and servohydraulic systems. In each case the load only approximately true with other materials. The Vickers-Brinell relationship is nearly linear up to at least 400, with the Vickers approximately 5 percent higher than the Brinell (actual values run from ≤ 2 to ≤ 11). The application system responds to control inputs from the load and deformation transducers. Important features in test machine design are the methods used for reducing friction, wear, and backlash. In older testing machines, test loads were determined from the machine itself (e.g., a pressure become more widely divergent owing to the flattening of the Brinell reading from the machine hydraulic pressure) so that machine friction ball. The Brinell-Rockwell relation is fairly satisfactory and is shown in made an important contribution to inaccuracy. The use of machine-independent transducers in modern testing has eliminated much of this source given on the same plot.

of error.

The hardness of wood is defined by the ASTM as the load in pounds Grips should not only hold the test specimen against slippage but

required to force a ball 0.444 in in diameter into the wood to a depth should also apply the load in the desired manner. Centering of the load is

of 0.222 in, the speed of penetration being 1/4 in/min. For a summary of great importance in compression testing, and should not be neglected in

tension testing if the material is brittle. Figure 5.1.24 shows the theoretical errors due to off-center loading; the results are directly applicable to compression tests using swivel loading blocks. Swivel (ball-and-socket) holders or compression blocks should be used with all except the most ductile materials, and in compression testing of brittle materials (concrete, stone, brick), any rough faces should be smoothly capped with plaster of paris and one-third portland cement. Serrated grips may be used to hold ductile materials or the shanks of other holders in tension; a taper of 1 in 6 on the wedge faces gives a self-tightening action without excessive jamming. Ropes are ordinarily held by wet eye splices, but braided ropes or small cords may be given several turns over a fixed pin and then clamped. Wire ropes should be zinced into forged sockets (solder and lead have insufficient strength). Grip selection for tensile testing is described in ASTM standards.

Accuracy and Calibration ASTM standards require that commercial machines have errors of less than 1 percent within the “loading range” when checked against acceptable standards of comparison at

Fig. 5.1.23. Hardness scales. least five suitably spaced loads. The “loading range” may be any range

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5-14 MECHANICS OF MATERIALS

through which the preceding requirements for accuracy are satisfied, Two standard forms of test specimens (ASTM) are shown in Figs.

except that it shall not extend below 100 times the least load to which 5.1.25 and 5.1.26. In wrought materials, and particularly in those which

the machine will respond or which can be read on the indicator. The use of calibration plots or tables to correct the results of an otherwise inaccurate machine is not permitted under any circumstances. Machines with errors less than 0.1 percent are commercially available (Tate-Emery and others), and somewhat greater accuracy is possible in the most refined research apparatus.

Fig. 5.1.25. Test specimen, 2-in (50-mm) gage length, 1/2-in (12.5-mm) diameter. Others available for 0.35-in (8.75-mm) and 0.25-in (6.25-mm) diameters. (ASTM).

Fig. 5.1.26. Charpy V-notch impact specimens. (ASTM.)

Fig. 5.1.24. Effect of centering errors on brittle test specimens.

Dead loads may be used to check machines of low capacity; accu- have been cold-worked, different properties may be expected in differ-

rately calibrated proving levers may be used to extend the range of ent directions with respect to the direction of the applied work, and the

available weights. Various elastic devices (such as the Morehouse prov- test specimen should be cut out from the parent material in such a way

ing ring) made of specially treated steel, with sensitive distortion-meas- as to give the strength in the desired direction. With the exception of

suring devices, and calibrated by dead weights at the NIST (formerly fatigue specimens and specimens of extremely brittle materials, surface

Bureau of Standards) are among the most satisfactory means of checking finish is of little practical importance, although extreme roughness tends

the higher loads. to decrease the ultimate elongation.

5.2 MECHANICS OF MATERIALS

by J. P. Vidosic

REFERENCES: Timoshenko and MacCullough, "Elements of Strength of Materi- tures," Lincoln Arc Welding Foundation. "Characteristics and Applications

als," Van Nostrand. Seeley, "Advanced Mechanics of Materials," Wiley. Timofte of Resistance Strain Gages," Department of Commerce, NBS Circ. 528,

shenko and Goodier, "Theory of Elasticity," McGraw-Hill. Phillips, "Introduc- 1954.

tion to Plasticity," Ronald. Van Den Broek, "Theory of Limit Design," Wiley.

Hetényi, "Handbook of Experimental Stress Analysis," Wiley. Dean and Doug- EDITOR'S NOTE: The almost universal availability and utilization of computers

las, "Semi-Conductor and Conventional Strain Gages," Academic. Robertson in engineering practice has led to the development of many forms of software

and Harvey, "The Engineering Uses of Holography," University Printing House, individually tailored to the solution of specific design problems in the area of

London. Sellers, "Basic Training Guide to the New Metrics and SI Units," Na- mechanics of materials. Their use will permit the reader to amplify and supple-

tional Tool, Die and Precision Machining Association. Roark and Young, "Form-ment a good portion of the formulary and tabular collection in this section, as well

mulas for Stress and Strain,” McGraw-Hill. Perry and Lissner, “The Strain Gage as utilize those powerful computational tools in newer and more powerful tech-

Primer," McGraw-Hill. Donnell, "Beams, Plates, and Sheets," Engineering So- niques to facilitate solutions to problems. Many of the approximate methods,

cieties Monographs, McGraw-Hill. Griffel, "Beam Formulas" and "Plate For- involving laborious iterative mathematical schemes, have been supplanted by the

mulas,” Ungar. Durelli et al., “Introduction to the Theoretical and Experimental computer.

Developments along those lines continue apace and bid fair to expand

Analysis of Stress and Strain," McGraw-Hill. "Stress Analysis Manual," De- the types of problems handled, all with greater confidence in the results obtained

partment of Commerce, Pub. no. AD 759 199, 1969. Blodgett , "Welded Struc- thereby.

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SIMPLE STRESSES AND STRAINS 5-15

Main Symbols paraffin; μ ↓ 0 for cork. For concrete, μ varies from 0.10 to 0.20 at Unit Stress working stresses and can reach 0.25 at higher stresses; μ for ordinary glass is about 0.25. In the absence of definitive data, μ for most structural metals can be taken to lie between 0.25 and 0.35. Extensive listings S_v or S_s pure shearing of Poisson's ratio are found in other sections; see Tables 5.1.3 and 6.1.9.

T true (ideal) stress

S_p proportional elastic limit

S_y yield point

S_M ultimate strength, tension

S_c ultimate compression

S_v vertical shear in beams

S_R modulus of rupture

Moment

M bending

M_t torsion

External Action

P force

G weight of body

W weight of load

V external shear Fig. 5.2.1

Modulus of Elasticity

Stress is an internal distributed force, or, force per unit area; it is the

E longitudinal

G shearing internal mechanical reaction of the material accompanying deformation

K bulk modulus. Stresses always occur in pairs. Stresses are normal [tensile stress ($\leq$) and compressive stress ($\geq$)]; and tangential, or shearing.

U_p modulus of resilience

U_R ultimate resilience

Geometric

l length

A area

V volume

v velocity

r radius of gyration

I rectangular moment of inertia

I_P or J polar moment of inertia

Deformation Fig. 5.2.2

$e, e_{\diamond}$ gross deformation

δ , δ/L unit deformation; strain Intensity of stress, or unit stress, S , lb/in² (kgf/cm²), is the amount of

d or ϕ unit, angular force per unit of area (Fig. 5.2.3). P is the load acting through the center

s unit, lateral of gravity of the area. The uniformly distributed normal stress is

ν Poisson's ratio

n reciprocal of Poisson's ratio $S = P/A$

r radius When the stress is not uniformly distributed, $S = dP/dA$.

f deflection A long rod will stretch under its own weight G and a terminal load P

(see Fig. 5.2.4). The total elongation e is that due to the terminal load

plus that due to one-half the weight of the rod considered as acting at

SIMPLE STRESSES AND STRAINS the end.

Deformations are changes in form produced by external forces or loads $e = (Pl \leq Gl/2)/(AE)$

that act on nonrigid bodies. Deformations are longitudinal, e , a lengthen-

ing ($\leq$) or shortening ($\geq$) of the body; and angular, ϕ , a change of angle The maximum stress is at the upper end.

between the faces. When a load is carried by several paths to a support, the different paths

Unit deformation (dimensionless number) is the deformation in unit

take portions of the load in proportion to their stiffness, which is con-

distance. Unit longitudinal deformation (longitudinal strain), $\delta = e/l$ trolled by material (E) and by design.

(Fig. 5.2.1). Unit angular-deformation $\tan \phi$ equals ϕ approx (Fig. EXAMPLE. Two pairs of bars rigidly connected (with the same elongation)

5.2.2). carry a load P_0 (Fig. 5.2.5). A_1 , A_2 and E_1 , E_2 and P_1 , P_2 and S_1 , S_2 are cross

The accompanying lateral deformation results in unit lateral defor- sections, moduli of elasticity,

loads, and stresses of the bars, respectively; $e =$

mation (lateral strain) $\delta = e/L$ (Fig. 5.2.1). For homogeneous, iso- elongation.

tropic material operating in the elastic region, the ratio $\delta = \delta$ is a constant $e = P_1 l / (E_1 A_1) = P_2 l / (E_2 A_2)$

and is a definite property of the material; this ratio is called Poisson's $P_0 = 2P_1 \leq 2P_2$

ratio ν . $S_2 = P_2 / A_2 = 1/2 [P_0 E_2 / (E_1 A_1 \leq E_2 A_2)]$

A fundamental relation among the three interdependent constants E , G , $S_1 = 1/2 [P_0 E_1 / (E_1 A_1 \leq E_2 A_2)]$

and ν for a given material is $E = 2G(1 \leq \nu)$. Note that ν cannot be Temperature Stresses When the deformation arising from change

larger than 0.5; thus the shearing modulus G is always smaller than the of temperature is prevented, temperature stresses arise that are propor-

elastic modulus E . At the extremes, for example, $\nu \downarrow 0.5$ for rubber and tional to the amount of deformation that is prevented. Let a ν coeffi-

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Fig. 5.2.7 Fig. 5.2.8

and Fig. 5.2.8 a common form of test piece that introduces bending stresses.

Let Fig. 5.2.9 represent the symmetric section of area A with a shearing force V acting through its centroid. If pure shear exists, $S_v = V/A$, and this shear would be uniformly distributed over the area A . When this

Fig. 5.2.3 Fig. 5.2.4 shear is accompanied by bending (transverse shear in beams), the unit shear coefficient of expansion per degree of temperature, α length of bar at temperature t_1 , and l_2 length at temperature t_2 . Then

$$l_2 = l_1[1 + \alpha(t_2 - t_1)]$$

If, subsequently, the bar is cooled to a temperature t_1 , the proportionate deformation is $s = \alpha(t_2 - t_1)$ and the corresponding unit stress $S = E\alpha(t_2 - t_1)$. For coefficients of expansion, see Sec. 4. In the case of steel, a change of temperature of 12°F (6.7 K , 6.7°C) will cause in general a unit stress of $2,340\text{ lb/in}^2$ (164 kgf/cm^2).

Fig. 5.2.9

S_v increases from the extreme fiber to its maximum, which may or may not be at the neutral axis OZ . The unit shear parallel to OZ at a point d distant from the neutral axis (Fig. 5.2.9) is

$$S_v =$$

$$\frac{V}{Ib}$$

$$\frac{\int yz\,dy}{I}$$

$$d$$

$$e$$

$$yz\,dy$$

$$yz\,dy$$

where z is the section width at distance y ; and I is the moment of inertia of the entire section about the neutral axis OZ . Note that $\int yz\,dy$ is the Fig. 5.2.5 first moment of the area above d with respect to axis OZ . For a rectangular cross section (Fig. 5.2.10a),

Shearing stresses (Fig. 5.2.2) act tangentially to surface of contact and do not change length of sides of elementary volume; they change the angle between faces and the length of diagonal. Two pairs of shearing

$$\frac{3}{2} \frac{V}{bh}$$

$$S_v =$$

$$\frac{3}{2} \frac{V}{bh}$$

$$1 \geq \frac{3}{2} \frac{V}{bh} \frac{y}{h}$$

$$h$$

$$2$$

stresses must act together. Shearing stress intensities are of equal magnitude $S_v(\text{max}) = \frac{3}{2} \frac{V}{bh}$ for $y = 0$

tude on all four faces of an element. $S_v \neq S_{\phi v}$ (Fig. 5.2.6). $2 bh \geq 2 A$

For a circular cross section (Fig. 5.2.10b),

$$4 V$$

$$S_v \neq$$

$$3 \pi r^2$$

$$1 \geq \pi \pi \pi y$$

$$r$$

$$2$$

$$4 V \neq 4 V$$

$$S_v (\max) \neq \pi \pi \text{ for } y \neq 0$$

$$3 \pi r^2 \geq 3 A$$

Fig. 5.2.6

In the presence of pure shear on external faces (Fig. 5.2.6), the resultant stress S on one diagonal plane at 45° is pure tension and on the other diagonal plane pure compression; $S \neq S_v \neq S_{\phi v}$. S on diagonal plane is called “diagonal tension” by writers on reinforced concrete. Failure under pure shear is difficult to produce experimentally, except under torsion and in certain special cases. Figure 5.2.7 shows an ideal case, Fig. 5.2.10

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SIMPLE STRESSES AND STRAINS 5-17

Table 5.2.1 Resilience per Unit of Volume Up

($S \neq$ longitudinal stress; $S_v \neq$ shearing stress; $E \neq$ tension modulus of elasticity; $G \neq$ shearing modulus of elasticity)

Tension or compression / $S \geq 2/E$

12 Torsion

Shear / $S \geq 2v / G$

12 Solid circular 14 2

/ $S \neq v / G$

Beams (free ends) $R_{12} \leq R_{22} \geq 1 S \geq 2v$

Rectangular section, bent in arc of circle; / $S \geq 2/E$

16 Hollow, radii R_1 and R_2

no shear $R_{12} \geq 4 G$

Ditto, circular section / $S \geq 2/E$

18 Springs

Concentrated center load; rectangular / $S \geq 2/E$

1 18 Carriage / $S \geq 2/E$

16

cross section Flat spiral, rectangular section / $S \geq 2/E$

1 24

1/4S 2/G

Ditto, circular cross section / S 2/E

1 24 Helical: axial load, circular wire v

/S 2/E Helical: axial twist 1/8S 2/E

Uniform load, rectangular cross section 5 36

3/32S 2/E Helical: axial twist, rectangular section 1/6S 2/E

1-beam section, concentrated center load

For a circular ring (thickness small in comparison with the major the elastic limit. For normal stress, resilience $\square$ work of deformation $\square$

diameter), $S_v(\max) \square 2V/A$, for $y \square 0$. average force times deformation $\square 1/2 P_e \square 1/2 AS \square SI/E \square 1/2 S 2V/E$.

For a square cross section (diagonal vertical, Fig. 5.2.10c), Modulus of resilience U_p (in $\square \text{ lb/in}^3$) [$\text{cm} \square \text{ kgf/cm}^3$], or unit resilience,

$\square \square \square$

is the elastic energy stored up in a cubic inch of material at the elastic

$V \sqrt{2} y \sqrt{2} y^2$

$S_v \square 1 \leq \geq 4$ limit. For normal stress,

$a^2 a a$

$V e U_p \square 1/2 S 2p /E$

$S_v (\max) \square 1.591$ for $y \square$

A 4 The unit resilience for any other kind of stress, as shearing, bending, torsion, is a constant times one-half the square of the stress divided by

For an I-shaped cross section (Fig. 5.2.10d),

the appropriate modulus of elasticity. For values, see Table 5.2.1.

$S_v (\max) \square$

$3 V$

$4 a$

$\square be 2 \geq (b \geq a)f^2$

$be 3 \geq (b \geq a)f^3$

$\square$ for $y \square 0$

Unit rupture work U_R , sometimes called ultimate resilience, is measured by the area of the stress-deformation diagram to rupture.

Elasticity is the ability of a material to return to its original dimensions $U_R \square 1/3 e_u(S_y \leq 2S_M)$ approx

after the removal of stresses. The elastic limit S_p is the limit of stress

within which the deformation completely disappears after the removal where e_u is the total deformation at rupture.

of stress; i.e., no set remains. For structural steel, $U_R \square 1/3 \square 27/100 \square [35,000 \leq (2 \square 60,000)] \square$

Hooke's law states that, within the elastic limit, deformation produced 13,950 in $\square \text{ lb/in}^3$ (982 cm $\square \text{ kgf/cm}^3$).

is proportional to the stress. Unless modified, the deduced formulas of EXAMPLE 1. A load $P \square 40,000 \text{ lb}$ compresses a wooden block of cross-sec-

mechanics apply only within the elastic limit. Beyond this, they are tional area $A \approx 10 \text{ in}^2$ and length $\approx 10 \text{ in}$, an amount $e \approx 4/100 \text{ in}$. Stress $S \approx 1/10$

modified by experimental coefficients, as, for instance, the modulus of $40,000 \approx 4,000 \text{ lb/in}^2$. Unit elongation $s \approx 4/100 \approx 10 \approx 1/250$. Modulus of elasticity

rupture. $E \approx 4,000 \approx 1/250 \approx 1,000,000 \text{ lb/in}^2$. Unit resilience $U_p \approx 1/2 \approx 4,000 \approx 4,000/$

The modulus of elasticity, lb/in^2 (kgf/cm^2), is the ratio of the increment $1,000,000 \approx 8 \text{ in} \approx \text{lb/in}^3$ ($0.563 \text{ cm} \approx \text{kgf/cm}^3$).

of unit stress to increment of unit deformation within the elastic limit. EXAMPLE 2. A weight $G \approx 5,000 \text{ lb}$ falls through a height $h \approx 2 \text{ ft}$; $V \approx$

The modulus of elasticity in tension, or Young's modulus, number of cubic inches required to absorb the shock without exceeding a stress of

$E \approx \text{unit stress/unit deformation} \approx Pl/(Ae) \approx 4,000 \text{ lb/in}^2$. Neglect compression of block. Work done by falling weight $\approx Gh \approx$

$5,000 \approx 2 \approx 12 \text{ in} \approx \text{lb} (2,271 \approx 61 \text{ cm} \approx \text{kgf})$ Resilience of block $\approx V \approx 8 \text{ in} \approx \text{lb} \approx$

The modulus of elasticity in compression is similarly measured. $5,000 \approx 2 \approx 12$. Therefore, $V \approx 15,000 \text{ in}^3$ ($245,850 \text{ cm}^3$).

The modulus of elasticity in shear or coefficient of rigidity, $G \approx Sv / \Delta$

Thermal Stresses A bar will change its length when its temperature

where Δ is expressed in radians (see Fig. 5.2.2).

The bulk modulus of elasticity K is the ratio of normal stress, applied to is raised (or lowered) by the amount $\Delta l / l_0 \approx \Delta(t_2 \approx 32)$. The linear

all six faces of a cube, to the change of volume. coefficient of thermal expansion Δ is assumed constant at normal tem-

Change of volume under normal stress is so small that it is rarely of

peratures and l_0 is the length at 32°F (273.2 K , 0°C). If this expansion

significance. For example, given a body with length l , width b , thick- (or contraction) is prevented, a thermal-time stress is developed, equal to

ness d , Poisson's ratio ν , and longitudinal strain ϵ , $V \approx lbd \approx \text{original } S \approx E_\Delta(t_2 \approx t_1)$, as the temperature goes from t_1 to t_2 . In thin flat plates

volume. The deformed volume $\approx (1 \approx \nu)l (1 \approx \nu)b(1 \approx \nu)d$. Ne- the stress becomes $S \approx E_\Delta(t_2 \approx t_1)/(1 \approx \nu)$; ν is Poisson's ratio. Such

glecting powers of ν , the deformed volume $\approx (1 \approx \nu \approx 2\nu)V$. The stresses can occur in castings containing large and small sections. Simi-

change in volume is $\approx (1 \approx 2\nu)V$; the unit volumetric strain is $\approx (1 \approx 2\nu)$. lar stresses also occur when heat flows through members because of the

Thus, a steel rod ($\nu \approx 0.3$, $E \approx 30 \approx 106 \text{ lb/in}^2$) compressed to a stress difference in temperature between one point and another. The heat

of $30,000 \text{ lb/in}^2$ will experience ≈ 0.001 and a unit volumetric strain flowing across a length b as a result of a linear drop in temperature Δt

of 0.0004 , or 1 part in 2,500. equals $Q \approx k A \Delta t / b \text{ Btu/h (cal/h)}$. The thermal conductivity k is in

The following relationships exist between the modulus of elasticity in $\text{Btu/(h)(ft}^2)(^\circ\text{F))/(in of thickness) [cal/(h)(m}^2)(k)/(m)]$. The thermal-flow

stress is then $S \approx E_\Delta Qb / (kA)$. Note, when Q is substituted the stress

tension or compression E , modulus of elasticity in shear G , bulk modu-

modulus of elasticity K , and Poisson's ratio ν : becomes $S = E_L \epsilon$ as above, only t is now a function of distance rather

than time.

$$E = 2G(1 + \nu)$$

$G = E/[2(1 + \nu)]$ EXAMPLE. A cast-iron plate 3 ft square and 2 in thick is used as a fire wall.

The temperature is 330°F on the hot side and 160°F on the other. What is the

average $(E \gg 2G)/(2G)$ thermal-flow stress developed across the plate?

$$K = E/[3(1 + 2\nu)]$$

$$\Delta T = (3K \gg E)/(6K) \quad S = E_L \epsilon = 13 \times 10^6 \times 6.5 \times 10^{-6} = 170$$

$$= 14,360 \text{ lb/in}^2 \text{ (1,010 kgf/cm}^2 \text{)}$$

Resilience U (in $\text{in} \cdot \text{lb}$) [$\text{cm} \cdot \text{kgf}$] is the potential energy stored up in a or $Q = 2.3 \times 9 \times 170 / 2 = 1,760 \text{ Btu} / \text{h}$

deformed body. The amount of resilience is equal to the work required and $S = 13 \times 10^6 \times 6.5 \times 10^{-6} = 1,760 \times 2 / 2.3 \times 9$

to deform the body from zero stress to stress S . When S does not exceed $14,360 \text{ lb/in}^2$ (1,010 kgf/cm^2)

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5-18 MECHANICS OF MATERIALS

COMBINED STRESSES $4,000 \leq 8,000$ $4,000 \geq 8,000$

$$60^\circ \text{ Sn} = (\geq 0.5000) \leq 0$$

$$2 \times 2$$

In the discussion that follows, the element is subjected to stresses lying

in one plane; this is the case of plane stress, or two-dimensional stress. $7,000 \text{ lb/in}^2$

Simple stresses, defined as such by the flexure and torsion theories, lie $4,000 \geq 8,000$

in planes normal or parallel to the line of action of the forces. Normal, as $60^\circ \text{ Ss} = (0.8660) \geq 0 \geq 1,732 \text{ lb/in}^2$

$$2$$

$$\sqrt{\quad}$$

as well as shearing, stresses may, however, exist in other directions. A

$$4,000 \leq 8,000 \quad 4,000 \geq 8,000 \quad 2$$

particle out of a loaded member will contain normal and shearing $S_{M,m} = 0$

stresses as shown in Fig. 5.2.11. Note that the four shearing stresses 2×2

must be of the same magnitude, if equilibrium is to be satisfied. $6,000 \times 2,000$

If the particle is "cut" along the plane AA, equilibrium will reveal $8,000$ and $4,000 \text{ lb/in}^2$ (564 and 282 kgf/cm^2)

that, in general, normal as well as shearing stresses act upon the plane 2×0

AC (Fig. 5.2.12). The normal stress on plane AC is labeled S_n , and at $\tan 2\theta = 0$ or $\theta = 90^\circ$ and 0°

$$4,000 \geq 8,000$$

$$\sqrt{\quad}$$

shearing S_s . The application of equilibrium yields

$$4,000 \geq 8,000/2$$

$$S_x \leq S_y \quad S_{sM,m} \leq 0$$

$$S_n \leq x \cos 2\theta \leq S_{xy} \sin 2\theta$$

$$2 \leq 2,000 \text{ lb/in}^2 \quad (141 \text{ kgf/cm}^2)$$

$$2$$

$$S \geq S_y$$

and $S_s = x \sin 2\theta \geq S_{xy} \cos 2\theta$ Mohr's Stress Circle The biaxial stress field with its combined 2 stresses can be represented graphically by the Mohr stress circle. For instance, for the particle given in Fig. 5.2.11, Mohr's circle is as shown in Fig. 5.2.14. The stress sign convention previously defined must be adhered to. Furthermore, in order to locate the point (on Mohr's circle) that yields the stresses on a plane θ° from the vertical side of the particle (such as plane AA in Fig. 5.2.11), $2\theta^\circ$ must be laid off in the same

Fig. 5.2.11 Fig. 5.2.12

A sign convention must be used. A tensile stress is positive while compression is negative. A shearing stress is positive when directed as on plane AB of Fig. 5.2.12; i.e., when the shearing stresses on the vertical planes form a clockwise couple, the stress is positive.

The planes defined by $\tan 2\theta = 2S_{xy}/(S_x - S_y)$, the principal planes, contain the principal stresses — the maximum and minimum normal stresses. These stresses are

Fig. 5.2.14

$$\sqrt{\quad}$$

$$S \leq S_y \quad S_x \geq S_y/2$$

$$S_{M,m} = x \leq S_{2xy}$$

2 direction from the radius to (S_x, S_{xy}) . For the previous example, Mohr's circle becomes Fig. 5.2.15.

The maximum and minimum shearing stresses are represented by the Eight special stress fields are shown in Figs. 5.2.16 to 5.2.23, along quantity with Mohr's circle for each.

$$\sqrt{\quad}$$

$$S_x \geq S_y/2$$

$$S_{sM,m} \leq S_{2xy}$$

$$2$$

and they act on the planes defined by

$$S_x \geq S_y$$

$$\tan 2\theta \geq$$

$$2S_{xy}$$

EXAMPLE. The steam in a boiler subjects a particular particle on the outer surface of the boiler shell to a circumferential stress of 8,000 lb/in² and a longitudinal stress of 4,000 lb/in² as shown in Fig. 5.2.13. Find the stresses acting on the

plane XX, making an angle of 60° with the direction of the 8,000 lb/in² stress. Find the principal stresses and locate the principal planes. Also find the maximum and minimum shearing stresses.

Fig. 5.2.15

Combined Loading Combined flexure and torsion arise, for instance, when a shaft twisted by a torque M_t is bent by forces produced by belts or gears. An element on the surface, such as ABCD on the shaft of Fig.

Fig. 5.2.13 5.2.24, is subjected to a flexure stress $S_x = Mc/I = 8Fl/(\pi d^3)$ and a

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PLASTIC DESIGN 5-19

compression) occurring at a point in two right-angle directions, and the change of the angle between them is γ_{xy} . The strain e at the point in any direction a at an angle θ with the x direction derives as

$$e \leq e_y \quad e \geq e_y$$

$$e_a = x \cos 2\theta \leq xy \sin 2\theta$$

$$2 \quad 2 \quad 2$$

Fig. 5.2.16 Fig. 5.2.17 Similarly, the shearing strain γ_{ab} (change in the original right angle between directions a and b) is defined by

$$\gamma_{ab} = (e_x \geq e_y) \sin 2\theta \leq \gamma_{xy} \cos 2\theta$$

Inspection easily reveals that the above equations for e_a and γ_{ab} are mathematically identical to those for S_n and S_s . Thus, once a sign convention is established, a Mohr circle for strain can be constructed and

Fig. 5.2.18 Fig. 5.2.19 used as the stress circle is used. The strain e is positive when an extension and negative when a contraction. If the direction associated with the first subscript a rotates counterclockwise during straining with respect to the direction indicated by the second subscript b , the shearing strain is positive; if clockwise, it is negative. In constructing the circle, positive extensional strains will be plotted to the right as abscissas and positive half-shearing strains will be plotted upward as ordinates.

Fig. 5.2.20 Fig. 5.2.21 For the strains shown in Fig. 5.2.26a, Mohr's strain circle becomes that shown in Fig. 5.2.26b. The extensional strain in the direction a , making an angle of θ_a with the x direction, is e_a , and the shearing strain is γ_{ab} counterclockwise. The strain 90° away is e_b . The maximum principal strain is e_M at an angle θ_M clockwise from the x direction. The other principal or minimum strain is e_m 90° away.

Fig. 5.2.22 Fig. 5.2.23

torsional shearing stress $S_{xy} = M_t c/J = 16M_t/(\pi d^3)$. These stresses will induce combined stresses. The maximum combined stresses will be $S_n = 1/2 (S_x \pm \sqrt{S_x^2 + 4S_{xy}^2})$

and $S_s \leq \frac{1}{2} \sqrt{S_x^2 + S_y^2} \leq 4S_{xy}$

The above situation applies to any case of normal stress with shear, as when a bolt is under both tension and shear. A beam particle subjected to both flexure and transverse shear is another case.

Fig. 5.2.26

Fig. 5.2.24 PLASTIC DESIGN

Early efforts in stress analysis were based on limit loads, that is, loads

Combined torsion and longitudinal loads exist on a propeller shaft. A which stress a member “wholly” to the yield strength. Euler’s famous

particle on this shaft will contain a tensile stress computed using S $\square$ paper on column action (“Sur la Force des Colonnes,” Academie des

F/A and a torsion shearing stress equal to $S_s \square M_{tc}/J$. The free body of a Sciences de Berlin, 1757) deals with the column problem this way.

particle on the surface of a vertical turbine shaft is subjected to direct More recently, the concept of limit loads, referred to as limit, or plastic,

compression and torsion. design, has found strong application in the design of certain structures.

The theory presupposes a ductile material, absence of stress raisers, and

fabrication free of embrittlement. Local load overstress is allowed, provided the structure does not deform appreciably.

To visualize the limit-load approach, consider a simple beam of uniform section subjected to a concentrated load of midspan, as depicted in

Fig. 5.2.27a. According to elastic theory, the outermost fiber on each side and at midspan — the section of maximum bending moment — will

Fig. 5.2.25 first reach the yield-strength value. Across the depth of the beam, the stress distribution will, of course, follow the triangular pattern, becom-

When combined loading results in stresses of the same type and ing zero at the neutral axis. If the material is ductile, the stress in the

direction, the addition is algebraic. Such a situation exists on an offset outermost fibers will remain at the yield value until every other fiber

link like that of Fig. 5.2.25. reaches the same value as the load increases. Thus the stress distribution

Mohr’s Strain Circle Strain equations can also be derived for plane- assumes the rectangular pattern before the plastic hinge forms and fail-

strain fields. Strains ϵ_x and ϵ_y are the extensional strains (tension or ure ensues.

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The problem is that of finding the final limit load. Elastic-flexure equal to one-half the moment at either end. A preferable situation, it

theory gives the maximum load — triangular distribution — as might be argued, is one in which the moments are the same at the three

stations — solid line. Thus, applying equilibrium to, say, the left half of

2 $S_y b h^2$ the beam yields a bending moment at each of the three plastic hinges of

$F_y I$

3 I

$w l^2$

$M_L I$

For the rectangular stress distribution, the limit load becomes 16

$S_y b h^2$

$F_L I$ DESIGN STRESSES

I

The ratio $F_L / F_y I \approx 1.50$ — an increase of 50 percent in load capability. If a machine part is to safely transmit loads acting upon it, a permissible

The ratio F_L / F_y has been named shape factor (Jenssen, Plastic Design in maximum stress must be established and used in the design. This is the

Welded Structures Promises New Economy and Safety, Welding Jour., allowable stress, the working stress, or preferably, the design stress. The

Mar. 1959). See Fig. 5.2.27b for shape factors for some other sections. design stress should not waste material, yet should be large enough to

The shape factor may also be determined by dividing the first moment prevent failure in case loads exceed expected values, or other uncertain-

of area about the neutral axis by the section modulus. ties react unfavorably.

The design stress is determined by dividing the applicable material property — yield strength, ultimate strength, fatigue strength — by a factor of safety. The factor should be selected only after all uncertainties have been thoroughly considered. Among these are the uncertainty with respect to the magnitude and kind of operating load, the reliability of the material from which the component is made, the assumptions involved in the theories used, the environment in which the equipment might operate, the extent to which localized and fabrication stresses might develop, the uncertainty concerning causes of possible failure, and the endangering of human life in case of failure. Factors of safety vary from industry to industry, being the result of accumulated experience with a class of machines or a kind of environment. Many codes, such as the ASME code for power shafting, recommend design stresses found safe in practice.

In general, the ductility of the material determines the property upon which the factor should be based. Materials having an elongation of over 5 percent are considered ductile. In such cases, the factor of safety is based upon the yield strength or the endurance limit. For materials with an elongation under 5 percent, the ultimate strength must be used because these materials are brittle and so fracture without yielding.

Factors of safety based on yield are often taken between 1.5 and 4.0.

For more reliable materials or well-defined design and operating condi-

tions, the lower factors are appropriate. In the case of untried materials or otherwise uncertain conditions, the larger factors are safer. The same values can be used when loads vary, but in such cases they are applied to the fatigue or endurance strength. When the ultimate strength determines the design stress (in the case of brittle materials), the factors of Fig. 5.2.27

safety can be doubled.

Thus, under static loading, the design stress for, say, SAE 1020,

A constant-section beam with both ends fixed, supporting a uni- which has a yield strength of 45,000 lb/in² (3,170 kgf/cm²) may be

formly distributed load, illustrates another application of the plastic- taken at 45,000/2, or 22,500 lb/in² (1,585 kgf/cm²), if a reasonably

load approach. The bending-moment diagram based on the elastic certain design condition exists.

A Class 30 cast-iron part might be de-

theory drawn in Fig. 5.2.28 (broken line) shows a moment at the center signed at 30,000/5 or 6,000 lb/in² (423 kgf/cm²). A 2017S-0 aluminum-

alloy component (13,000 lb/in² endurance strength) could be computed

at a design stress of 13,000/2.5 or 5,200 lb/in² (366 kgf/cm²) in the

usual fatigue-load application.

BEAMS

For properties of structural steel and wooden beams, see Sec. 12.2.

Notation

I □ rectangular moment of inertia

I_p □ polar moment of inertia

I/c □ section modulus

M □ bending moment

P, W ♦ □ concentrated load

Q or V □ total vertical shear

R □ reaction

S □ unit normal stress

S_s or S_v □ transverse shearing stress

Fig. 5.2.28 W □ total distributed load

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BEAMS 5-21

f □ deflection $x \times x \times h \times h \times 3$

h □ □ □ $\geq$, if h is in pounds per foot and weight of beam is

i □ slope 1 2 3 6 6l

l □ distance between supports $h \times x \times h \times 2$

neglected. The vertical shear V □ $R_1 \geq \square \square \geq$. Note again that V □

r □ radius of gyration $l \ 2 \ 6 \ 2l$

r_c = radius of curvature

w = distributed load per longitudinal unit

d

dx

$\int h dx$

6

$h x^3$

$6I$

$\int \int \geq$

hI

6

$h x^2$

$2I$

A simple beam rests on supports at its ends which permit rotation. A Table 5.2.2 gives the reactions, bending-moment equations, vertical

cantilever beam is fixed (no rotation) at one end. When computing reac- shear equations, and the deflection of some of the more common types

tions and moments, distributed loads may be replaced by their resultants of beams.

acting at the center of gravity of the distributed-load area. Maximum Safe Load on Steel Beams See Table 5.2.3 To obtain

Reactions are the forces and/or couples acting at the supports and maximum safe load (or maximum deflection under maximum safe load)

holding the beam in place. In general, the weight of the beam should be for any of the conditions of loading given in Table 5.2.5, multiply the

accounted for. corresponding coefficient in that table by the greatest safe load (or

The bending moment (pound-feet or pound-inches) (kgf $\cdot$ m) at any deflection) for distributed load for the particular section under consider-

section is the algebraic sum of the external forces and moments acting ation as given in Table 5.2.4.

on the beam on one side of the section. It is also equal to the moment of The following approximate factors for reducing the load should be

the internal-stress forces at the section, $M = \int \sigma dA/y$. A bending mo- used when beams are long in comparison with their breadth:

ment that bends a beam convex downward (tensile stress on bottom

fiber) is considered positive, while convex upward (compression on bot- Ratio of unsupported (lateral)

tom) is negative. length to flange width or

The vertical shear V (lb) (kgf) effective on a section is the algebraic breadth 20 30 40 50 60 70

sum of all the forces acting parallel to and on one side of the section,

$V = \int \tau F$. It is also equal to the sum of the transverse shear stresses acting Ratio of greatest safe load to

calculated load 1 0.9 0.8 0.7 0.6 0.5

on the section, $V = \int \tau S_s dA$.

Moment and shear diagram may be constructed by plotting to scale the particular entity as the ordinate for each section of the beam. Such Theory of Flexure A bent beam is shown in Fig. 5.2.31. The con- diagrams show in continuous form the variation along the length of the concave side is in compression and the convex side in tension. These are beam. divided by the neutral plane of zero stress A-B-A. The intersection of Moment-Shear Relation The shear V is the first derivative of moment with respect to distance along the beam, $V = dM/dx$. This relationship does not, however, account for any sudden changes in moment.

Fig. 5.2.31

It is assumed that a beam is prismatic, of a length at least 10 times its depth, and that the external forces are all at right angles to the axis of the beam and in a plane of symmetry, and that flexure is slight. Other assumptions are: (1) That the material is homogeneous, and obeys Fig. 5.2.29 Hooke's law. (2) That stresses are within the elastic limit. (3) That every layer of material is free to expand and contract longitudinally and laterally under stress as if separate from other layers. (4) That the tensile and compressive moduli of elasticity are equal. (5) That the cross section remains a plane surface. (The assumption of plane cross sections is strictly true only when the shear is constant or zero over the cross section, and when the shear is constant throughout the length of the beam.)

It follows then that: (1) The internal forces are in horizontal balance. (2) The neutral axis contains the center of gravity of the cross section, where there is no resultant axial stress. (3) The stress intensity varies directly with the distance from the neutral axis.

The moment of the elastic forces about the neutral axis, i.e., the stress moment or moment of resistance, is $M = SI/c$, where S is an elastic unit stress at outer fiber whose distance from the neutral axis is c ; and I is the rectangular moment of inertia about the neutral axis. I/c is the section

Fig. 5.2.30

modulus.

EXAMPLES. Figure 5.2.29 illustrates a simple beam subjected to a uniform This formula is for the strength of beams. For rectangular beams, $M =$

$\frac{1}{6} S b h^2$, where b = breadth and h = depth; i.e., the elastic strength of $\times w/x \times 2 \times w$

load. $M = R_1 \times \geq w \times \geq$ and $V = R_1 \geq w \times \geq w$. Note also

2 2 2 2 beam sections varies as follows: (1) for equal width, as the square of the that $V =$

d
 dx
 $\square wlx$
 2
 $\geq \square$
 wx^2
 2
 wl
 2
 $\geq wx$.

depth; (2) for equal depth, directly as the width; (3) for equal depth and width, directly as the strength of the material; (4) if span varies, then for Figure 5.2.30 is a simple beam carrying a uniformly varying load; $M \propto R_1 x \propto$ equal depth, width, and material, inversely as the span.

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Table 5.2.2 Beams of Uniform Cross Section, Loaded Transversely

W W Wc_1 Wc
 $R_2 \propto W$ $R_1 \propto R$ $R_1 \propto R_2$
 $M_x \propto Wx^2$ 2 2 l l
 $M_{max} \propto Wl$, ($x \propto 1$) Wx Wc_1 x Wcx_1
 $Q_x \propto W$ $M_x \propto M_x$ $\diamond$ $\square$
 2 l l
 f
 Wl^3
 $3EI$
 $(\max) M_{max}$ $\square$
 Wl
 4
 $\square$ $\square$ x
 l
 2
 M_{max} $\square$
 Wcc_1
 l
 $\square$, ($x_1 \propto c_1$ or $x \propto c$)
 W Wc_1 Wc
 $Q_x \propto Q_x$ $\square$, Q_{x1} $\square$
 2 l l

f
 W
 E
 $(\max) f$
 Wc
 $3E$
 $c(l \leq c_1)$
 3
 $3/2$
 $(\max)$
 $\text{Max } f \text{ occurs at } x = \sqrt{c(l \geq c_1)}/3$
 $5 \ 11 \ W \ W$
 $R_1 \leq W, R_2 \leq W \ R_1 \leq R, R \leq R_1 \leq W$
 $16 \ 16 \ 2 \ 2 \ 2 \ R_2 \leq W$
 Mx
 5
 16
 $Wx \ Mx$
 W
 2
 x
 l
 1
 4
 $Mx \geq Wc \leq \text{const}$
 $QW \text{ to } R_1 \leq W$
 x
 $QR_1 \text{ to } R_2 \leq 0$
 $5 \ 11 \ x_1 \geq Wl \ x \ 3 \ QR_2 \text{ to } W \leq W$
 $Mx_1 \leq Wl \geq Mx_1 \geq$
 $32 \ 16 \ 2 \ 2 \ 4$
 x
 $Wcl \ 2$
 $3 \ Wl \ f_1 \leq (\max)$
 $M_{\max} \geq Wl, x_1 \leq M_{\max}, x \leq E$
 x
 $16 \ 2 \ 8 \ 2$
 $Wc \ 2 \ 3l$
 $5 \ 11 \ W \ W \ f_2 \leq c \leq (\max)$
 $Qx \leq W, Qx_1 \geq W \ Qx, Qx_1 \geq E$

16 16 2 2

11 W I3

$Q_{\max} \leq W, f_y (\max)$

16 EI 192

$\bar{x} \bar{y}$

I

2

to $x \leq I_y$

W I3

f_y

EI 768

If a beam is cut in halves vertically, the two halves laid side by side center of gravity, and these two axes are always at right angles to each

each will carry only one-half as much as the original beam. other. The principal axes are axes with respect to which the moment of

Tables 5.2.6 to 5.2.8 give the properties of various beam cross sec- inertia is, respectively, a maximum and a minimum, and for which the

tions. For properties of structural-steel shapes, see Sec. 12.2. product of inertia is zero. For symmetrical sections, axes of symmetry

Oblique Loading It should be noted that Table 5.2.6 includes certain are always principal axes. For unsymmetrical sections, like a rolled angle

cases for which the horizontal axis is not a neutral axis, assuming the section (Fig. 5.2.32), the inclination of the principal axis with the X axis

common case of vertical loading. The rectangular section with the diag- may be found from the formula $\tan 2\theta = 2I_{xy} / (I_y - I_x)$, in which θ

onal as a horizontal axis (Table 5.2.6) is such a case. These cases must angle of inclination of the principal axis to the X axis, I_{xy} the product

be handled by the principles of oblique loading. of inertia of the section with respect to the X and Y axes, I_y moment of

Every section of a beam has two principal axes passing through the inertia of the section with respect to the Y axis, I_x moment of inertia of

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BEAMS 5-23

Table 5.2.2 Beams of Uniform Cross Section, Loaded Transversely (Continued)

W wI 3 3

$R_2 \leq W \leq R_1$ $R_1 \leq W \leq R_1$

2 2 8 8

wx 2

$M_x \leq W wI 5 5$

2 R2 $\leq R_2 \leq W \leq wI$

2 2 8 8

□ □

wl 2

Mmax □ ≥ , (x □ l) wx wx 3

2 Mx □ (l ≥ x) Mx □ l ≥ x

2 2 4

□ □

Qx □ ≥ wx

Q max □ ≥ wl, (x □ l) wl 2 9 3

Mmax □ , (x □ 1/2l) Mmax □ wl 2, x □ l

8 128 8

W l3

f □ (max) wl wl 2

EI 8 Qx □ ≥ wx Mmax □ ≥ , (x □ l)

2 8

wl 3

Q max □ , (x □ 0) Qx □ wl ≥ wx

2 8

W 5l 3 5

f □ (max) Q max □ ≥ wl

EI 384 8

W l3

f □ (max)

EI 185

W wl W wl 1 2

R1 □ □ , R2 □ □ R2 □ W □ total load R1 □ W, R2 □ W

2 2 2 2 3 3

Mx □ ≥

wl 2

2

□ 1

6

x

l

x2

≥ ≤ 2

l

□ Mx □ ≥

W x3

3 l2

Mx □

Wx

3

 $\square 1 \geq$

x2

l2

 $\square$ $M_{\max} \square \geq$

1

12

 $wl 2, (x \square 0, \text{ or } x \square l) M_{\max} \square \geq$

Wl

3

 $M_{\max} \square$

2

 $9\sqrt{3}$ $Wl, \square x \square$ $\sqrt{3}$

1

 $\square$ $Qx \square$

wl

2

 $\geq wx$ $Wx 2$ $Qx \square \geq 2$

l

 $Qx \square W \square 1$

3

x2

 ≥ 2

l

 $\square$ $wl Q_{\max} \square \geq W 2$ $Q_{\max} \square \square Q_{\max} \square \geq W, (x \square l)$ $2 W l 3 3$ $f \square (\max)$ $W l 3 E l 15 W l 3$ $f \square (\max) f \square 0.01304 (\max)$ $E l 384 E l$

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Table 5.2.2 Beams of Uniform Cross Section, Loaded Transversely (Continued)

W W W W W 4W

R1 □ , R2 □ R1 □ , R2 □ R1 □ , R2 □

2 2 2 2 5 5

$M_x \square W_x \square 1$

2

x

$\geq \leq 2$

l

2 x2

3l

$\square M_x \square W_x \square 1$

2

2 x2

3 l2

$\square M_x \square W_x \square 1$

5

x2

≥ 2

3l

□

$M_{max} \square$

Wl

12

,□ x□

1

2

$l \square M_{max} \square$

Wl

6

,□ x□

1

2

$l \square M_{max} \square \geq$

2

15

Wl at support 2

$Q_x \square W \square 1$

2

2x

l

2×2
 ≤ 2
 l
 $Q \times W \leq 1$
 2
 2×2
 ≥ 2
 l
 $Q \times W \leq 1$
 5
 $\times 2$
 ≥ 2
 l
 Q
 $W \leq 4W$
 $Q \max \leq (x \leq 0) Q \max \leq (x \leq 0) Q \max \geq$
 $2 \times 2 \times 5$
 $W \leq 3 \times W \leq 16W \leq 3$
 $f \leq (\max) f \leq (\max) f$
 $EI \leq 320 EI \leq 60 \times 1,500 \sqrt{5EI}$
 $0.00477W \leq 3$
 $Q \leq (\max)$
 EI

Concentrated load $W \leq$

Uniformly dist. load $W \leq w \leq$

 $c \leq 2(3c \leq 2c \leq 1) \leq 3$
 $R \leq W \leq W$
 $2 \leq 3 \leq 8$
 $(2c \leq 2 \leq 6cc \leq 1 \leq 3c \leq 12)c \leq 5$
 $R \leq W \leq W$
 $2 \leq 3 \leq 8$
 $R \leq 1$
 W
 2
 $w \leq$
 $Q, R \leq 2$
 2
 W
 2
 $w \leq$
 2

M2 □ W♠
cc1(2c ≦ c1)
2l 2
≦W □□ l
8
Mx □
Wx
2
□ 1≧
c
x
x
l
□ , (x → c) MW♠ □ W♠
cc12 (3c ≦ 2c1)
2l 3
≦W
(3c1 ≧ c)c
8l
Wx 2 W♠ l 2 5c ≧ 3c1
Mx □ ≧ , (x □ c) (a) ↑ 2
W 4c1 3c ≦ 2c1
□ □
2l
Mmax □
Wl
4
□ 1
2
2c
l
□ □ ,c□
√2 ≧ 1
2
□ l Mc max □
R12
2W
l, x□
R1l
W
W♠ l 2(3c1 ≧ 5c)

$Qx \square$

W

$\geq wx (x \rightarrow c) (b) \updownarrow$

$W 4c(2c \leq 6cc1 \leq 3c12)$

$\square \square$

2

$Qx \square \geq wx (x \square c) (R1 \geq W\diamond)2 R1 \geq W\diamond$

$Mc1 \max \square W\diamond c \leq l, x \square l$

$2W W$

Deflection under $W\diamond$

$W\diamond c 2c13(4c \leq 3c1) W cc12(3c \leq c1)$

$f \square \leq$

$EI 12l 3 EI 48l$

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BEAMS 5-25

Table 5.2.2 Beams of Uniform Cross Section, Loaded Transversely (Continued)

Concentrated load $W\diamond$

Uniformly dist . load $W \square wl; c \updownarrow c1$

$c1 W c \updownarrow c1$

$R1 \square W\diamond \leq 2$

$l 2 (3c \leq c1)c21 W$

$R1 \square W\diamond \leq$

$c W l 3 2$

$R2 \square W\diamond \leq$

$l 2 (c \leq 3c1)c 2 W$

$R2 \square W\diamond \leq$

$W\diamond c \geq c l 3 2$

(a) $\updownarrow 1$

$W 2c cc21 Wl$

$\square \square$

$M_{\max} \square M1 \square W\diamond \leq$

$x1 R2 l R2 l l 2 12$

$M_{\max} \square R2 \square 2, x1 \square$

$2 2W W$ Deflection under $W\diamond$

(b)

$W\diamond$

W

$c1 \geq c$

$2c f \square$

1

 $\square W_{\diamond}$

c 3c13

3l 3

 $\leq W$

c 2c12

 $\square$ $\square \square$

EI 24l

W cc1

 $M_{\max} \square W_{\diamond} \leq , (x1 \square c1)$

2 l

Deflection of beam under $W_{\diamond}$: $f \square W_{\diamond} \leq$ $l^2 \leq cc1$

8cc1

 $W \square c 2c21$

3EI

Table 5.2.3 Uniformly Distributed Loads on Simply Supported Rectangular Beams 1-in Wide*
(Laterally Supported Sufficiently to Prevent Buckling)

[Calculated for unit fiber stress at 1,000 lb/in² (70 kgf/cm²): nominal size]

Total load in pounds (kgf)† including the weight of beam

Depth of beam, in (cm)§

Span, ft

(m)‡ 6 7 8 9 10 11 12 13 14 15 16

5 800 1,090 1,420 1,800 2,220 2,690 3,200 3,750 4,350 5,000 5,690

6 670 910 1,180 1,500 1,850 2,240 2,670 3,130 3,630 4,170 4,740

7 570 780 1,010 1,290 1,590 1,920 2,280 2,680 3,110 3,570 4,060

8 500 680 890 1,120 1,390 1,680 2,000 2,350 2,720 3,130 3,560

9 440 600 790 1,000 1,230 1,490 1,780 2,090 2,420 2,780 3,160

10 400 540 710 900 1,110 1,340 1,600 1,880 2,180 2,500 2,840

11 360 490 650 820 1,010 1,220 1,450 1,710 1,980 2,270 2,590

12 330 450 590 750 930 1,120 1,330 1,560 1,810 2,080 2,370

13 310 420 550 690 850 1,030 1,230 1,440 1,680 1,920 2,190

14 290 390 510 640 790 960 1,140 1,340 1,560 1,790 2,030

15 270 360 470 600 740 900 1,070 1,250 1,450 1,670 1,900

16 250 340 440 560 690 840 1,000 1,170 1,360 1,560 1,780

17 230 320 420 530 650 790 940 1,100 1,280 1,470 1,670

18 220 300 400 500 620 750 890 1,040 1,210 1,390 1,580

19 210 290 380 470 590 710 840 990 1,150 1,320 1,500

20 200 270 360 450 560 670 800 940 1,090 1,250 1,420

22 180 250 320 410 500 610 730 850 990 1,140 1,290
 24 160 230 290 370 460 560 670 780 910 1,040 1,180
 26 150 210 270 340 420 520 610 720 840 960 1,090
 28 140 190 250 320 390 480 570 670 780 890 1,010
 30 130 180 240 300 370 450 530 630 730 830 950

* This table is convenient for wooden beams. For any other fiber stress $S_{\diamond}$, multiply the values in table by $S_{\diamond}/1,000$. See Sec. 12.2 for properties of wooden beams of commercial sizes.

† To change to kgf, multiply by 0.454.

‡ To change to m, multiply by 0.305.

§ To change to cm, multiply by 2.54.

the section with respect to the X axis. When this principal axis has been loads makes with the y axis, $M \cos \phi$ is the component of bending

found, the other principal axis is at right angles to it. moment causing bending about the principal axis which has been design-

Calling the moments of inertia with respect to the principal axes $I_{\diamond x}$ nated as the X axis, $M \sin \phi$ is the component of bending moment

and $I_{\diamond y}$, the unit stress existing anywhere in the section at a point whose causing bending about the principal axis which has been designated as

coordinates are x and y (Fig. 5.2.33) is $S = My \cos \phi / I_{\diamond x} \pm Mx \sin \phi / I_{\diamond y}$, the Y axis. The sign of the two terms for unit stress may be determined

in which M is bending moment with respect to the section in question, by inspection in the usual way, and the result will be tension or com-

ϕ is the angle which the plane of bending moment or the plane of the pression as determined by the algebraic sum of the two terms.

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Table 5.2.4 Approximate Safe Loads in Pounds (kgf) on Steel Beams,* Simply Supported, Single Span

Allowable fiber stress for steel, 16,000 lb/in² (1,127 kgf/cm²) (basis of table)

Beams simply supported at both ends.

L is distance between supports, ft (m) a is interior area, in² (cm²)

A is sectional area of beam, in² (cm²) d is interior depth, in (cm)

D is depth of beam, in (cm) w is total working load, net tons (kgf)

Greatest safe load, lb Deflection, in

Shape of section Load in middle Load distributed Load in middle Load distributed

890AD 1,780AD wL3 wL3

Solid rectangle

L L 32 AD 2 52 AD 2

890(AD $\geq$ ad) 1,780(AD $\geq$ ad) wL3 wL3

Hollow rectangle

L L 32(AD 2 $\geq$ ad 2) 52(AD 2 $\geq$ ad 2)

667AD 1,333AD wL3 wL3

Solid cylinder

L L 24AD 2 38AD 2

667(AD $\geq$ ad) 1,333(AD $\geq$ ad) wL3 wL3

Hollow cylinder

L L 24(AD 2 $\geq$ ad 2) 38(AD 2 $\geq$ ad 2)

1,795AD 3,390AD wL3 wL3

I beam

L L 58AD 2 93AD 2

In general, it may be stated that when the plane of the bending moment coincides with one of the principal axes, the other principal axis is the neutral axis. This is the ordinary case, in which the ordinary formula for unit stress may be applied. When the plane of the bending moment does not coincide with one of the principal axes, the above formula for oblique loading may be applied.

Internal Moment Beyond the Elastic Limit

Ordinarily, the expression $M \leq SI/c$ is used for stresses above the elastic limit, in which case S becomes an experimental coefficient S_R , the modulus of rupture, and the formula is empirical. The true relation is obtained by applying to the cross section a stress-strain diagram from a tension and compression test, as in Fig. 5.2.34. Figure 5.2.34 shows the side of a beam of depth d under flexure beyond its elastic limit; line Fig. 5.2.32 Fig. 5.2.33 1 - 1 shows the distorted cross section; line 3 - 3, the usual rectilinear

Table 5.2.5 Coefficients for Correcting Values in Table 5.2.4 for Various Methods of Support and of Loading, Single Span

Max relative

deflection under

Max relative max relative safe

Conditions of loading safe load load

Beam supported at ends:

Load uniformly distributed over span 1.0 1.0

Load concentrated at center of span 1/2 0.80

Two equal loads symmetrically concentrated 1/4c

Load increasing uniformly to one end 0.974 0.976

Load increasing uniformly to center 3/4 0.96

Load decreasing uniformly to center 3/2 1.08

Beam fixed at one end, cantilever:

Load uniformly distributed over span /

14 2.40

Load concentrated at end /

18 3.20

Load increasing uniformly to fixed end/

38 1.92

Beam continuous over two supports equidistant from ends:

Load uniformly distributed over span

1. If distance $a \rightarrow 0.2071l$ $l^2/(4a^2)$

2. If distance $a \downarrow 0.2071l$ l

$l \geq 4a$

3. If distance $a \approx 0.2071l$ 5.83

Two equal loads concentrated at ends $l/(4a)$

NOTE: l = length of beam; c = distance from support to nearest concentrated load; a = distance from support to end of

beam.

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BEAMS 5-27

Table 5.2.6 Properties of Various Cross Sections*

(I = moment of inertia; I/c = section modulus; $r = \sqrt{I/A}$ = radius of gyration)

N.A.

$bh^3/12$ $bh^3/3$ $b^3h/12$ $bh^2/2$

$I = (h^3 \cos^2 \alpha + b^3 \sin^2 \alpha)/12$

$12 \leq (b^2 \leq h^2) \leq 12$

I

c

$bh^2/6$

6

$bh^2/3$

3

$b^2h/2$

$6\sqrt{b^2 \leq h^2}$

bh

6

$\frac{1}{12} h^3 \cos^2 \alpha + \frac{1}{12} b^3 \sin^2 \alpha$

$h \cos \alpha \leq b \sin \alpha$

$\frac{1}{12}$

$\frac{1}{12} h^3 \cos^2 \alpha + \frac{1}{12} b^3 \sin^2 \alpha$

$r = 0.289h \leq 0.577h$

$\sqrt{12} \sqrt{b^2 \leq h^2} \leq 12$

$b^4 \geq h^4 \geq h^4$ $b^3 \geq h^3$ 2

$I = (H^3 \geq h^3) ; c = h$

$12 \leq 12 \leq 36$ 3

$$I_b H^3 \geq h^3 I_H^4 \geq h^4 \sqrt{2} H^4 \geq h^4 b h^2$$

$$c_6 H^6 H^{12} H^{24}$$

$$\sqrt{12}(H \geq h) \sqrt{12} \sqrt{12}$$

$$H^3 \geq h^3 H^2 \leq h^2 H^2 \leq h^2 h$$

$$r \square$$

$$\sqrt{18}$$

N.A.

$$b h^3 \sqrt{3} \sqrt{4} \sqrt{1} \leq 2 \sqrt{2} \sqrt{4}$$

$$I \square R R$$

$$12 \sqrt{16} \sqrt{6}$$

$$I b h^2 \sqrt{3} \sqrt{3}$$

$$\square 58 / R^3 R 0.6906 R^3$$

$$c \sqrt{12} \sqrt{16}$$

$$\sqrt{24} R$$

$$h^5$$

$$r \square 0.475 R$$

$$\sqrt{6}$$

NOTE: Square, axis same as first rectangle, side $\square h$; $I \square h^4 / 12$; $I/c \square h^3 / 6$; $r \square 0.289h$.

Square, diagonal taken as axis: $I \square h^4 / 12$; $I/c \square 0.1179h^3$; $r \square 0.289h$.

relation of stress to strain; and line 2 – 2, an actual stress-strain diagram, applied to the cross section of the beam, compression above and tension below. The neutral axis is then below the gravity axis. The outer material may be expected to develop greater ultimate strength than in simple stress, because of the reinforcing action of material nearer the neutral axis that is not yet overstrained. This leads to an equalization of stress over the cross section. SR exceeds the ultimate strength SM in tension as follows: for cast iron, $SR \square 2SM$; for sandstone, $SR \square 3SM$; for concrete, $SR \square 2.2SM$; for wood (green), $SR \square 2.3SM$.

In the case of steel I beams, failure begins practically when the elastic limit in the compression flange is reached.

Because of the support of adjoining material, the elastic limit in flexure

S_p is also greater than in tension, depending upon the relation of breadth

Fig. 5.2.34 to depth of section. For the same breadth, the difference decreases with

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Table 5.2.6 Properties of Various Cross Sections* (Continued)

$$\sqrt{A} \square \sqrt{24} \downarrow 2$$

$$A \square I \square 6R^2 \geq a^2 R$$

$$\text{Equilateral Polygon } I \square (6R^2 \geq a^2) \square$$

$A = \text{area} = 24cr$

$\sqrt{48}$

$R = \text{rad circumscribed} = \frac{a}{12r} \leq \frac{a}{2}$

circle $\frac{(12r)^2}{a^2} \leq 1$

$48 \cdot 180^\circ$

$r = \text{rad inscribed circle} = R \cos$

n

$n = \text{no. sides} = \frac{AR}{2}$

$\frac{1}{n}$ (approx)

$a = \text{length of side} = 4$

AR

(approx)

Axis as in preceding 4

section of octagon

$6b^2 \leq 6bb_1 \leq b^2 + 3b_1^2 \mid 6b^2 \leq 6bb_1 \leq b^2 + 12b_1^2 \mid \sqrt{12b^2} \leq 12bb_1 \leq 2b^2 + 12b_1^2$

$l = h = r$

$36(2b \leq b_1) \leq 12(3b \leq 2b_1) \leq 6(2b \leq b_1)$

$1 \leq 3b \leq 2b_1$

$c = h$

$3 \leq 2b \leq b_1$

$\sqrt{12(BH \leq bh)}$

$BH^3 \leq bh^3 \mid BH^3 \leq bh^3$

$l = r$

12

$l \mid BH^3 \leq bh^3$

$c = 6H$

$\sqrt{12(BH \geq bh)}$

$BH^3 \geq bh^3 \mid BH^3 \geq bh^3$

$l = r$

12

$l \mid BH^3 \geq bh^3$

$c = 6H$

$\sqrt{Bd \leq bd \leq a(h \leq h)}$

$l = \frac{1}{3} (Bc^3 \geq B_1 h^3 \leq bc^3 \geq b_1 h^3) \mid$

r

$1 \leq aH^2 \leq B_1 d^2 \leq b_1 d_1^2 (2H \geq d_1) \mid 1$

c_1

$2 \leq aH \leq B_1 d \leq b_1 d_1$

$l = \frac{1}{3} (Bc^3 \geq bh^3 \leq ac^3)$

$1 \leq aH^2 \leq bd^2$

$c_1 \leq$
 $2 a_H \leq b_d$
 $c_2 \leq H \leq c_1$
 $\sqrt{B_d} \leq a(H \leq d)$
 I
 r
 $\sqrt{A} \leq 2 \leq 4$
 $d_4 \leq r_4 A \leq d_3 \leq r_3 A \leq r_d$
 $I \leq r^2 \leq r$
 $64 d_4 c_{32} d_4$
 $\leq 0.05 d_4 \text{ (approx)} \leq 0.1 d_3 \text{ (approx)}$

increase of height. No difference will occur in the case of an I beam, or Deflection of Beams with hard materials. When a beam is subjected to bending, the fibers on one side elongate, Wide plates will not expand and contract freely, and the value of E while the fibers on the other side shorten (Fig. 5.2.35). These changes in

will be increased on account of side constraint. As a consequence of length cause the beam to deflect. All points on the beam except those

lateral contraction of the fibers of the tension side of a beam and lateral directly over the support fall below their original position, as shown in

swelling of fibers at the compression side, the cross section becomes Figs. 5.2.31 and 5.2.35.

distorted to a trapezoidal shape, and the neutral axis is at the center of The elastic curve is the curve taken by the neutral axis. The radius of

gravity of the trapezoid. Strictly, this shape is one with a curved perime- curvature at any point is ter, the radius being r_c / ν , where r_c is the radius of curvature of the

neutral line of the beam, and ν is Poisson's ratio. $r_c \leq EI/M$

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BEAMS 5-29

Table 5.2.6 Properties of Various Cross Sections* (Continued)

 $\sqrt{A} \leq$
 $I \leq D_4 \leq d_4 \leq \sqrt{R^2} \leq r^2 \leq \sqrt{D^2} \leq d^2$
 $I \leq (D_4 \leq d_4) \leq$
 $64 c_{32} D_2 d_4$
 $\leq 4 \leq R_4 \leq r_4$
 $\leq (R \leq r_4) \leq$
 $4 d_4 R$
 $\leq 1/4 A(R^2 \leq r^2) \leq 0.8 d_{m2} s \text{ (approx)}$
 $\leq 0.05(D_4 \leq d_4) s$

(approx) when is very small

 d_m
 $d_m \leq 1/2 (D \leq d)$

$$s \approx 1/2 (D \gg d)$$

$$\frac{1}{2}$$

$$I$$

$$\sqrt{A} \approx$$

$$\frac{1}{8} \approx 0.1908r^3 I \sqrt{9} \approx 2 \gg 64$$

$$I \approx r^4 \gg c^2 r \approx 0.264r$$

$$8 \cdot 9 \approx 6$$

$$I$$

$$\frac{1}{8} \approx 0.1098r^4 \approx 0.2587r^2$$

$$c^1$$

$$c^1 \approx 0.4244r$$

$$\sqrt{A} \approx \sqrt{\frac{1}{2}} (R \gg r)$$

$$4 R^2 \leq Rr \leq r^2 I \approx 2I$$

$$I \approx 0.1098(R^4 \gg r^4) c^1 \approx$$

$$3 \approx R \leq r^2$$

$$0.283R^2 r^2 (R \gg r)$$

$$\gg c^2 \approx R \gg c^1 \approx 0.31r^1 \text{ (approx)}$$

$$R \leq r$$

$$\approx 0.3tr^3 \text{ (approx)}$$

$$t$$

when is very small

$$r^1$$

$$\frac{1}{3}a^3b \approx \frac{1}{2}a^2b$$

$$I \approx 0.7854a^3b \approx 0.7854a^2b r$$

$$4 c^4 \approx 2$$

$$\sqrt{\frac{1}{2}ab} \gg a b$$

$$\frac{1}{3} I \approx I$$

$$I \approx (a b \gg a^3 b^1 \approx a(a \leq 3b)t r$$

$$4 c^4 \approx 1$$

$$2\sqrt{a} \leq b$$

$$\frac{1}{2} \text{ (approx)} a a \leq 3b$$

$$\frac{1}{2} a (a \leq 3b)t \approx$$

$$4$$

(approx) (approx)

$$\frac{1}{2}$$

$$1 \cdot 3 \approx 4 I$$

$$I \approx d \leq b(h^3 \gg d^3) \leq b^3 (h \gg d) r$$

$$12 \cdot 16 d^2$$

$$\frac{1}{2} \leq 2b(h \gg d)$$

$$I$$

c

1

6h

 $\frac{1}{3} \times 4$

16

 $d \leq b(h^3 \leq d^3) \leq b^3 (h \geq d) \times 4$

(approx)

 $\frac{1}{3} \times$ $\sqrt{\frac{1}{3}}$ $t \leq B^3 \leq Bh^2 \times 2 \times I$ $\frac{1}{3}$ $I \leq B^2 h \leq h^3 \times r$ $4 \times 16 \times 2 \times 3 \times B$ $2 \leq h \times t$ $h \leq H \geq \frac{1}{2} \times B \times 4$ $I \times 2I$ $c \times H \leq t$

A beam bent to a circular curve of constant radius has a constant bending moment.

Replacing r_c in the equation by its approximate geometrical value, $1/r_c \approx d^2 y/dx^2$, the fundamental equation from which the elastic curve of a bent beam can be developed and the deflection of any beam obtained is,

$$M \approx EI \frac{d^2 y}{dx^2} \text{ (approx)}$$

Substituting the value of M , in terms of x , and integrating once, gives the slope of the tangent to the elastic curve of the beam at point x ;

Fig. 5.2.35

 $\tan i \approx dy/dx \approx 0$ x

$$M \frac{dx}{EI}$$

Since i is usually small, $\tan i \approx i$,

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Table 5.2.6 Properties of Various Cross Sections* (Continued)

 $\sqrt{t(2B \leq 5.2H)}$ $64 \times 3I$ $I \leq (b \times h^2 \geq b^2 \times h^3/2)$, where $r \approx$ $105 \times 1 \times 1$

1

 $\frac{1}{3}$

$$h_1 \leq t/2 \quad (H \leq t) \quad b_1 \leq 1/4 \quad (B \leq 2.6t)$$

$$h_2 \leq 1/2 \quad (H \geq t) \quad b_2 \leq 1/4 \quad (B \geq 2.6t)$$

I 2I

Corrugated sheet iron, $c \leq H \leq t$

parabolically curved

Approximate values of least radius of gyration r

$$r \approx 0.36336D \quad 0.295D \quad D/4.58 \quad D/3.54 \quad D/6$$

$$r \approx D/4.74 \quad D/5 \quad BD/[2.6(B \leq D)] \quad D/4.74$$

* Some of the cross sections depicted in this table will be encountered most often in machinery as castings, forgings, or individual sections

assembled and joined mechanically (or welded). A number of the sections shown are obsolete and will be encountered mainly in older equipment

and/or building structures.

expressed in radians. A second integration gives the vertical deflection criterion for design, e.g., of machine tools, for which the relative posi-

of any point of the elastic curve from its original position. tions of tool and workpiece must be maintained while the cutting loads

EXAMPLE. In the cantilever beam shown in Fig. 5.2.35, the bending moment

are applied during operation. Similarly, large steam-turbine shafts sup-

at any section $x \geq P(l \geq x) \leq EI \, d^2y/(dx)^2$. Integrate and determine constant by ported on two end bearings must maintain alignment and tight critical

the condition that when $x = 0$, $dy/dx = 0$. Then $EI \, dy/dx \leq P/x \leq 1/2 \, Px^2$. clearances between the rotating blade assemblies and the stationary

Integrate again; and determine constant by the condition that when $x = 0$, $y = 0$. stator blades during operation. When more than one beam shares a load,

Then $EIy \leq 1/2 \, Plx^2 \leq Px^3/6$. This is the equation of the elastic curve. When $x =$ each beam will assume a portion of the load that is proportional to its

I , $y = f \leq Pl^3/(3EI)$. In general, the two constants of integration must be stiffness. Superposition may be used in connection with both stresses and

determined simultaneously. deflections.

Deflection in general, f , may be expressed by the equation $f \leq Pl^3/$ EXAMPLE. (Fig. 5.2.36). Two wooden stringers — one (A) 8 in in cross

(mEI), where m is a coefficient. See Tables 5.2.2 and 5.2.4 for values of section and 20 ft in span, the other (B) 8 in in 8 in in 16 ft — carrying the center

f for beams of various sections and loadings. For coefficients of deflec- load $P_0 = 22,000$ lb are required, the load carried by each stringer. The deflections

tion of wooden beams and structural steel shapes, see Sec. 12.2. f of the two stringers must be equal. Load on A $= P_1$, and on B $= P_2$. $f \leq$

Since I varies as the cube of the depth, the stiffness, or inverse deflec- $P_1I_1^{3/4}/(48EI_1) \leq P_2I_2^{3/4}/(48EI_2)$. Then $P_1/P_2 \leq I_2^{3/4}/(I_1^{3/4}I_2) \leq 4$. $P_0 \leq P_1 \leq P_2 \leq$

tion, of various beams varies, other factors remaining constant, in- $4P_2 \leq P_2$, whence $P_2 \leq 22,000/5 \leq 4,400$ lb (1,998 kgf) and $P_1 \leq 4 \leq$

versely as the load, inversely as the cube of the span, and directly as the $4,400 \times 17,600 \text{ lb (7,990 kgf)}$.

cube of the depth. This deflection is due to bending moment only. In general, however, the bending of beams involves transverse shearing stresses which cause shearing strains and thus add to the total deflection. The contribution of shearing strain to overall deflection becomes significant only when the beam span is very short. These strains may affect substantially the strength as well as the deflection of beams. When deflection due to transverse shear is to be accounted for, the differential equation of the elastic curve takes the form

$$EI$$

$$d^2y$$

$$dx^2$$

$$EI \frac{d^2y}{dx^2}$$

$$dx^2$$

$$d^2y$$

$$dx^2$$

$$EI \frac{d^2y}{dx^2}$$

$$kEI$$

$$AG$$

$$d^2M$$

$$dx^2$$

where k is a factor dependent upon the beam cross section. Sergius

Sergev, in "The Effect of Shearing Forces on the Deflection and Strength of Beams" (Univ. Wash. Eng. Exp. Stn. Bull. 114) gives k

1.2 for rectangular sections, 10/9 for circular sections, and 2.4 for I Fig. 5.2.36

beams. He also points out that in the case of a deep, rectangular-section

cantilever, carrying a concentrated load at the free end, the deflection Relation between Deflection and Stress

due to shear may be up to 3.1 percent of that due to bending moment; if

this beam supports a uniformly distributed load, it may be up to 4.1 Combine the formula $M = SI/c = Pl/n$, where n is a constant, $P =$ load,

percent. A deep, simple beam deflection may increase up to 15.6 per- and $I =$ span, with formula $f = Pl^3/(mEI)$, where m is a constant. Then

cent when carrying a uniformly distributed load and up to 12.5 percent $f = C_{\diamond\diamond} SI^2/(Ec)$

when the load is concentrated at midspan.

Design of beams may be based on strength (stress) or on stiffness if where $C_{\diamond\diamond}$ is a new constant n/m . Other factors remaining the same,

deflection must be limited. Deflection rather than stress becomes the the deflection varies directly as the stress and inversely as E . If the span is

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Table 5.2.7

Beam Load n m $C \diamond \diamond$

Cantilever Concentrated at end 1 3/

13

Cantilever Uniform 2 8/

14

Simple Concentrated at center 4 48 1/12

Simple Uniform 8 384/5 5/48

Fixed ends Concentrated at center 8 192 1/24

Fixed ends Uniform 12 384 1/32

One end fixed, one end supported Concentrated at center 16/3 768/7 7/144

One end fixed, one end supported Uniform 128/9 185 1/13

Simple Uniformly varying, 6 60 1/10

maximum at center

constant, a shallow beam will submit to greater deformations than a moment diagram $\int M dx$ up to that point; and a slope diagram may be

deeper beam without exceeding a safe stress. If depth is constant, a derived from the moment diagram in the same manner as the moment

beam of double span will attain a given deflection with only one-quarter diagram was derived from the shear diagram.

the stress. Values of n , m , and $C \diamond \diamond$ are given in Table 5.2.7 (for other If I is not constant, draw a new curve whose ordinates are M/I and

values, see Table 5.2.2). use these M/I ordinates just as the M ordinates were used in the case where I was constant; that is, $\int (M/I) dx \propto E (f \leq C)$. The ordinate at any

Graphical Relations point of the slope curve is thus proportional to the area of the M/I curve

Referring to Fig. 5.2.37, the shear V acting at any section is equal to the to the right of that point.

Again, since $iE \propto E df/dx$.

total load on the right of the section, or

$V \propto \int w dx$

$\propto iE dx \propto E df \propto E (f \leq C)$

and thus the ordinate f to the elastic curve at any point is proportional to

Since $w dx$ is the product of w , a loading intensity (which is expressed the area of the slope diagram $\int i dx$ up to that point. The equilibrium

as a vertical height in the load diagram), and dx , an elementary length polygon may be used in drawing the deflection curve directly from the

along the horizontal, evidently $w dx$ is the area of a small vertical strip M/I diagram.

of the load diagram. Then $\int w dx$ is the summation of all such vertical Thus, the five curves of load, shear, moment, slope, and deflection

strips between two indefinite points. Thus, to obtain the shear in any are so related that each curve is derived from the previous one by a

process of graphical integration, and with proper regard to scales the deflection is thereby obtained.

The vertical distance from any point A (Fig. 5.2.38) on the elastic curve of a beam to the tangent at any other point B equals the moment of the area of the $M/(EI)$ diagram from A to B about A. This distance, the tangential deviation t_{AB} , may be used with the slope-area relation and the geometry of the elastic curve to obtain deflections. These theorems, together with the equilibrium equations, can be used to compute reactions in the case of statically indeterminate beams.

Fig. 5.2.37

Fig. 5.2.38

section mn, find the area of the load diagram up to that section, and draw a second diagram called the shear diagram, any ordinate of which is proportional to the shear, or to the area in the load diagram to the right of mn. Since $V = dM/dx$,

$$\int V dx = M$$

$$y_B = t_{AB} = \text{moment area}$$

$$M$$

$$EI$$

$$\int B$$

$$A$$

$$A$$

$$\frac{1}{EI} \int \int \int \int$$

By similar reasoning, a moment diagram may be drawn, such that the ordinate at any point is proportional to the area of the shear diagram to the right of that point. Since $M = EI \frac{d^2 f}{dx^2}$, $\frac{1}{EI} \int \int \int \int$

$$EI \frac{d^4 f}{dx^4} = \frac{1}{48EI}$$

ordinate at any point is proportional to the area of the shear diagram to the right of that point. Since $M = EI \frac{d^2 f}{dx^2}$, $\frac{1}{EI} \int \int \int \int$

$$\int \text{area}$$

$$M$$

$$\int C$$

$$1$$

$$\frac{1}{EI}$$

$$\frac{1}{EI^2}$$

$$\int$$

$$B EI B EI \frac{1}{48EI}$$

$$M dx = EI \left(\frac{df}{dx} \leq C \right) = EI \left(\frac{1}{EI} \leq C \right) y_D = \int \int \int \int$$

$$I$$

4

 $\geq t_{DC}$

if I is constant. Here C is a constant of integration. Thus i , the slope or $\frac{1}{EI} \int \frac{1}{2} P l^2 dx = \frac{1}{EI} \int \frac{1}{2} P l^2 dx = \frac{1}{EI} \int \frac{1}{2} P l^2 dx$

 $\frac{1}{EI} \int \frac{1}{2} P l^2 dx = \frac{1}{EI} \int \frac{1}{2} P l^2 dx = \frac{1}{EI} \int \frac{1}{2} P l^2 dx$

grade of the elastic curve at any point, is proportional to the area of the $\frac{1}{EI} \int \frac{1}{2} P l^2 dx = \frac{1}{EI} \int \frac{1}{2} P l^2 dx = \frac{1}{EI} \int \frac{1}{2} P l^2 dx$

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5-32 MECHANICS OF MATERIALS

Resilience of Beams If the two moving loads are of unequal weight, the condition for maxi-

The external work of a load gradually applied to a beam, and which maximum moment is that the maximum moment will occur under the heavy

increases from zero to P , is $\frac{1}{2} P f$ and equals the resilience U . But, from wheel when the center of the beam bisects the distance between the

the formulas $P = nSI/(cl)$ and $f = nSI^2/(mcE)$, where n and m are resultant of the loads and the heavy wheel. Figure 5.2.41 shows this

constants that depend upon loading and supports, S = fiber stress, c = position and the shear and moment diagrams.

distance from neutral axis to outer fiber, and l = length of span. Substi- When several wheel loads constituting a system occur, the several sus-

tute for P and f , and pected wheels must be examined in turn to determine which will cause

 $\frac{1}{EI}$

the greatest moment. The position for the greatest moment that can occur

 $n^2 k^2 S^2 V$

under a given wheel is, as stated, when the center of the span bisects the

 $U =$

$m c^2 E$ distance between the wheel in question and the resultant of all the loads

then on the span. The position for maximum shear at the support will be

where k is the radius of gyration and V the volume of the beam. For when one wheel is passing off the span.

values of U , see Table 5.2.1.

The resilience of beams of similar cross section at a given stress is proportional to their volumes. The internal resilience, or the elastic deformation energy in the material of a beam in a length x is dU , and

 $U = \frac{1}{2} \int \frac{M^2 dx}{EI} = \frac{1}{2} \int M di$

where M is the moment at any point x , and di is the angle between the tangents to the elastic curve at the ends of dx . The values of resilience and deflection in special cases are easily developed from this equation.

Rolling Loads

Rolling or moving loads are those loads which may change their position on a beam. Figure 5.2.39 represents a beam with two equal concentrated moving loads, such as two wheels on a crane girder, or the wheels of a

Fig. 5.2.40

Fig. 5.2.39

truck on a bridge. Since the maximum moment occurs where the shear is zero, it is evident from the shear diagram that the maximum moment will occur under a wheel. $x \downarrow a/2$:

□ □

Fig. 5.2.41

$2x \ a$

$R_1 \ \square \ P \ 1 \ \geq \ \leq$

|| Constrained Beams

$M_2 \ \square$

PI

2

$\square 1 \geq \leq$

a

I

$2x \ a$

$||$

≥ 2

$4x \ 2$

I

□ Constrained beams are those so held or “built in” at one or both ends that the tangent to the elastic curve remains fixed in direction. These

□ □

beams are held at the ends in such a manner as to allow free horizontal $2x \ a$ motion, as illustrated by Fig. 5.2.42. A constrained beam is stiffer than a

$R_2 \ \square \ P \ 1 \ \leq \ \geq$

|| simple beam of the same material, because of the modification of the

$M_1 \ \square$

PI

2

$\square \ a$

$1 \geq \geq 2 \ \leq$

I

$2a \ 2$

I

$2x \ 3a$

$||$

$4x \ 2$

≥ 2

l

□ moment by an end resisting moment. Figure 5.2.43 shows the two most common cases of constrained beams. See also Table 5.2.2.

M2 max when $x = 1/4a$

M1 max when $x = 3/4a$

Mmax □

Pl

2

□1 ≧

a 2

2l

□ □P

2l

l ≧

a 2

2

□ □

EXAMPLE. Two wheel loads of 3,000 lb each, spaced on 5-ft centers, move on a span of $l = 15$ ft, $x = 1.25$ ft, and $R_2 = 2,500$ lb. ♦ Mmax □ M2 □

Fig. 5.2.42 Fig. 5.2.43

2,500 □ 6.25 (1,135 □ 1.90) □ 15,600 lb □ ft (2,159 kgf □ m).

Continuous Beams

Figure 5.2.40 shows the condition when two equal loads are equally distant on opposite sides of the center. The moment is equal under the A continuous beam is one resting upon several supports which may or

two loads. may not be in the same horizontal plane. The general discussion for

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beams holds for continuous beams. $S_v A = V$, $SI/c = M$, and $d^2 f/dx^2 =$ Figure 5.2.46 shows the values of the functions for a uniformly

$M/(EI)$. The shear at any section is equal to the algebraic sum of the loaded continuous beam resting on three equal spans with four supports.

components parallel to the section of all external forces on either side of Continuous beams are stronger and much stiffer than simple beams.

the section. The bending moment at any section is equal to the moment However, a small, unequal subsidence of piers will cause serious

of all external forces on either side of the section. The relations stated

above between shear and moment diagrams hold true for continuous

beams. The bending moment at any section is equal to the bending

moment at any other section, plus the shear at that section times its arm,

plus the product of all the intervening external forces times their respective arms. To illustrate (Fig. 5.2.44):

$$V_x \leq R_1 \leq R_2 \leq R_3 \geq P_1 \geq P_2 \geq P_3$$

$$M_x \leq R_1(l_1 \leq l_2 \leq x) \leq R_2(l_2 \leq x) \leq R_3 x$$

$$\geq P_1(l_2 \leq c \leq x) \geq P_2(b \leq x) \geq P_3 a$$

$$M_x \leq M_3 \leq V_3 x \geq P_3 a$$

Table 5.2.8 gives the value of the moment at the various supports of a uniformly loaded continuous beam over equal spans, and it also gives the values of the shears on each side of the supports. Note that the shear Fig. 5.2.45 is of opposite sign on either side of the supports and that the sum of the two shears is equal to the reaction.

Fig. 5.2.44

Figure 5.2.45 shows the relation between the moment and shear diagrams for a uniformly loaded continuous beam of four equal spans (see Table 5.2.8). Table 5.2.8 also gives the maximum bending moment which will occur between supports, and in addition the position of this moment and the points of inflection (see Fig. 5.2.46). Fig. 5.2.46

Table 5.2.8 Uniformly Loaded Continuous Beams over Equal Spans

(Uniform load per unit length $\leq w$; length of equal span $\leq l$)

Shear on each

side of support.

L $\leq$ left, R $\leq$ right.

Reaction at any Distance to point Distance to point

Notation of support is L $\leq$ R Moment Max of max moment, of inflection,

Number of support over each moment in measured to right measured to right

supports of span L R support each span from support from support

2 1 or 2 0/

12 0 0.125 0.500 None

3 1 0/

38 0 0.0703 0.375 0.750

2/

58/

58/

18 0.0703 0.625 0.250

4 1 0/

4 10 0 0.080 0.400 0.800

2/

6 10/

5 10/

1 10 0.025 0.500 0.276, 0.724

5 1 0/

11 28 0 0.0772 0.393 0.786
 2/
 17 28/
 15 28/
 3 28 0.0364 0.536 0.266, 0.806
 3 13/28/
 13 28/
 2 28 0.0364 0.464 0.194, 0.734
 6 1 0/
 15 38 0 0.0779 0.395 0.789
 2/
 23 38/
 20 38/
 4 38 0.0332 0.526 0.268, 0.783
 3 18/38 19/38 3/38 0.0461 0.500 0.196, 0.804
 7 1 0/
 41 104 0 0.0777 0.394 0.788
 2/
 63 104/
 55 104/
 11 104 0.0340 0.533 0.268, 0.790
 3 49/104 51/104 8/104 0.0433 0.490 0.196, 0.785
 4 53/104 53/104 9/104 0.0433 0.510 0.215, 0.804
 8 1 0/
 56 142 0 0.0778 0.394 0.789
 2/
 86 142/
 75 142/
 15 142 0.0338 0.528 0.268, 0.788
 3 67/142/
 70 142/
 11 142 0.0440 0.493 0.196, 0.790
 4 72/142 71/142 12/142 0.0405 0.500 0.215, 0.785

Values apply to: wl wl wl 2 wl 2 l

NOTE: The numerical values given are coefficients of the expressions at the foot of each column.

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Table 5.2.9 Beams of Uniform Strength (in Bending)

Elevation

Beam Cross section and plan Formulas

1. Fixed at one end, load P concentrated at other end

Rectangle: Elevation: 1, 6P

width (b) top, straight $y^2 \propto x$

bSs

constant, line; bottom,

$\sqrt{bS}$

6PI

depth (g) parabola. 2, $h \propto$

variable complete pa- s

rabola

Deflection at A:

Plan: rectangle

$f \propto$

8P

bE

$\propto I$

h

3

Rectangle: Elevation: rec- 6P

width (y) tangle $y \propto x$

h 2Ss

variable,

6PI

depth (h) $b \propto$

constant h 2Ss

Plan: triangle Deflection at A:

$f \propto$

6P

bE

$\propto I$

h

3

Rectangle: Elevation: cubic 6P

width (z) parabola $y^3 \propto x$

kSs

variable,

depth (y) $z \propto ky$

$\sqrt{kS}$

variable 3 6 PI

Plan: cubic pa- $h \propto$

z

$\frac{1}{2} k(\text{const.})$ parabola s

y b $\frac{1}{2} kh$

Circle: diam Elevation: cubic $\frac{3}{2} P$

$y^3 \frac{1}{2} x$

(y) variable parabola $\frac{1}{2} Ss$

$\frac{3}{2} \frac{3}{2} PI$

Plan: cubic pa- d $\frac{1}{2}$

parabola $\frac{1}{2} Ss$

2. Fixed at one end, load of total magnitude P uniformly distributed

$\sqrt{\frac{1}{2}} bIS$

Rectangle Elevation: $\frac{3}{2} P$

width (b) triangle $y \frac{1}{2} x$

constant,

$\sqrt{\frac{1}{2}} bS$

$\frac{3}{2} PI$

depth (y) h $\frac{1}{2}$

variable s

Plan: rectangle f $\frac{1}{2} 6$

P

bE

$\frac{1}{2} \frac{1}{2} I$

h

$\frac{3}{2}$

Rectangle: Elevation: rec- $\frac{3}{2} Px \frac{2}{2}$

width (y) triangle $y \frac{1}{2}$

$ISs h^2$

variable, $\frac{3}{2} PI$

depth (h) b $\frac{1}{2}$

$Ss h^2$

constant

Plan: two para- Deflection at A:

bolic curves

with vertices at

free end

f $\frac{1}{2}$

$\frac{3}{2} P$

bE

$\frac{1}{2} \frac{1}{2} I$

h

3

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Table 5.2.9 Beams of Uniform Strength (in Bending) (Continued)

Elevation

Beam Cross section and plan Formulas

2. Fixed at one end, load of total magnitude P uniformly distributedRectangle: Elevation: semi- $3Px^2$ width (z) cubic parabola $y^3 \propto$ kS s l

variable,

depth (y) $z \propto ky$ $\sqrt{kS}$ variable, $3 \cdot 3PI$ Plan: semicubic $h \propto$ z $\propto k$ parabola s $y \propto b \propto kh$ Circle: diam Elevation: semi- $16P^2$ $y^3 \propto x$ (y) variable cubic parabola $\propto ISs$ $\sqrt{\propto S}$ $3 \cdot 16 PI$ Plan: semicubic $d \propto$

parabola s

3. Supported at both ends, load P concentrated at point C $\sqrt{S} \propto b \propto x$ Rectangle: Elevation: two $3P$ width (b) parabolas, ver- $y \propto$ s

constant, tics at points

 $\sqrt{2bS}$ depth (y) of support $3PI$ $h \propto$

variable

 $\propto \propto$ s $P \propto 3$

Plan: rectangle f

$2Eb h$

Rectangle: Elevation: rec- $3P$

width (y) tangle y x

$S sh2$

variable,

$3PI$

depth (h) Plan: two trian- b

constant gles, vertices $2Ss h2$

at points of $3PI3$

f

support $8Eb h3$

Rectangle: Elevation: two $6P(l \geq p)$

width (b) parabolas, ver- $y2 \leq x$

$bISs$

constant, tices at points

$6Pp$

depth (y or of support $y12 \leq x1$

$y1$) variable $bISs$

$\sqrt{bIS}$

$6P(l \geq p)p$

Plan: rectangle h

s

Load P moving across span

Rectangle: Elevation: $x2 y2$

≤ 1

width (b)

constant,

depth (y)

ellipse

Major axis $\leq l$

Minor axis $\leq 2h$

$\leq l$

2

$2 3PI$

$2bSs$

variable

$\sqrt{2bS}$

$3PI$

Plan: rectangle h

s

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Table 5.2.9 Beams of Uniform Strength (in Bending) (Continued)

Elevation

Beam Cross section and plan Formulas

4. Supported at both ends, load of total magnitude P uniformly distributed

Rectangle: Elevation: $x^2 y^2$ $\leq \frac{1}{2}$ $\frac{1}{2}$ width (b) ellipse $\frac{2}{3} \pi I$

I

constant,

 $\frac{2}{3} 4bSs$

depth (y)

 $\sqrt{\frac{4}{3} bS}$ variable $\frac{3}{2} \pi I$ h^2

s

Deflection at O:

Plan: rectangle $\frac{1}{2} \pi I^3$ f^2 $\frac{64}{3} EI$ $\frac{3}{2} P$ $\frac{16}{3} bE$ $\frac{1}{2} \pi I$

h

3

Rectangle:

width (y)

Elevation: rec-

tangle y^2 $\frac{3}{2} P$ $Ss h^2$ $\frac{1}{2} \pi x^2$ x^2

I

variable,

 $\frac{3}{2} \pi I$

depth (h) Plan: two parab- b^2
 constant σ as with τ - $4S_s h^2$
 tics at center
 of span

changes in sign and magnitude of the bending stresses. reactions, and shears.

Maxwell's Theorem When a number of loads rest upon a beam, the deflection at any point is equal to the sum of the deflections at this point due to each of the loads taken separately. Maxwell's theorem states that if unit loads rest upon a beam at two points A and B, the deflection at A due to the unit load at B equals the deflection at B due to the unit load at A.

Castigliano's theorem states that the deflection of the point of application of an external force acting on a beam is equal to the partial derivative of the work of deformation with respect to this force. Thus, if P is the force, f the deflection, and U the work of deformation, which equals the resilience,

$$dU/dP = f$$

According to the principle of least work, the deformation of any struc-

Fig. 5.2.47 Fig. 5.2.48

ture takes place in such a manner that the work of deformation is a minimum.

Beams of Uniform Strength

TORSION

Beams of uniform strength vary in section so that the unit stress S remains constant, the I/c varies as M. For rectangular beams, of breadth b and depth d, $I/c = bd^2/6$; and $M = Sbd^2/6$. Thus, for a cantilever beam Pp. Elements of the surface becomes helices of angle ϕ , and a radius

of rectangular cross section, under a load P, $Px = Sbd^2/6$. If b is constant, d^2 varies with x, and the profile of the shape of the beam will be a parabola, as Fig. 5.2.47. If d is constant, b will vary as x and the beam moment of inertia; G = shearing modulus of elasticity. It is assumed

will be triangular in plan, as shown in Fig. 5.2.48. that the cross sections remain plane surfaces. The strain on the cross

Shear at the end of a beam necessitates a modification of the forms section is wholly tangential, and is zero at the center of the section.

determined above. The area required to resist shear will be P/S_v in a Note that $I_d = r^2$.

cantilever and R/S_v in a simple beam. The dotted extensions in Figs. In the case of a circular cross section, the stress S_v increases directly as

5.2.47 and 5.2.48 show the changes necessary to enable these cantilevers to resist shear. The extra material and cost of fabrication, however, make many of the forms impractical. The polar moment of inertia I_p for any section may be obtained from

ever, make many of the forms impractical. $I_p \geq I_1 + I_2$, where I_1 and I_2 are the rectangular moments of inertia of

Table 5.2.9 shows some of the simple sections of uniform strength. In the section about any two lines at right angles to each other, through the center of gravity. none of these, however, is shear taken into account.

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TORSION 5-37

The external twisting moment M_t is balanced by the internal resisting which has the same shape as the bar and then inflated. The resulting

moment. three-dimensional surface provides the following: (1) The torque trans-

For strength, $M_t \propto S_v I_p / r$. mitted is proportional to twice the volume under the inflated membrane,

For stiffness, $M_t \propto a G I_p / l$. and (2) the shear stress at any point is proportional to the slope of the

The torsional resilience $U \propto 1/2 P \Delta \propto S^2 v I_p l / (2r^2 G) \propto a^2 G I_p / (2l)$. curve measured perpendicular to that slope. In recent years, several

other mathematical techniques have become widely used, especially with the aid of faster computational methods available from electronic computers.

By using finite-difference methods, the differential operators of the governing equations are replaced with difference operators which are related to the desired unknown values at a gridwork of points in the outline of the cross section being investigated.

The finite-element method, commonly referred to as FEM, deals with a spatial approximation of a complex shape which is then analyzed to determine deformations, stresses, etc. By using FEM, the exact structure is replaced with a set of simple structural elements interconnected at a finite number of nodes. The governing equations for the approximate

Fig. 5.2.49 structure can be solved exactly. Note that inasmuch as there is an exact solution for an approximate structure, the end result must be viewed, and the results thereof used, as approximate solutions to the real structure.

The state of stress on an element taken from the surface of the shaft, Using a finite-difference approach to Poisson's partial differential

as in Fig. 5.2.50, is pure shear. Pure tension exists at right angles to one equation, which defines the stress functions for solid and hollow shafts

45° helix and pure compression at right angles to the opposite helix. with generalized contours, along with Prandl's membrane analogy, Isa-

Reduced formulas for shafts of various sections are given in Table kower has developed a series of practical design charts (ARRADCOM-

5.2.11. When the ratio of shaft length to the largest lateral dimension in MISD Manual UN 80-5, January 1981, Department of the Army). Di-

the cross section is less than approximately 2, the end effects may drastically affect the torsional stresses calculated. shearing stress have been generated. Information for circular shafts with

Failure under torsion in brittle materials is a tensile failure at right

rectangular and circular keyways, external splines and milled flats, as

angles to a helical element on the surface. Plastic materials twist off well as rectangular and X-shaped torsion bars, is presented.

squarely. Fibrous materials separate in long strips. Assuming the stress distribution from the point of maximum stress to

Torsion of Noncircular Sections When a section is not circular, the

the corner to be parabolic, Bach derived the approximate expression,

unit stress no longer varies directly as the distance from the center. $S_s = M / (2b^2h)$ for a rectangular section, b by h , where $h \gg b$. For

Cross sections become warped, and the greatest unit stress usually closer results, the shearing stresses for a rectangular section (Fig. 5.2.51)

occurs at a point on the perimeter of the cross section nearest the axis of may be expressed $S_A = M / (\alpha_A b^2h)$ and $S_B = M / (\alpha_B b^2h)$. The angle of

twist; thus, there is no stress at the corners of square and rectangular twist for these shafts is $\theta = M / (\alpha_G b^3h)$. The factors α_A , α_B , and α are

sections. The analyses become complex for noncircular sections, and functions of the ratio h/b and are given in Table 5.2.10.

the methods for solution of design problems using them most often In the case of composite sections, such as a tee or angle, the torque that

admit only of approximations. can be resisted is $M_t = G \theta \sum hb^3$; the summation applies to each of the

rectangles into which the section can be divided. The maximum stress

occurs on the component rectangle having the largest b value. It is

computed from

$$S_A = M_t \alpha_A / (\alpha_A b^2h^3)$$

Torque, deflection, and work relations for some additional sections

are given in Table 5.2.11.

Fig. 5.2.50

Torsion problems have been solved for many different noncircular

cross sections by utilizing the membrane analogy, due to Prandtl, which

makes use of the fact that the mathematical treatment of a twisted bar is

governed by the same equations as for a membrane stretched over a hole Fig. 5.2.51

Table 5.2.10 Factors for Torsion of Rectangular Shafts (Fig. 5.2.51)

b/b 1.00 1.50 1.75 2.00 2.50 3.00 4.00 5.00 6.00 8.00 10.0 ∞

α_A 0.208 0.231 0.239 0.246 0.258 0.267 0.282 0.291 0.299 0.307 0.312 0.333

α_B 0.208 0.269 0.291 0.309 0.336 0.355 0.378 0.392 0.402 0.414 0.421

0.141 0.196 0.214 0.229 0.249 0.263 0.281 0.291 0.299 0.307 0.312 0.333

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Table 5.2.11 Torsion of Shafts of Various Cross Sections

(For strength and stiffness of shafts, see Sec. 8.2)

Angular twist, ϕ

Torsional

(length ℓ in, radius r in)

resisting

moment In terms of tor- In terms of max Work of torsion

Cross section M_t sional moment shear (V volume)

$\phi = 3 M_t / 32 M_t S_{vmax} = 1 / 1 S_{2v} \max$

$d S_v = 2 V$

$16 GIP = d^4 G / d^4 G$

(Note 1)

$\phi = D^4 \geq d^4 32 M_t S_{vmax} = 1 / 1 S_{v2} \max D^2 \leq d^2$

$S_v = 2 V$

$16 D = (D^4 \geq d^4) G G D^4 G D^2$

(Note 2)

$\phi = 2 / 16 b^2 \leq h^2 M_t S_{vmax} b^2 \leq h^2 1 S_{2v} \max b^2 \leq h^2$

$b h S_v V$

$16 \phi = b^3 h^3 G G b h^2 8 G h^2$

($h \rightarrow b$) (Note 3)

$2/9 b^2 h^5$

$b^2 \leq h^2 M_t S_{vmax} b^2 \leq h^2 4 S_{2v} \max b^2 \leq h^2$

$v = 3.6 \cdot 0.8 \cdot V$

($h \rightarrow b$) $b^3 h^3 G G b h^2 45 G h^2$

(Note 4)

$2/9 \cdot 3$

$1 M_t S_{vmax} = 1 / 8 S_{2v} \max$

$/ h S_v = 7.2 / 1.6 V$

$h^4 G G h^4 5 G$

(Note 5)

$b^2 = 1 M_t S_{vmax} = 1$

$S = 46.2 / 2.31$

$20 v = b^4 G G b$

$b^3 = 1 M_t S_{vmax} = 1$

$S_v = 0.967 / 0.9$

1.09 b4 G G b

*When $h/b \square 1 \ 2 \ 4 \ 8$

Coefficient 3.6 becomes $\square 3.56 \ 3.50 \ 3.35 \ 3.21$

Coefficient 0.8 becomes $\square 0.79 \ 0.78 \ 0.74 \ 0.71$

NOTES: (1) S_{vmax} at circumference. (2) S_{vmax} at outer circumference. (3) S_{vmax} at A ; $S_v B \square 16$
 $M_t / \square bh^2$. (4) S_{vmax} at middle of side h ; in middle of b ,

$S_v \square 9 M_t / 2bh^2$. (5) S_{vmax} at middle of side.

COLUMNS when both are fixed, $n \square 4$; and when one end is fixed with the other

free, $n \square 1/4$. The slenderness ratio that separates long columns from

Members subjected to direct compression can be grouped into three short ones depends upon the modulus of elasticity and the yield strength

classes. Compression blocks are so short (slenderness ratios below 30) of the column material.

When Euler's formula results in $(P_{cr} / A) \rightarrow S_y$,

that bending of member is unlikely. At the other limit, columns so strength rather than buckling causes failure, and the column ceases to be

slender that bending is primary, are the long columns defined by Euler's long. In round numbers, this critical slenderness ratio falls between 120

theory. The intermediate columns, quite common in practice, are called and 150. Table 5.2.12 gives additional facts concerning long columns.

short columns.

Short Columns The stress in a short column may be considered

Long columns and the more slender short columns usually fail by partly due to compression and partly due to bending. A theoretical

buckling when the critical load is reached. This is a matter of instability;

equation has not been derived. Empirical, though rational, expressions

that is, the column may continue to yield and deflect even though the are, in general, based on the assumption that the permissible stress must

load is not increased above critical. The slenderness ratio is the unsup- be reduced below that which could be permitted were it due to compres-

ported length divided by the least radius of gyration, parallel to which it sion only. The manner in which this reduction is made determines the

can bend. type of equation as well as the slenderness ratio beyond which the

Long columns are handled by Euler's column formula, equation does not apply. Figure 5.2.52 illustrates the situation. Some

$P_{cr} \square n \square 2EI / l^2 \square n \square 2EA / (l/r)^2$ typical formulas are given in Table 5.2.13.

EXAMPLE. A machine member unsupported for a length of 15 in has a square

The coefficient n accounts for end conditions. When the column is pivoted cross section 0.5 in on a side. It is to be subjected to compression. What maximum

at both ends, $n \square 1$; when one end is fixed and other rounded, $n \square 2$; safe load can be applied centrally, according to the AISC formula? At the com-

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COLUMNS 5-39

Table 5.2.12 Strength of Round-ended Columns according to Euler's Formula*

Low- Medium-

Wrought carbon carbon

Material Cast iron iron† steel steel

Ultimate compressive strength, lb/in² 107,000 53,400 62,600 89,000Allowable compressive stress, lb/in² 7,100 15,400 17,000 20,000

(maximum)

Modulus of elasticity 14,200,000 28,400,000 30,600,000 31,300,000

Factor of safety 8 5 5 5

Smallest I allowable at worst section, in⁴ PI 2 PI 2 PI 2 PI 2

17,500,000 56,000,000 60,300,000 61,700,000

Limit of ratio, l/r 50.0 60.6 59.4 55.6

Rectangle ($r = b\sqrt{1/12}$), l/b → 14.4 17.5 17.2 16.0Circle ($r = 1/4 d$), l/d → 12.5 15.2 14.9 13.9Circular ring of small thickness ($r = d\sqrt{1/8}$), l/d → 17.6 21.4 21.1 19.7

* P = allowable load, lb; l = length of column, in; b = smallest dimension of a rectangular section, in; d = diameter of a circular section, in; r = least radius of gyration of section.

† This material is no longer manufactured but may be encountered in existing structures and machinery.

Table 5.2.13 Typical Short-Column Formulas

Formula Material Code Slenderness ratio

 $S_w \leq 17,000 \geq 0.485 \frac{l}{r}$

r

2

Carbon steels AISC l/r ↓ 20

 $S_w \leq 16,000 \geq 70 \left(\frac{l}{r} \right)$ Carbon steels Chicago l/r ↓ 120 $S_w \leq 15,000 \geq 50 \frac{l}{r}$

r

Carbon steels AREA l/r ↓ 150

l

 $S_w \leq 19,000 \geq 100 \left(\frac{l}{r} \right)$ Carbon steels Am. Br. Co. 60 ↓ ↓ 120

r

 $Scr^* \leq 135,000 \geq 15.9$

c

l

r

2

Alloy-steel tubing ANC

1

 $\sqrt{cr}$

↓ 65

□□

Sw □ 9,000 ≧ 40

l

r

Cast iron NYC

l

r

↓ 70

Scr* □ 34,500 ≧ □□ 245

√c

l

r

2017ST Aluminum ANC

1

√cr

↓ 94

Scr* □ 5,000 ≧ □□ 0.5

c

l

r

2

Spruce ANC

l

√cr

↓ 72

Scr* □ Sy □ □ □□

1≧

Sy

4n□ 2E

l

r

2

Steels Johnson

l

r

√ S

2n□ 2E

y

Sy l

$S_{cr} \dagger \square \uparrow$ critical

$\square \sqrt{\square}$

Steels Secant

$E_c I P r$

$1 \leq \sec$

$r^2 r 4AE$

* S_{cr} theoretical maximum, c end fixity coefficient,

$c = 2$, both ends pivoted; $c = 2.86$, one pivoted, other fixed;

$c = 4$, both ends fixed; $c = 1$, one end fixed, one end free.

$\dagger e$ is initial eccentricity at which load is applied to center of column cross section.

puted load, what size section (also square) would be needed, if it were to be two bending moments, M_1 due to longitudinal load ($\leq$ for compression

designed according to the AREA formula? and $\geq$ for tension), and M_2 due to transverse load. $M = M_2 = M_1$. Here

$l/r = 15/0.5/\sqrt{12} = 104$ short column $M_1 = P f$ and $f = C S_b / 2 / (E_c)$.

$P/A = 17,000 \geq 0.485 (104)^2 = 11,730$

or $P = 0.25 = 11,730 = 2,940 \text{ lb} (1,335 \text{ kgf})$

and

2,940

a^2

$= 15,000 \geq 50 = 15la$

$\sqrt{12}$

$= 15,000 \geq$

2,600

a

thus $a^2 \geq 0.173a \geq 0.196 = 0$ or $a = 0.536 \text{ in} (1.36 \text{ cm})$

Combined Flexure and Longitudinal Force Figure 5.2.53 shows a

bar under flexure due to transverse and longitudinal loads. The maxi-

mum fiber stress S is made up of S_0 , due to the direct action of load P ,

and S_b , due to the entire bending moment M . M is the algebraic sum of Fig. 5.2.52

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FOR THE CASE OF LONGITUDINAL COMPRESSION. $S_b I/c = M_2 \leq$

$C P S_b I 2 / (E_o)$, or $S_b = M_2 c / I \geq C P I 2 / E$. The maximum stress is $S =$

$S_b \leq S_0$ compression. The constant C for the case of Fig. 5.2.53 is de-

rived from the equations $P \Delta l / 4 = S_b I / c$ and $f = P \Delta l^3 / (48 E I)$. Solving for

f ; $f = 1/12 S_b I 2 / (E_c)$, or $C = 1/12$. For a beam supported at the ends and

uniformly loaded, $C = 5/48$. Other cases can be similarly calculated.

FOR THE CASE OF LONGITUDINAL TENSION. $M = M_2 \geq P f$, and $S_b =$

$M_2 c / I \leq C P I 2 / E$. The maximum stress is $S = S_b \leq S_0$, tension.

Fig. 5.2.53

ECCENTRIC LOADS

Fig. 5.2.54

When short blocks are loaded eccentrically in compression or in tension,

i.e., not through the center of gravity (cg), a combination of axial and CHIMNEY PROBLEM. Weight of chimney $\approx 563,000$ lb; $e \approx 1.56$ ft; OD of

bending stress results. The maximum unit stress S_M is the algebraic sum chimney ≈ 10 ft 8 in; ID ≈ 6 ft 6 1/2 in. Overturning moment $\approx Pe \approx 878,000$

of these two unit stresses. $f_t \approx \text{lb}$, $r_1/r \approx 0.6$. $e/r \approx 0.29$. This gives $z/r \approx 2$. Therefore, the entire area of the

In Fig. 5.2.54 a load P acts in a line of symmetry at the distance e base is under compression. Area under compression ≈ 55.8 ft²; $I \approx 546$; $S \approx$

from cg; $r \approx$ radius of gyration. The unit stresses are (1) S_c , due to P , as $563,000/55.8 \approx (878,000 \approx 5.33)/546 \approx 18,700$ (max) and 1,500 (min) lb com-

if it acted through cg, and (2) S_b , due to the bending moment of P acting pression per ft². From Table 5.2.15, by interpolation, $S_{\max}/S_{\text{av}} \approx 1.85$. $\diamond S_{\max} \approx$

$(563,000/55.8) \approx 1.85 \approx 18,685$ lb/ft² (91,313 kgf/m²).

with a leverage of e about cg. Thus unit stress S at any point y is

$S \approx S_c \pm S_b$ The kern is the area around the center of gravity of a cross section

$\approx P/A \approx Pe_y/I$ within which any load applied will produce stress of only one sign

$\approx S_c$ ($1 \approx ey/r^2$) throughout the entire cross section. Outside the kern, a load produces stresses of different sign. Figure 5.2.57 shows kerns (shaded) for y is positive for points on the same side of cg as P , and negative on the various sections.

opposite side. For a rectangular cross section of width b , the maximum

For a circular ring, the radius of the kern $r \approx D[1 \leq (d/D)^2]/8$.

stress $S_M \approx S_c$ ($1 \leq 6e/b$). When P is outside the middle third of width b and is a compressive load, tensile stresses occur.

For a circular cross section of diameter d , $S_M \approx S_c$ ($1 \leq 8e/d$). The stress due to the weight of the solid will modify these relations.

NOTE. In these formulas e is measured from the gravity axis, and gives tension when e is greater than one-sixth the width (measured in the same direction as e), for rectangular sections; and when greater than one-eighth the diameter for solid circular sections.

If, as in certain classes of masonry construction, the material cannot withstand tensile stress and thus no tension can occur, the center of moments (Fig. 5.2.55) is taken at the center of stress. For a rectangular section, P acts at distance k from the nearest edge. Length under compression $\approx 3k$, and $S_M \approx 2/3P/(hk)$. For a circular section, $S_M \approx [0.372 \leq 0.056(k/r)]P/k \sqrt{rk}$, where $r \approx$ radius and $k \approx$ distance of P from circumference. For a circular ring, $S \approx$ average compressive stress on cross section produced by P ; $e \approx$ eccentricity of P ; $z \approx$ length of

diameter under compression (Fig. 5.2.56). Values of z/r and of the ratio of S_{max} to average S are given in Tables 5.2.14 and 5.2.15. Fig. 5.2.55 Fig. 5.2.56 Fig. 5.2.57

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Table 5.2.14 Values of the Ratio z/r (Fig. 5.2.56)

r_1	r	0.0	0.5	0.6	0.7	0.8	0.9	1.0
e/r	0.25	2.00	0.25					
	0.30	1.82	0.30					
	0.35	1.66	1.89	1.98	0.35			
	0.40	1.51	1.75	1.84	1.93	0.40		
	0.45	1.37	1.61	1.71	1.81	1.90	0.45	
	0.50	1.23	1.46	1.56	1.66	1.78	1.89	2.00
	0.55	1.10	1.29	1.39	1.50	1.62	1.74	1.87
	0.60	0.97	1.12	1.21	1.32	1.45	1.58	1.71
	0.65	0.84	0.94	1.02	1.13	1.25	1.40	1.54
	0.70	0.72	0.75	0.82	0.93	1.05	1.20	1.35
	0.75	0.59	0.60	0.64	0.72	0.85	0.99	1.15
	0.80	0.47	0.47	0.48	0.52	0.61	0.77	0.94
	0.85	0.35	0.35	0.35	0.36	0.42	0.55	0.72
	0.90	0.24	0.24	0.24	0.24	0.24	0.32	0.49
	0.95	0.12	0.12	0.12	0.12	0.12	0.12	0.25

Table 5.2.15 Values of the Ratio S_{max}/S_{avg}

(In determining S average, use load P divided by total area of cross section)

r_1	r	0.0	0.5	0.6	0.7	0.8	0.9	1.0
e/r	0.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	0.05	1.20	1.16	1.15	1.13	1.12	1.11	1.10
	0.10	1.40	1.32	1.29	1.27	1.24	1.22	1.20
	0.15	1.60	1.48	1.44	1.40	1.37	1.33	1.30
	0.20	1.80	1.64	1.59	1.54	1.49	1.44	1.40
	0.25	2.00	1.80	1.73	1.67	1.61	1.55	1.50
	0.30	2.23	1.96	1.88	1.81	1.73	1.66	1.60
	0.35	2.48	2.12	2.04	1.94	1.85	1.77	1.70
	0.40	2.76	2.29	2.20	2.07	1.98	1.88	1.80

0.45 3.11 2.51 2.39 2.23 2.10 1.99 1.90 0.45
 0.50 3.55 2.80 2.61 2.42 2.26 2.10 2.00 0.50
 0.55 4.15 3.14 2.89 2.67 2.42 2.26 2.17 0.55
 0.60 4.96 3.58 3.24 2.92 2.64 2.42 2.26 0.60
 0.65 6.00 4.34 3.80 3.30 2.92 2.64 2.42 0.65
 0.70 7.48 5.40 4.65 3.86 3.33 2.95 2.64 0.70
 0.75 9.93 7.26 5.97 4.81 3.93 3.33 2.89 0.75
 0.80 13.87 10.05 8.80 6.53 4.93 3.96 3.27 0.80
 0.85 21.08 15.55 13.32 10.43 7.16 4.50 3.77 0.85
 0.90 38.25 30.80 25.80 19.85 14.60 7.13 4.71 0.90
 0.95 96.10 72.20 62.20 50.20 34.60 19.80 6.72 0.95
 1.00 ∞ ∞ ∞ ∞ ∞ ∞ ∞ ∞ 1.00

For a hollow square (H and h = lengths of outer and inner sides), the CURVED BEAMS kern is a square similar to Fig. 5.2.57a, where

The application of the flexure formula for a straight beam to the case of

$r_{min} =$

$H/1$

$6\sqrt{2}$

$\frac{H}{12} \left(1 - \frac{h^2}{H^2} \right)$

$1 \leq$

h

H

2

$\frac{H}{12} \left(1 - \frac{h^2}{H^2} \right)$

$1 \leq$

h

H

2 a curved beam results in error. When all “fibers” of a member have the same center of curvature, the concentric or common type of curved beam exists (see Fig. 5.2.58). Such a beam is defined by the Winkler-Bach

For a hollow octagon R_o and R_i = radii of circles circumscribing theory. The stress at a point y units from the centroidal axis is

$\frac{M}{A R^2} \left(1 - \frac{y^2}{R^2} \right)$

the outer and inner sides; thickness of wall $t = 0.9239(R_o - R_i)$, the

kern is an octagon similar to Fig. 5.2.57c, where $0.2256R$ becomes M/y

$S \leq 1$

$0.2256R_o [1 \leq (R_i/R_o)^2]$. $A R^2 (R \leq y)$

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M is the bending moment, positive when it increases curvature; Y is may be applied. This force must then be eliminated by equating it to

positive when measured toward the convex side; A is the cross-sectional zero at the end.

area; R is the radius of the centroidal axis; Z is a cross-section property EXAMPLE. A quadrant of radius R is fixed at one end as shown in Fig.

defined by 5.2.59b. The force F is applied in the radial direction at the free end B. Find the

$Z \cong$

1

A

$\int y$

$R \leq y$

dA

deflection of B.

By moment area:

$y = R \sin \theta \times R(1 \cong \cos \theta)$

Analytical expressions for Z of certain sections are given in Table 5.2.16. $ds = R d\theta$ $M = FR \sin \theta$

Also Z can be found by graphical integration methods (see any advanced

$B \otimes x =$

FR^3

EI

$\int_0$

$\pi/2$

$\sin^2 d\theta =$

$\int FR^3$

$4 EI$

$B \otimes y =$

FR^3

EI

$\int \pi/2$

0

$\sin \theta (1 \cong \cos \theta) d\theta \cong$

FR^3

$2EI$

$\sqrt{1} \leq 4$

$FR^3 \pi^2$

and $B \otimes$

$2EI$

at $\theta \times \tan \cong 1 \cong$

FR^3

$2EI$

$4EI$

$\sigma_{FR} = 3$

$\sigma \tan \theta \geq 1$

2

$\theta = 32.5^\circ$

By Castigliano:

$B_{\theta x} =$

σU

σF

σF_x

$\sigma 0$

$\sigma/2 F R^2 \sin^2 \theta$

$2EI$

$\sin^2 \theta = d^2 \theta$

$\sigma_{FR} = 3$

$4EI$

Fig. 5.2.58

strength book). The neutral surface shifts toward the center of curvature,

$B_{\theta y} =$

σU

σF_y

σF_y

$\sigma 0$

$\sigma/2 [FR \sin \theta \geq F R (1 \geq \cos \theta)]^2 R d\theta$

y

$2EI$

or inside fiber, an amount equal to $e = ZR/(Z \leq 1)$. The Winkler-Bach FR^3

theory, though practically satisfactory, disregards radial stresses as well $2EI$

as lateral deformations and assumes pure bending. The maximum stress F_y , assumed downward, is equated to zero, after the integration and differen-

occurring on the inside fiber is $S = M h_i / (A e R_i)$, while that on the out- tiation are performed to find $B_{\theta y}$. The remainder of the computation is exactly as

side fiber is $S = M h_o / (A e R_o)$. in the moment-area method.

EXAMPLE. A split steel ring of rectangular cross section is subjected to a diametral force of 1,000 lb as shown in Fig. 5.2.59a. Compute the stress at the point 0.5 in from the outside fiber on plane mm. Also compute the maximum stress.

$R \leq C$

$Z \geq 1 \leq \ln$

$h R \geq C$

$10 \leq 2$

$\geq 1 \leq \ln = 0.0133$

$4 \leq 2$ Fig. 5.2.59

S1.5

M

AR

$\frac{1}{R} \leq \frac{1}{y}$

y

Z ($R \leq y$)

$\frac{1}{R} \leq \frac{1}{y}$

F

A Eccentrically Curved Beams These beams (Fig. 5.2.60) are

$\frac{1}{R} \geq \frac{1}{1,000}$ $\frac{1}{10}$

$\frac{1}{R} \geq \frac{1}{8}$ $\frac{1}{10}$

$\frac{1}{R} \leq \frac{1}{1}$

1.5

$0.0133(10 \leq 1.5)$

$\frac{1}{R} \leq \frac{1}{1}$

1,000

8

bounded by arcs having different centers of curvature. In addition, it is possible for either radius to be the larger one. The one in which the section depth shortens as the central section is approached may be

$\frac{1}{R} \geq \frac{1}{1,250} \leq \frac{1}{125} \frac{1}{R} \geq \frac{1}{1,125}$ lb/in² (compr.)(79 kgf/cm²)

$\frac{1}{R} \leq \frac{1}{1}$

called the arch beam. When the central section is the largest, the beam is

$\frac{1}{R} \geq \frac{1}{1,000} \geq \frac{1}{2,000}$ of the crescent type. Crescent I denotes the beam of larger outside radius

SM $\frac{1}{R} \leq \frac{1}{1}$

$\frac{1}{R} \geq \frac{1}{0.0133}$ ($10 \geq 2$) 8 and crescent II of larger inside radius. The stress at the central section of

$\frac{1}{R} \geq \frac{1}{2,230} \leq \frac{1}{125} \frac{1}{R} \geq \frac{1}{2,355}$ lb/in² (166 kgf/cm²) such beams may be found from $S = \frac{KM}{I}$. In the case of rectangular

ZR $\frac{1}{R} \geq \frac{1}{0.0133}$ $\frac{1}{10}$ cross section, the equation becomes $S = \frac{6KM}{(bh)^2}$ where M is the

or $e = \frac{1}{R} \leq \frac{1}{0.131}$ bending moment, b is the width of the beam section, and h its height.

$Z \leq 1$ $\frac{1}{R} \geq \frac{1}{0.0133} \leq 1$

The stress factors K for the inner boundary, established from photoelastic

Mh i F $\frac{1}{R} \geq \frac{1}{1,000} \frac{1}{10} \frac{1}{R} \geq \frac{1}{1.87}$ $\frac{1}{1,000}$

and SM $\frac{1}{R} \leq \frac{1}{1} \leq \frac{1}{1}$ data, are given in Table 5.2.17. The outside radius is denoted by R o and

AeR i A $\frac{1}{R} \geq \frac{1}{0.131} \frac{1}{8}$ 8 the inside by R i . The geometry of crescent beams is such that the stress $\frac{1}{R} \geq \frac{1}{2,355}$ lb/in² (166 kgf/cm²) can be larger in off-center sections. The stress at the central section deter-

The deflection in curved beams can be computed by means of the mined above must then be multiplied by the position factor k, given in

moment-area theory. If the origin of axes is taken at the point whose Table 5.2.18. As in the concentric beam, the neutral surface shifts

deflection is wanted, it can be shown that the component displacements slightly toward the inner boundary (see Vidosic, Curved Beams with

in the x and y directions are Eccentric Boundaries, Trans. ASME, 79, pp., 1317 – 1321).

$$\phi_x \approx 0$$

$$\int_M y \, ds$$

$$EI$$

$$\text{and } \phi_y \approx 0$$

$$\int_M x \, ds$$

$$EI$$

The resultant deflection is then equal to $\phi_0 \approx \sqrt{\phi_x^2 + \phi_y^2}$ in the direction defined by $\tan \theta = \phi_y / \phi_x$. Deflections can also be found conveniently

by use of Castigliano's theorem. It states that in an elastic system the displacement in the direction of a force (or couple) and due to that force (or couple) is the partial derivative of the strain energy with respect to

the force (or couple). Stated mathematically, $\phi_z = \partial U / \partial F_z$. If a force does not exist at the point and/or in the direction desired, a dummy force Fig. 5.2.60

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Table 5.2.16 Analytical Expressions for Z

Section Expression

$$R \leq C$$

$$Z \geq 1 \leq \ln$$

$$h \geq C$$

$$Z \geq 1 \leq 2 \ln \sqrt{\frac{R}{h}}$$

$$R$$

$$r$$

$$R$$

$$r$$

$$\geq R$$

$$r$$

$$2$$

$$\geq 1$$

$$Z \geq 1 \leq$$

$$R$$

$$A$$

$$\frac{1}{t} \ln(R \leq C) \leq (b \geq t) \ln(R \geq C) \geq b \ln(R \geq C) \frac{1}{t}$$

$$A \frac{1}{t} C \geq (b \geq t) C \leq b C \frac{1}{2}$$

$$Z \geq 1 \leq$$

R

A

 $\square b \ln$ $R \leq C2$ $R \geq C2$ $\leq (t \geq b) \ln$ $R \leq C1$ $R \geq C1$ $\square$ $A \square 2[(t \geq b)C1 \leq bC2]$

Table 5.2.17 Stress Factors for Inner Boundary at Central Section Table 5.2.18 Crescent-Beam Position Stress Factors

(See Fig. 5.2.60) (See Fig. 5.2.60)

1. For the arch-type beams Angle k

$h R_o \leq R_i \square$,

(a) $K \square 0.834 \leq 1.504$ if $\downarrow 5$. deg Inner Outer

$R_o \leq R_i h$

$h R \leq R_i 10 1 \leq 0.055H/h 1 \leq 0.03H/h$

(b) $K \square 0.899 \leq 1.181$ if $5 \downarrow o \downarrow 10$. $20 1 \leq 0.164H/h 1 \leq 0.10H/h$

$R_o \leq R_i h 30 1 \leq 0.365H/h 1 \leq 0.25H/h$

(c) In the case of larger section ratios use the equivalent beam solution. $40 1 \leq 0.567H/h 1 \leq 0.467H/h$

2. For the crescent I-type beams $(0.5171 \geq 1.382H/h)/12$

$50 1.521 \geq 1 \leq 0.733H/h$

1.382

$h R \leq R_i$

(a) $K \square 0.570 \leq 1.536$ if $o \downarrow 2$. $(0.2416 \geq 0.6506H/h)/12$

$R_o \leq R_i h 60 1.756 \geq 1 \leq 1.123H/h$

0.6506

$h R_o \leq R_i$

(b) $K \square 0.959 \leq 0.769$ if $2 \downarrow \downarrow 20$. $(0.4817 \geq 1.298H/h)/12$

$R_o \leq R_i h 70 2.070 \geq 1 \leq 1.70H/h$

$\square$

0.6492

(c) $K \square 1.092$

h

$R_o \leq R_i$

$\square 0.0298$

if

$R_o \leq R_i$

h

$$\rightarrow 20.80 \cdot 2.531 \geq$$

$$(0.2939 \geq 0.7084H/h) /$$

$$0.3542$$

$$12$$

$$1 \leq 2.383H/h$$

3. For the crescent II-type beams $90 \leq 3.933H/h$

$h \leq R_o \leq R_i$ NOTE: All formulas are valid for $0 \leq H/h \leq 0.325$. Formulas for the inner bound-

(a) $K \leq 0.897 \leq 1.098$ if $\theta \leq 8^\circ$. ary, except for 40° deg, may be used to $H/h \leq 0.36$. H = distance between centers.

$$R_o \leq R_i \leq h$$

$$(b) K \leq 1.119 \leq h$$

$$R_o \leq R_i$$

$$\leq 0.0378$$

$$\text{if } \theta \leq 8^\circ$$

$$R_o \leq R_i$$

$$h$$

$\theta \leq 20^\circ$. IMPACT

$$(c) K \leq 1.081 \leq h$$

$$R_o \leq R_i$$

$$\leq 0.0270$$

$$\text{if}$$

$$R_o \leq R_i$$

$$h$$

$$\rightarrow 20.$$

A force or stress is considered suddenly applied when the duration of load application is less than one-half the fundamental natural period of vibration of the member upon which the force acts. Under impact, a

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compression wave propagates through the member at a velocity c on another. Forces due to gravity, inertia, magnetism, etc., which act

$\sqrt{E/\rho}$, where ρ is the mass density. As this compression wave travels over the entire volume of a body, are called body forces. Both surface

back and forth by reflection from one end of the bar to the other, a and body forces can be best handled if resolved into three orthogonal

maximum stress is produced which is many times larger than what it components. Surface forces are thus designated X, Y, and Z, while body

would be statically. An exact determination of this stress is most difficult- forces are labeled X, Y, and Z.

cult. However, if conservation of kinetic and strain energies is applied, In general, there exists a normal stress σ and a shearing stress τ at

the impact stress is found to be each point of a loaded member. It is convenient to deal with components

$$\sqrt{\sigma^2 + \tau^2}$$

of each of these stresses on each of six orthogonal planes that bound the

W 3W point element. Thus there are at each point six stress components, σ_x , σ_y ,

$$S_{\phi} = S$$

$W_b \leq W \leq W_b \sigma_z$, $\sigma_{yx} = \sigma_{xy}$, $\sigma_{xz} = \sigma_{zx}$, and $\sigma_{yz} = \sigma_{zy}$. Similarly, if the normal unit strain

The weight of the striking mass is here denoted by W, that of the struck is designated by the letter $\bar{W}$ and shearing unit strain by γ , the six compo-

bar by W_b , while S is the static stress, W/A (A is the cross-sectional area nents of strain are defined by

of the bar). Above is the case of sudden impact. When the ratio W/W_b is $\sigma_x = \sigma_u/\sigma_x$, $\sigma_y = \sigma_v/\sigma_y$, $\sigma_z = \sigma_w/\sigma_z$

small, the stress computed by the above equation may be erroneous. A $\sigma_{xy} = \sigma_u/\sigma_y \leq \sigma_v/\sigma_x$, $\sigma_{yz} = \sigma_v/\sigma_z \leq \sigma_w/\sigma_y$

better solution of this problem may result from and $\sigma_{xz} = \sigma_u/\sigma_z \leq \sigma_w/\sigma_x$

W 2 The elastic displacements of particles on the body in the x, y, and z

$$S_{\phi} = S \leq S \leq$$

W_b 3 directions are identified as the u, v, and w components, respectively.

Since metals have the usually assumed elastic as well as isotropic

If a weight W falls a distance h before striking a bar of mass W_b, properties, Hooke's law holds.

Therefore, the interrelationships be-

energy conservation will yield the relation tween stress and strain can easily be obtained.

$$S_{\phi} = S \sqrt{1 + \frac{2h}{e}}$$

$$1 \leq 1 \leq$$

$$2h$$

$$e$$

$$3W$$

$$3W \leq W_b$$

$$\sigma_x =$$

$$1$$

$$E_x$$

$$[\sigma \geq \sigma_y \leq \sigma_z]$$

The elongation $e = \Delta l = SI/E$. When the striking mass W is assumed 1

rigid, the elasticity factor is taken equal to 1. Thus the equation becomes $\sigma_y = [\sigma_y \geq \sigma_x \leq \sigma_z]$

$$E$$

$$S_{\phi} = S (1 \leq \sqrt{1 + 2h/e})$$

$$\sigma_z = [\sigma_z \geq \sigma_x \leq \sigma_y]$$

If, in addition, h is taken equal to zero (sudden impact), the radical equals E

1, and so the stress becomes $S_{\phi} = 2S$. Since Hooke's law is applicable, $\sigma_{xy} = \sigma_{xy}/G$, $\sigma_{xz} = \sigma_{xz}/G$

the relations and $\sigma_{yz} = \sigma_{yz} / G$

$e \leq e$ ($1 \leq \sqrt{1} \leq 2h/e$) and $e \leq 2e$ The general case of strain can be obtained by superposing the elonga-

are also true for the same conditions. tion strains upon the shearing strains.

The expression may be converted, by using $v^2 = 2gh$, to Problems depending upon theories of elasticity are considerably sim-

plified if the stresses are all parallel to one plane or if all deformations

$S \leq S$ [$1 \leq \sqrt{1} \leq v^2/(eg)$] occur in planes perpendicular to the length of the member. The first case This might be called the energy impact form. If the natural frequency f_n of is one of plane stress, as when a thin plate of uniform thickness is sub-

the bar is used, the stress equation is jected to central, boundary forces parallel to the plane of the plate. The

second is a case of plane strain, such as a gate subjected to hydrostatic

$S \leq S$ ($1 \leq \sqrt{1} \leq 0.204hf^2/n$) pressure, the intensity of which does not vary along the gate's length.

In general, the maximum impact stress in a beam and a shaft can be All particles therefore displace at right angles to the length, and so cross

approximated from the simplified falling-weight equation. It is neces- sections remain plane. In plane-stress problems, three of the six stress

sary, though, to substitute the maximum deflection y for e , in the case of components vanish, thus leaving only σ_x , σ_y , and σ_{xy} . Similarly, in

beams, and for the angle of twist ϕ in the case of shafts. Of course $S \leq$ plane strain, only σ_x , σ_y , and σ_{xy} will not equal zero; thus the same three

M_c/I and $M_t c/J$, respectively. Thus stresses σ_x , σ_y , and σ_{xy} remain to be considered. Plane problems can

$\sigma \sqrt{\sigma}$

thus be represented by the element shown in Fig. 5.2.61. Equilibrium

$2h$

$S \leq S$ $1 \leq 1$ considerations applied to this particle result in the differential equations

y (or ϕ) of equilibrium which reduce to

For a more exact solution, elastic yield in each member must be consid- $\sigma \sigma_x \sigma$

ered. The theory then yields $\sigma_{xy} \leq X \leq 0$

$\sigma_x \sigma_y$

$S \leq S$ $\sigma \sqrt{1} \leq 1$

$2h$

y

$\sigma 35W$

$35W \leq 17Wb$

σ and

$\sigma \sigma_y$

σ_y

$\sigma \sigma_{xy}$

σ_x

$\leq Y \leq 0$

for a simply supported beam struck in the middle by a weight W . Since the two differential equations of equilibrium are insufficient to

find the three stresses, a third equation must be used. This is the compatibility equation relating the three strain components. It is

THEORY OF ELASTICITY

$$\epsilon_x^2 + \epsilon_y^2 + \epsilon_{xy}^2$$

Loaded members in which the stress distribution cannot be estimated $\leq$

$$\epsilon_y^2 + \epsilon_x^2 + \epsilon_{xy}^2$$

fail of solution by elementary strength-of-material methods. To such

cases, the more advanced mathematical principles of the theory of elas- If strains are expressed in terms of the stresses, the compatibility equa-

ticity must be applied. When this is not possible, experimental stress tion becomes

analysis has to be used. Because of the complexity of solution, only

some of the more practical problems have been solved by the theory of

elasticity. The more general concepts and methods are presented.

$$\epsilon_x^2$$

$$\epsilon_x^2$$

$$\epsilon_x^2$$

$$\leq 2$$

$$\epsilon_y$$

$$\epsilon (\epsilon_x \leq \epsilon_y) \leq 0$$

Two kinds of forces may act on a body. Surface forces are distributed Now, in any two-dimensional problem, the compatibility equation along

over the surface as the result of, for instance, the pressure of one body with the differential equilibrium equations must be simultaneously

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solved for the three unknown stresses. This is accomplished using stress CYLINDERS AND SPHERES functions, which permit the integration and satisfy boundary conditions

A thin-wall cylinder has a wall thickness such that the assumption of in each particular situation.

constant stress across the wall results in negligible error. Cylinders having internal-diameter-to-thickness (D/t) ratios greater than 10 are usually considered thin-walled. Boilers, drums, tanks, and pipes are often treated as such. Equilibrium equations reveal the circumferential, or hoop, stress to be $S = pr/t$ under an internal pressure p (see Fig. 5.2.62). If the cylinder is closed at the ends, a longitudinal stress of $pr/(2t)$ is developed. The tensile stress developed in a thin hollow sphere subjected to internal pressure is also $pr/(2t)$.

Fig. 5.2.61

In three-dimensional problems, the third dimension must be considered. This results in three differential equations of equilibrium, as well as three compatibility equations. The six stress components can thus be found. The complexity involved in the solution of these equations is such, however, that only a few special cases have been solved.

In certain problems, such as rotating circular disks, polar coordinates become more convenient. In such cases, the stress components in a two-dimensional field are the radial stress σ_r , the tangential stress σ_θ , and the shearing stress $\tau_{r\theta}$. In terms of these stresses the polar differential equations become are subjected to external pressure, collapse becomes the mode of failure.

The shell is assumed perfectly round and of uniform thickness, the material obeys Hooke's law, the radial stress is negligible, and the normal stress distribution is linear. Other, lesser assumptions are also made. Using the theory of elasticity, R. G. Sturm (Univ. Ill. Eng. Exp. Stn. Bull., no 12, Nov. 11, 1941) derived the collapsing pressure as

and the radial stress distribution is linear. Other, lesser assumptions are also made. Using the theory of elasticity, R. G. Sturm (Univ. Ill. Eng. Exp. Stn. Bull., no 12, Nov. 11, 1941) derived the collapsing pressure as

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The body force per unit volume is represented by R . The compatibility equation in polar coordinates is

The body force per unit volume is represented by R . The compatibility equation in polar coordinates is

The body force per unit volume is represented by R . The compatibility equation in polar coordinates is

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The body force per unit volume is represented by R . The compatibility equation in polar coordinates is

The body force per unit volume is represented by R . The compatibility equation in polar coordinates is

w

2l

$\frac{1}{3} 2y^3$

3

$2c^2 y$

5

$\frac{1}{15}$

The origin of coordinates is at the center of the beam, 2c is the beam depth, and 2l is the span length. Thus the maximum stress at $x = 0$ and $y = c$ is $\sigma_x = \frac{wl}{2c^2} \frac{wc^3}{15l}$.

The first term represents the stress as

$\frac{wl}{15l}$

obtained by the elementary flexure theory; the second is a correction. The second term becomes negligible when c is small compared to l.

The important case of a flat plate of unit width with a circular hole of diameter 2a at its center, subjected to a uniform tensile load, has been solved using polar coordinates. If S is the uniform stress at some distance from the hole, r is measured from the center of the hole, and θ is the angle of r with respect to the longitudinal axis of the member, the stresses are

$\sigma_r =$

S

2

$\left(1 - \frac{a^2}{r^2}\right)$

$1 - \frac{a^2}{r^2}$

r^2

S

2

$1 - \frac{a^2}{r^2}$

$3a^4$

r^4

$4a^2$

≥ 2

r

$\cos 2\theta$

$\left(1 - \frac{a^2}{r^2}\right)$

Fig. 5.2.63 Radial external pressure with simply supported edges.

$S a^2 S 3a^4$

$\left(1 - \frac{a^2}{r^2}\right) \geq 1 - \frac{a^2}{r^2} \cos 2\theta$ When the cylinder is stiffened with rings, the shell may be assumed to

$2r^2 r$

□ □

be divided into a series of shorter shells, equal in length to the ring

$S = 3a/4$ spacing. The previous equation can then be applied to a ring-to-ring

$$\frac{r_2}{r_1} \geq 1 \geq \frac{1}{\sin 2\alpha}$$

$2/r_4 - r_2$ length of cylinder. However, the flexural rigidity of the combined

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stiffener and shell. The necessary to withstand the pressure is $E t_c$ shown in Fig. 5.2.67b and the equation is integrated, the general tangen-

$W_s D^3 L_s / 24$. W_s is the pressure, L_s the length between rings, and I_c the tial and radial stress relations, called the Lamé equations, are derived.

combined moment of inertia of the ring and that portion of the shell

$$\frac{r_2}{r_1} \frac{p_1}{p_2} \geq \frac{r_2}{r_1} \frac{p_2}{p_1} \leq \left(\frac{p_1}{p_2} \geq \frac{p_2}{p_1} \right) \frac{r_2}{r_1} \frac{r_2}{r_1} / r^2$$

assumed acting with the ring. S_t □

$$\frac{r_2}{r_1} \geq \frac{r_2}{r_1}$$

$$\frac{r_2}{r_1} \frac{p_1}{p_2} \geq \frac{r_2}{r_1} \frac{p_2}{p_1} \geq \left(\frac{p_1}{p_2} \geq \frac{p_2}{p_1} \right) \frac{r_2}{r_1} \frac{r_2}{r_1} / r^2$$

and S_r □

$$\frac{r_2}{r_1} \geq \frac{r_2}{r_1}$$

When the external pressure $p_2 = 0$, the equations reduce to

S_t □

$$\frac{r_2}{r_1} \frac{p_1}{p_2}$$

$$\frac{r_2}{r_1} \geq \frac{r_2}{r_1}$$

$$\frac{1}{r_1} \leq$$

$$\frac{r_2}{r_1}$$

$$\frac{r_2}{r_1}$$

$$\frac{1}{r_1}$$

and

$$\frac{r_2}{r_1} p$$

$$S_r = \frac{1}{2} \frac{1}{r_1} \frac{1}{r_2}$$

$$\frac{r_2}{r_1} \geq \frac{r_1}{r_2}$$

$$\frac{1}{r_2}$$

$$\frac{1}{r_1} \geq \frac{1}{r_2}$$

$$\frac{r_2}{r_1}$$

$$\frac{1}{r_1}$$

Fig. 5.2.64 Radial external pressure with fixed edges.

In some instances, cylinders collapse only after a stress in excess of the elastic limit has been reached; that is, plastic range stresses are present. In such cases the same equation applies, but the modulus of elasticity must be modified.

When the average stress S_a is less than the proportional limit S_p , and

the maximum stress (direct, plus bending) is S , the modified modulus

$$E \leq E \leq 1$$

1

4

$$S \geq S_l$$

$$S_u \geq S_a$$

$$S \leq S_l$$

2

Fig. 5.2.66 Radial and end external pressure with fixed edges.

S_u is the modulus of rupture. When the average stress is larger than the At the inner boundary the tangential elongation ϵ_t is equal to

proportional limit, the modified modulus is taken as the tangent at the average stress. $\epsilon_t = (S_t \geq S_r) / E$

The increase in the bore radius δr_1 resulting therefrom is

$$\delta r_1 =$$

$$r_1 p_1$$

$$E_h$$

$$1 \leq r_{21} / r_{22}$$

$$1 \geq r_{21} / r_{22}$$

$$\leq 1$$

Similarly a solid shaft of r radius under external pressure p_2 will have its radius decreased by the amount

$$r p_2$$

$$\delta r = (1 \geq 1)$$

$$E_s$$

In the case of a press or shrink fit, $p_1 = p_2 = p$. The sum of δr_1 and δr_2 absolute is the radial interference; twice this sum is the diametral interference δ or

$$\delta = 2 r_1 p \leq 1$$

$$E_h$$

$$1 \leq r_{21} / r_{22}$$

$$1 \geq r_{21} / r_{22}$$

$$\leq 1 \leq$$

$$1 \geq 1$$

$$E_s$$

$$1$$

If the hub and shaft materials are the same, $E_h = E_s = E$, and

$$4 r_1 r_{22} p$$

$$E (r_{22} \geq r_{21})$$

If the equation is solved for p and this value is substituted in Lamé's

Fig. 5.2.65 Radial and end external pressure with simply supported edges. equation, the maximum tangential stress on the inner surface of the hub

is found to be

In thick-walled cylinders (Fig. 5.2.67a) the circumferential, hoop, or $E\sigma$

tangential stress S_t is not uniform. In addition a radial stress S_r is present. $S_t \propto (r_2^2 \leq r \leq r_1)$

When equilibrium is applied to the annulus taken out of Fig. 5.2.67a and $4r_1 r \leq 2$

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FLAT PLATES 5-47

PRESSURE BETWEEN BODIES

WITH CURVED SURFACES

(See Hertz, "Gesammelte Werke," vol. 1, pp. 159 et seq., Barth.)

Two Spheres The radius A of the compressed area is obtained from the formula $A^3 \propto 0.68P(c_1 \leq c_2)/(1/r_1 \leq 1/r_2)$, in which P is the compressing force, c_1 and c_2 ($\propto 1/E_1$ and $1/E_2$) are reciprocals of the respective moduli of elasticity, and r_1 and r_2 are the radii. (Reciprocal of Poisson's ratio is assumed to be $n \propto 10/3$.) The greatest contact pressure in the middle of the compressed surface will be $S_{\max} \propto 1.5(P/A^2)$, and $S_{3\max} \propto 0.235P(1/r_1 \leq 1/r_2)^2/(c_1 \leq c_2)^2$

The total deformation of the two spheres will be Y , which is obtained from

$Y^3 \propto 0.46P^2(c_1 \leq c_2)^2(1/r_1 \leq 1/r_2)$

For $c_1 \propto c_2 \propto 1/E$, i.e., two spheres with the same modulus of elasticity, it follows that $A^3 \propto 1.36 P/E(1/r_1 \leq 1/r_2)$, $S_{3\max} \propto 0.059PE^2(1/r_1 \leq 1/r_2)^2$, and $Y^3 \propto 1.84P^2(1/r_1 \leq 1/r_2)/E^2$. If the radii of these spheres are Fig. 5.2.67 also equal, $A^3 \propto 0.68Pr/E \propto 0.34Pd/E$; $S_{3\max} \propto 0.235PE^2/r^2 \propto 0.94PE^2/d^2$; and $Y^3 \propto 3.68P^2/(E^2r) \propto 7.36P^2/(E^2d)$.

EXAMPLE. The barrel of a field gun has an outside diameter of 9 in and a bore

Sphere and Flat Plate In this case $r_1 \propto r$ and $r_2 \propto \infty$, and the above

of 4.7 in. An internal pressure of 16,000 lb/in² is developed during firing. What

maximum stress occurs in the barrel? An investigation of Lamé's equations for formulas become $A^3 \propto 0.68Pr(c_1 \leq c_2) \propto 1.36Pr/E$, and

internal pressure reveals the maximum stress to be the tangential one on the inner $S_{3\max} \propto 0.235P/[r^2(c_1 \leq c_2)^2] \propto 0.059PE^2/r^2$

surface. Thus, $Y^3 \propto 0.46P^2(c_1 \leq c_2)^2/r \propto 1.84P^2/(E^2r)$

$r_1 \leq r \leq r_2$ 16,000 (2.352 $\leq$ 4.52)

$S_t \propto p_1$ Two Cylinders The width b of the rectangular pressure surface is

$r_2 \geq r_1$ 4.52 $\geq$ 2.352

obtained from $(b/4)^2 \propto 0.29P(c_1 \leq c_2)/l[(1/r_1) \leq (1/r_2)]$, where r_1 and $\propto$ 28,000 lb/in² (1,972 kgf/cm²) r_2 are the radii, and l the length

Oval Hollow Cylinders In Fig. 5.2.68, let a and b be the semiminor

$$S_{2\max} = [4P/(bI)]^2 = 0.35P(1/r_1 \leq 1/r_2)/(c_1 \leq c_2)$$

and semimajor axes. The bending moments at A and C will then be For cylinders with the same moduli of elasticity, $c_1 = c_2 = 1/E$, and

$$M_0 = pa^2/2 \geq p l x / (2 S) \geq p l y / (2 S) \quad (b/4)^2 = 0.58P/EI[(1/r_1) \leq (1/r_2)]; \text{ and } S_{2\max} = 0.175PE(1/r_1 \leq 1/r_2)/l.$$

$$M_1 = M_0 \geq p(a^2 \geq b^2)/2 \text{ When } r_1 = r_2 = r, (b/4)^2 = 0.29Pr/(EI), \text{ and } S_{2\max} = 0.35PE/(lr).$$

Cylinder and Flat Plate Here $r_1 = r$, $r_2 = \infty$, and the above formulas

where I_x and I_y are the moments of inertia of the arc AC about the x and y reduce to $(b/4)^2 = 0.29Pr(c_1 \leq c_2)/l = 0.58Pr/(EI)$, and

axes, respectively. The bending moment at any point will be

$$S_{2\max} = 0.35P/[lr(c_1 \leq c_2)] = 0.175PE/(lr)$$

$$M = M_0 \geq pa^2/2 \leq p x^2/2 \leq p y^2/2$$

For application to ball and roller bearings and to gear teeth, see Sec. 8.

Thick Hollow Spheres With an internal pressure p , where $p \uparrow$

$T/0.65$.

$$r_2 = r_1 [(T \leq 0.4p)/(T \geq 0.65p)]^{1/3} \text{ FLAT PLATES}$$

The maximum tensile stress is on the inner surface, in the direction of The analysis of flat plates subjected to lateral loads is very involved

the circumference. With an external pressure p , where $p \uparrow T/1.05$, because plates bend in all vertical planes. Strict mathematical deriva-

tions have therefore been accomplished only in some special cases.

$$r_2 = r_1 [T/(T \geq 1.05p)]^{1/3} \text{ Most of the available formulas contain some amount of rational empiri-}$$

In both cases T is the true stress. cism. Plates may be classified as (1) thick plates, in which transverse

shear is important; (2) average-thickness plates, in which flexure stress predominates; (3) thin plates, which depend in part upon direct tension; and (4) membranes, which are subject to direct tension only. However, exact lines of demarcation do not exist.

The flat-plate formulas given apply primarily to symmetrically loaded average-thickness plates of constant thickness. They are valid only if the maximum deflection is small relative to the plate thickness; usually, $y \leq 0.4t$. In the mathematical analyses, allowance for stress redistribution, because of slight local yielding, is usually not made. Since this yielding, especially in ductile materials, is beneficial, the formulas generally err on the side of safety. Certain cases of symmetrically loaded circular and rectangular plates are presented in Figs. 5.2.69 and 5.2.70.

The maximum stresses are calculated from

$$w R^2 P C$$

$$SM = k SM = k \text{ or } SM = k$$

$$t_2 t_2 t_2$$

The first equation is for a uniformly distributed load w , lb/in²; the second supports a concentrated load P , lb; and the third a couple C , per

unit length, uniformly distributed along the edge. Combinations of these

Fig. 5.2.68 loadings may be treated by superposition. The factors k and k_1 are given

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Table 5.2.19 Coefficients k and k_1 for Circular Plates

($\nu = 0.3$)

Case k k_1

1 1.24 0.696

2 0.75 0.171

3 6.0 4.2

R/r

1.25 1.5 2 3 4 5

Case k k_1 k k_1 k k_1 k k_1 k k_1 k k_1

4 0.592 0.184 0.976 0.414 1.440 0.664 1.880 0.824 2.08 0.830 2.19 0.813

5 0.105 0.0025 0.259 0.0129 0.481 0.057 0.654 0.130 0.708 0.163 0.730 0.176

6 1.10 0.341 1.26 0.519 1.48 0.672 1.88 0.734 2.17 0.724 2.34 0.704

7 0.195 0.0036 0.320 0.024 0.455 0.081 0.670 0.171 1.00 0.218 1.30 0.238

8 0.660 0.202 1.19 0.491 2.04 0.902 3.34 1.220 4.30 1.300 5.10 1.310

9 0.135 0.0023 0.410 0.0183 1.04 0.0938 2.15 0.293 2.99 0.448 3.69 0.564

10 0.122 0.00343 0.336 0.0313 0.740 0.1250 1.21 0.291 1.45 0.417 1.59 0.492

11 0.072 0.00068 0.1825 0.005 0.361 0.023 0.546 0.064 0.627 0.092 0.668 0.112

12 6.865 0.2323 7.448 0.6613 8.136 1.493 8.71 2.555 8.930 3.105 9.036 3.418

13 6.0 0.196 6.0 0.485 6.0 0.847 6.0 0.940 6.0 0.801 6.0 0.658

14 0.115 0.00129 0.220 0.0064 0.405 0.0237 0.703 0.062 0.933 0.092 1.13 0.114

15 0.090 0.00077 0.273 0.0062 0.710 0.0329 1.54 0.110 2.23 0.179 2.80 0.234

in Tables 5.2.19 and 5.2.20; R is the radius of circular plates or one side of rectangular plates, and t is the plate thickness.

[In Figs. 5.2.69 and 5.2.70, $r = R$ for circular plates and $r =$ smaller side rectangular plates.]

The maximum deflection for the same cases is given by

$w_R = \frac{PR^2}{4CR^2}$

$y_M = \frac{k_1}{k} y_M = \frac{k_1}{k} y_M$ and $y_M = \frac{k_1}{k}$

$E t^3$ $E t^3$ $E t^3$

The factors k_1 are also given in the tables. For additional information, including shells, refer to ASME Handbook, "Metals Engineering: Design," McGraw-Hill.

Fig. 5.2.70 Rectangular and elliptical plates. [R is the longer dimension except in cases (21) and (23).]

in the proposal of various theories of failure. Figure 5.2.71 shows the

principal stresses on a triaxially stressed element. It is assumed, for simplicity, that $S_1 \rightarrow S_2 \rightarrow S_3$. Compressive stresses are negative.

Fig. 5.2.69 Circular plates. Cases (4), (5), (6), (7), (8), and (13) have central 1. Maximum-stress theory (Rankine) assumes failure occurs when the

hole of radius r ; cases (9), (10), (11), (12), (14), and (15) have a central piston of largest principal stress reaches the yield stress in a tension (or compres-

radius r to which the plate is fixed. sion) specimen. That is, $S_1 \leq S_y$.

2. Maximum-shear theory (Coulomb) assumes yielding (failure)

occurs when the maximum shearing stress equals that in a simple ten-

THEORIES OF FAILURE

sion (or compression) specimen at yield. Mathematically, $S_1 \geq S_3 \leq S_y$.

Material properties are usually determined from tests in which speci- $\leq S_y$.

mens are subjected to simple stresses under static or fluctuating loads. 3. Maximum-strain-energy theory (Beltrami) assumes failure occurs

The attempt to apply these data to bi- or triaxial stress fields has resulted when the energy absorbed per unit volume equals the strain energy per

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PLASTICITY 5-49

Table 5.2.20 Coefficients k and k_1 for Rectangular and Elliptical Plates

($\nu = 0.3$)

R/r

1.0 1.5 2.0 3.0 4.0

Case k k_1 k k_1 k k_1 k k_1 k k_1

16 0.287 0.0443 0.487 0.0843 0.610 0.1106 0.713 0.1336 0.741 0.1400

17 0.308 0.0138 0.454 0.0240 0.497 0.0277 0.500 0.028 0.500 0.028

18 0.672 0.140 0.768 0.160 0.792 0.165 0.798 0.166 0.800 0.166

19 0.500 0.030 0.670 0.070 0.730 0.101 0.750 0.132 0.750 0.139

20 0.418 0.0209 0.626 0.0582 0.715 0.0987 0.750 0.1276 0.750

21* 0.418 0.0216 0.490 0.0270 0.497 0.0284 0.500 0.0284 0.500 0.0284

22 0.160 0.0221 0.260 0.0421 0.320 0.0553 0.370 0.0668 0.380 0.0700

23* 0.160 0.0220 0.260 0.0436 0.340 0.0592 0.430 0.0772 0.490 0.0908

24 1.24 0.70 1.92 1.26 2.26 1.58 2.60 1.88 2.78 2.02

25 0.75 0.171 1.34 0.304 1.63 0.379 1.84 0.419 1.90 0.431

* Length ratio is r/R in cases 21 and 23.

unit volume in a tension (or compression) specimen at yield. Mathemat- above holds for fluctuating stresses, provided that principal stresses at the

ically, $S_1 \leq S_2 \leq S_3 \geq 2\sqrt{(S_1 S_2 \leq S_2 S_3 \leq S_3 S_1)} \leq S_y$. maximum load are used and the endurance strength in simple bending is

4. Maximum-distortion-energy theory (Huber, von Mises, Hencky) as- substituted for the yield strength.

sumes yielding occurs when the distortion energy equals that in simple

EXAMPLE. A steel shaft, 4 in in diameter, is subjected to a bending moment tension at yield. The distortion energy, that portion of the total energy of 120,000 in $\square$ lb, as well as a torque. If the yield strength in tension is 40,000 which causes distortion rather than volume change, is lb/in², what maximum torque can be applied under the (1) maximum-shear theory

1 $\leq \square$ 2 and (2) the distortion-energy theory?

$U_d \square (S_1 \leq S_2 \leq S_3 \geq S_1 \ S_2 \geq S_2 \ S_3 \geq S_3 \ S_1)$

$3E \ Mc \ 120,000 \ \square^2$

$S_x \ \square \ \square$

$I \ 12.55$

Thus failure is defined by

$TC \ T \square^2$

$S_1 \leq S_2 \leq S_3 \geq (S_1 \ S_2 \leq S_2 \ S_3 \leq S_3 \ S_1) \ \square \ S_y \ \square \ 19,100 \text{ lb/in}^2 \ S_{xy} \ \square \ \square \ 0.0798T$

$J \ 25.1$

$\sqrt{\square} \ \square$

5. Maximum-strain theory (Saint-Venant) claims failure occurs when $S_x \ S_x^2$

the maximum strain equals the strain in simple tension at yield or $S_1 \geq$ and $S_{M,m} \ \square \ \square \leq S_{xy}$

2

$2 \ 2$

$\sqrt{\square} \ \square$

$\square(S_2 \leq S_3) \ \square \ S_y \cdot 19,100^2$

6. Internal-friction theory (Mohr). When the ultimate strengths in ten- $S_M \geq S_m \ \square \ S_y$ or $2 \leq (0.0798T)^2 \ (1)$

2

sion and compression are the same, this theory reduces to that of maxi- $\square \ (40,000)^2$

imum shear. For principal stresses of opposite sign, failure is defined by or $T \ \square \ 221,000 \text{ in} \ \square \text{ lb}$ $(254,150 \text{ cm} \ \square \text{ kgf})$

$S_1 \geq (S_{uc}/S_u) \ S_2 \ \square \ \geq S_{uc}$; if the signs are the same $S_1 \ \square \ S_u$ or $\geq S_{uc}$, $S_2 \ M \leq S_m^2 \geq S \ M \ S_m \ \square \ S_y^2 \ (2)$

where S_{uc} is the ultimate strength in compression. If the principal

substituting and simplifying,

stresses are both either tension or compression, then the larger one, say

$\sqrt{\square} \ \square$

$19,100^2$

S_1 , must equal S_u when S_1 is tension or S_{uc} when S_1 is compression. $(9,550)^2 \leq 3 \leq (0.0798T)^2 \ \square \ (40,000)^2$

A graphical representation of the first four theories applied to a biaxial 2

stress field is presented in Fig. 5.2.72. Stresses outside the bounding or $T \ \square \ 255,000 \text{ in} \ \square \text{ lb}$ $(293,250 \text{ cm} \ \square \text{ kgf})$

lines in the case of each theory mean failure (yield or fracture). A

comparison with experimental data proves the distortion-energy theory

(4) best for ductile materials of equal tension-compression properties. PLASTICITY

When these properties are unequal, the internal friction theory (6) appears best. In practice, judging by some accepted codes, the maximum- The reaction of materials to stress and strain in the plastic range is not fully defined. However, some concepts and theories have been proposed.

Ideally, a purely elastic material is one complying explicitly with Hooke's law. In a viscous material, the shearing stress is proportional to the shearing strain. The purely plastic material yields indefinitely, but only after reaching a certain stress. Combinations of these are the elasto-viscous and the elastoplastic materials.

Engineering materials are not ideal, but usually contain some of the elastoplastic characteristics. The total strain ϵ_t is the sum of the elastic strain ϵ_o plus the plastic strain ϵ_p , as shown in Fig. 5.2.73, where the stress-strain curve is approximated by two straight lines. The natural strain, which is at the same time the total strain, is $\epsilon = \epsilon_o + \epsilon_p$

l_o

l

$dl/l = \epsilon$

$\ln(l/l_o)$. In this equation, l is the instantaneous length, while l_o is the

Fig. 5.2.71 Fig. 5.2.72 original length. In terms of the normal strain, the natural strain becomes

$\epsilon = \ln(1 + \epsilon_o)$. Since it is assumed that the volume remains constant,

shear theory (2) is generally used for ductile materials, and the maximum- $l/l_o = A_o/A$, and so the natural stress becomes $S = P/A = (P/A_o)(1 + \epsilon_o)$

mum-stress theory (1) for brittle materials. ϵ_o). A_o is the original cross-sectional area. If the natural stress is plotted

Fatigue failures cannot be related, theoretically, to elastic strength against strain on log-log paper, the graph is very nearly a straight line.

and thus to the theories described. However, experimental results justify The plastic-range relation is thus approximated by $S = K\epsilon^n$, where the

this, at least to a limited extent. Therefore, the theory evaluation given proportionality factor K and the strain-hardening coefficient n are deter-

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mined from best fits to experimental data. Values of K and n determined

by Low and Garofalo (Proc. Soc. Exp. Stress Anal., vol. IV, no. 2, 1947)

and $\epsilon_3 =$

$S(1 +$

e

$\geq n)/n$

$K^{1/n}$

$\epsilon \geq$

3

4

 $S_1 \geq S_1$

are given in Table 5.2.21. The maximum-shear theory, which is applicable to a ductile material under com-

bined stress, is acceptable here. Thus rupture will occur at

$$S_1 \geq S_3 \geq S_u, \text{ and}$$

$$\sqrt{\frac{S_1^2 + S_3^2}{2}} \geq \sqrt{\frac{S_u^2}{2}}$$

$$2 \geq 1/2$$

$$S_e \geq S_1 \geq S_3 \geq 2 \geq 3$$

$$\leq \leq S_{12} \geq S_1 \geq S_u$$

$$2 \geq 2 \geq 4 \geq 4$$

$$S_1 \geq S_3 \geq S_u \geq S_u$$

$$[(3/4)^{1/2} S_u]^{(1 \geq n)/n}$$

$$K^{1/n}$$

3

S

4 u

3

4

$$1 \leq 0.229/0.458 \geq 85,000$$

143,000

1/0.229

$$\geq 0.0475 \text{ in/in (0.0475 cm/cm)}$$

$$p_d \geq 2t \geq S_u$$

Since $S_u \geq S_1 \geq S_3$, then $p \geq$

$$2t \geq d$$

$$2 \geq 0.375 \geq 85,000$$

or $p \geq 1,550 \text{ lb/in}^2 \text{ (109 kgf/cm}^2 \text{)}$

Fig. 5.2.73 41

The geometry of Fig. 5.2.73 can be used to arrive at a second approx- ROTATING DISKS
imate relation

$$\geq \geq$$

Rotating circular disks may be of various profiles, of constant or vari-

H

$S \geq S_o \leq (\rho p \geq \rho_o) \tan \geq \geq S_o \geq 1 \geq \leq \rho p$ H able thickness, with or without centrally and noncentrally located holes,

E and with radial, tangential, and shearing stresses.

where $H \geq \tan \geq$ is a kind of plastic modulus. Solution starts with the differential equations of equilibrium and

The deformation theory of plastic flow for the general case of combined compatibility and the subsequent application of appropriate boundary stress is developed using the above concepts. Certain additional assumptions for the derivation of working-stress equations.

assumptions involved include: principal plastic-strain directions are the same as principal stress directions; the elastic strain is negligible compared with the plastic strain; the variation of stress with thickness can be assumed to be negligible, and symmetry eliminates the shearing stress. In the rotating case, the disk weight strains — $(\epsilon_1 \gg \epsilon_2)$, $(\epsilon_2 \gg \epsilon_3)$, $(\epsilon_3 \gg \epsilon_1)$ — to the principal shearing is neglected, but its inertia force becomes the body-force term in the stresses — $(S_1 \gg S_2)/2$, $(S_2 \gg S_3)/2$, $(S_3 \gg S_1)/2$ — are equal. The relations between the principal strains and stresses in terms of the simple tension quantities become

3 equations

$$\epsilon_r = \frac{1}{E} \left[\frac{S_1 + S_2}{2} - \nu \frac{S_3}{2} \right]$$

$$\epsilon_1 = \frac{1}{E} \left[\frac{S_1 + S_2}{2} - \nu \frac{S_3}{2} \right]$$

$$\epsilon_2 = \frac{1}{E} \left[\frac{S_2 + S_3}{2} - \nu \frac{S_1}{2} \right]$$

$$\epsilon_3 = \frac{1}{E} \left[\frac{S_3 + S_1}{2} - \nu \frac{S_2}{2} \right]$$

$$\epsilon_3 = \frac{1}{E} \left[\frac{S_3 + S_1}{2} - \nu \frac{S_2}{2} \right]$$

If these equations are added, the plastic-flow theory is expressed: where ν = Poisson's ratio; ρ = mass density, lb/in³; ω = angular

speed, rad/s; R = outside disk radius; and r = radius to point in ques-

$$S = \frac{1}{2} \left[(S_1 - S_2)^2 + (S_2 - S_3)^2 + (S_3 - S_1)^2 \right]^{1/2}$$

tion.

$\frac{1}{2}(\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2)$ The largest stresses occur at the center of the solid disk and are

In the above equation 3 equations

$$\epsilon_r = \frac{1}{E} \left[\frac{S_1 + S_2}{2} - \nu \frac{S_3}{2} \right]$$

$$\sqrt{\frac{1}{2}[(S_1 - S_2)^2 + (S_2 - S_3)^2 + (S_3 - S_1)^2]} = S_e$$

and $\frac{1}{2}(\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2)$ = ϵ_e A disk with a central hole of radius r_h (no external forces) is subjected

to the following stresses:

are the effective, or significant, stress and strain, respectively,

EXAMPLE. An annealed, stainless-steel type 430 tank has a 41-in inside di-

$$3 \text{ in}$$

$$8 \text{ in}$$

$$\frac{1}{2}$$

$$R^2 - r_h^2$$

$$\frac{1}{2} R^2 - r_h^2 \leq r^2 \leq \frac{1}{2} R^2 + r_h^2$$

$$r$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

ameter and has a wall 0.375 in thick. The ultimate strength of the stainless steel is 85,000 lb/in². Compute the maximum strain as well as the pressure at fracture. $3 \leq R_2 r_2 \leq 3$
 The tank constitutes a biaxial stress field where $S_1 = p d / (2t)$, $S_2 = p d / (4t)$, and $r_2 \leq r_2 \leq 2h \geq r$
 $8 r_3 \leq$

$S_3 = 0$. Taking the power stress-strain relation
 The maximum radial stress $\sigma_r|_M$ occurs at $r = \sqrt{R r_h}$, and
 $S_e = K \epsilon^n$ or $\sigma/S = S_e (1 + n)/n / K^{1/n}$

σ
 $S_e (1 + n)/n$
 $3 \leq$

thus $\sigma_1 =$
 3

$S = 0$, $\sigma_r|_M = \sigma_2 (R \geq r_h)^2$
 $K^{1/n} 4^{1/8}$

Table 5.2.21 Constants K and n for Sheet Materials

Material Treatment K, lb/in² n

0.05%C rimmed steel Annealed 77,100 0.261

0.05%C killed steel Annealed and tempered 73,100 0.234

Decarburized 0.05%C steel Annealed in wet H₂ 75,500 0.284

0.05/0.07% phos. low C Annealed 93,330 0.156

SAE 4130 Annealed 169,400 0.118

SAE 4130 Normalized and tempered 154,500 0.156

Type 430 stainless Annealed 143,000 0.229

Alcoa 24-S Annealed 55,900 0.211

Reynolds R-301 Annealed 48,450 0.211

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EXPERIMENTAL STRESS ANALYSIS 5-51

The largest tangential stress $\sigma_\theta|_M$ exists at the inner boundary, and other property is changing rapidly. In cases of sudden or abrupt section

$\sigma_\theta|_M =$
 $3 \leq$
 4
 σ
 $\sigma_2 R_2 \leq$
 $1 \geq 2$
 r
 $3 \leq h$

ρ changes, it is best to fair in across the change; the material density

should, however, be adjusted to give a total mass equal to the actual. Six to ten stations are often sufficient.

As the hole radius r_h approaches zero, the tangential stress assumes a value twice that at the center of a rotating solid disk, given above. The modulus of elasticity has a significant effect, and its exact value

value twice that at the center of a rotating solid disk, given above. at the temperature of each station should be used. The coefficients of

Stresses in Turbine Disks Explicit solutions for cases other than thermal expansion are usually averaged for the temperature between the

those cited are not available; so approximate solutions, such as those station and at which no thermal stress occurs.

proposed by Stodola, Thomson, Hetényi, and Robinson, are necessary. The first two Eqs. (5.2.2) and the last two must be worked simulta-

Manson uses the calculus of finite differences. See commentary under neously.

previous discussion of torsion for alternate methods of approximate At the outer boundary, loads external to the disk may be imposed,

solution. The problem illustrated below is a prime example of the ele- e.g., the radial stress σ_r , R from the centrifuged pull of a bucket. At the

gance of the combination of approximate methods and electronic com- center, the disk may be shrunk on a shaft with the fit pressures causing a

puters, which allow a rapid solution to be obtained. The speed with radial external push at this boundary.

which the repetitive calculations are done allows equally rapid solutions Numerical solutions are most expeditiously accomplished by use of a

with changes in design variables. table with column-to-column procedures. This technique lends itself

The customary, simplifying assumptions of axial symmetry — no readily to programmable computers or calculators.

variation of stress in the thickness direction and a completely elastic Disks with Noncentral Holes This case has not been solved explic-

stress situation — are made. The differential equations of equilibrium itly, but approximations are useful (e.g., Armstrong, Stresses in Rotat-

and comparability are rewritten in finite-difference form. ing Tapered Disks with Noncentral Holes, Ph.D. dissertation, Iowa

Solution of the finite-difference equations, appreciation of their linear State University, 1960). The area between the holes is considered re-

nature, and successive application of them yield the stresses at any moved and replaced by uniform spokes, each one with a cross-sectional

station in terms of those at a boundary station such as r_0 . The equations area equal to the original minimum spoke area and with a length equal

thus derived are to the diameter of the noncentral holes. The higher stress in such a spoke results in an additional extension, which is then applied to the outer

$\sigma_r, n \approx A_r, n \approx t, r_0 \leq B_r, n$ (5.2.1) annulus according to thin-ring theory and based on the average radius of

$\sigma_t, n \approx A_t, n \approx t, r_0 \leq B_t, n$ the ring. The additional stress is considered constant and is added to the

The finite-difference expressions yield Eqs. (5.2.2), which permit the tangential stress which would be present in a disk of the same dimen-

coefficients at station n to be computed from those at station $n \geq 1$. sions but filled (that is, no noncentral holes).

The stress in the substitute spoke is computed by adjusting the stress

$A_{r,n} \leq K_n A_{r,n \geq 1} \leq L_n A_{t,n \geq 1}$ at the hole-center radius in the solid or filled disk in proportion to the

$A_{t,n} \leq K_n A_{r,n \geq 1} \leq L_n A_{t,n \geq 1}$ areas, or $S_{sp} = \sigma_{r,h} (A_g / A_{sp})$, where $\sigma_{r,h}$ is the radial stress in the filled

$B_{r,n} \leq K_n B_{r,n \geq 1} \leq L_n B_{t,n \geq 1} \leq M_n$ (5.2.2)

disk at the radius of the hole circle, A_g is the gross circumferential area

$B_{t,n} \leq K_n B_{r,n \geq 1} \leq L_n B_{t,n \geq 1} \leq M_n$ at the same radius of the filled disk, and A_{sp} is the area of the substitute

The coefficients at the first station can be established by inspection. spoke. The increase in total strain is $\epsilon = \sigma_{r,h} / [E(A_g / A_{sp} \geq 1) l_{sp}]$, where

For a solid disk, for instance, where both stresses are equal to the tan- l_{sp} is the length of the substitute spoke.

gential stress at the center, the coefficients in Eqs. (5.2.1) are $A_{r,n} =$ The spoke-effect correction to be applied to the tangential stress is

$A_{t,n} = 1$ and $B_{r,n} = B_{t,n} = 0$. In the case of the disk with a central hole, therefore $\sigma_c = \sigma E / r_c$, where r_c is the average outer-rim or annulus

where $\sigma_{r,h} = 0$, $A_{r,h} = B_{r,h} = B_{t,h} = 0$ and $A_{t,rl} = 1$. Knowing radius. This is added to the tangential stress found at the corresponding

these, all others can be found from Eqs. (5.2.2). radius in the filled disk. The final step is to adjust the tangential and

At the outer boundary, $\sigma_{r,R} = A_{r,R} \leq B_{r,R}$ and $\sigma_{t,r0} = (\sigma_{r,R} \geq$ radial stresses as determined for stress concentrations caused by the

$B_{r,R} / A_{r,R}$. The radial and tangential stresses at each station are succes- holes in the actual disk. The factors for this adjustment are those in an

sively obtained, knowing $\sigma_{t,r0}$ and all the coefficients, using Eqs. (5.2.1). infinite plate of uniform thickness having the same size hole. The

The remaining coefficients in Eqs. (5.2.2), extracted from the finite- method is claimed to yield stresses within 5 percent of those measured

difference equations, are defined below, where E is Young's modulus at photoelastically at points of highest stress.

the temperature of the point in question, h is the profile thickness, α is

the thermal coefficient of expansion, ΔT is the temperature increment EXPERIMENTAL STRESS ANALYSIS

above that at which the thermal stress is zero, ν is Poisson's ratio, ω is

angular velocity of disk, and ρ is the mass density of disk material. Analytical methods of stress analysis can reach limits of applicability.

Many experimental techniques have been suggested and tried; several

$C_n = r_n / h_n$ have been developed to a state of great usefulness, e.g., photoelasticity,

$C_n = \rho n / E n \leq (1 \leq \rho n) (r_n \geq 1) / (2 E n r_n)$

strain-gage measurement, brittle coating, birefringent coating, and ho-

$$D_n = \frac{1}{2}(n_x - n_y)h$$

lography.

$$D_n = \frac{1}{E} \left(\frac{1}{2} (n_x - n_y) \right) (n_x - n_y) / (2E n_y)$$

$$F_n = n_x - n_y \quad h = 1 \text{ Photoelasticity}$$

$F_n = \left(\frac{1}{2} (n_x - n_y) \right) \geq \left(\frac{1}{2} (n_x - n_y) \right) (n_x - n_y) / (2E n_y)$ Most transparent materials exhibit temporary double refraction, or bire-

$G_n = \frac{1}{2} (n_x - n_y)h$ fringing, when stressed. Light is resolved into components along the

$$G_n = \left(\frac{1}{2} (n_x - n_y) \right) \geq \left(\frac{1}{2} (n_x - n_y) \right) (n_x - n_y) / (2E n_y)$$

two principal plane directions. The effect is temporary as long as the

$$H_n = \frac{1}{2} \left(\frac{1}{2} (n_x - n_y) \right) (n_x - n_y) / (2E n_y)$$

elastic stress is not exceeded and is in direct proportion to the applied

$$H_n = \frac{1}{2} \left(\frac{1}{2} (n_x - n_y) \right) (n_x - n_y) / (2E n_y)$$

load. The stress magnitude can be established by the amount of compo-

$$K_n = \left(\frac{1}{2} (n_x - n_y) \right) / (C n_y)$$

nent wave retardation, as given in the white and black band field (fringe

$$L_n = \left(\frac{1}{2} (n_x - n_y) \right) / (C n_y)$$

pattern) obtained when a monochromatic light source is used. The po-

$$L_n = \left(\frac{1}{2} (n_x - n_y) \right) / (C n_y)$$

lariscope, consisting of the light source, the polarizer, the model in a

$$L_n = \left(\frac{1}{2} (n_x - n_y) \right) / (C n_y)$$

loading frame, an analyzer (same as polarizer), and a screen or camera,

$$M_n = \left(\frac{1}{2} (n_x - n_y) \right) / (C n_y)$$

is used to produce and evaluate the fringe effect. Quarter-wave plates

$$M_n = \left(\frac{1}{2} (n_x - n_y) \right) / (C n_y)$$

may be placed on either side of the model, making the light components

Situations need not be equally spaced between the two boundaries. It through the model independent of the absolute orientation of polarizer

is best to space them more closely where the profile, temperature, or and analyzer. The polarizer is a plane polariscope and yields the direc-

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tions of principal stresses (the isoclinics); the analyzer is a circular pal planes at each point, two families of orthogonal curves are drawn.

polariscope yielding the fringes (isochromatics) as well. Care must be exercised in the drawing of trajectories for practical accu-

Figure 5.2.74 shows the fringe pattern and the 20° isoclinics of a disk racy.

loaded radially at four places. Stress Separation If knowledge of each principal stress is required,

The isochromatics in the fringe pattern depict the difference between the photoelastic data must be treated to separate the stresses from the

principal stresses. At free boundaries where the normal stress is zero, difference given by the data. If the sum of the two stresses is also

the difference automatically becomes the tangential stress. Starting at obtained somehow, a simultaneous solution of the sum and difference

such a boundary and proceeding into the interior, the stresses can be values will yield each principal stress. One can also start at a boundary

separated by numerical calculation. where the normal stress value is zero. There, the photoelastic reading

gives the principal stress parallel to the boundary. Starting with the single value, methods have been developed which can be used to proceed with the separation. Typical of the former are lateral-extensometer, iteration, and membrane-analogy techniques; typical of the latter are the slope-equilibrium, shear-difference, graphical-integration, alternating-summation methods, and oblique incidence. Often, however, the surface stresses are the maximum valued ones. (See Frocht, "Photoelasticity," McGraw-Hill.)

EXAMPLE. The fringe pattern of a Homalite disk 1.31 in in diam, 0.282 in thick, and carrying four radial loads of 155 lb each is shown in Fig. 5.2.74.

A closed solution is not known. However, by counting the fringe order at any point, the stress can be determined photoelastically. For instance, the dark spot at the center marks a fringe of zero order, as do the disk edges except in the immediate vicinity of the concentrated loads. The point at the center, which remained dark throughout the loading, is an isotropic point (zero stress difference and normal stresses are equal in all directions). Counting out from the center toward the load, the first "circular" fringe is of order 3. Therefore, anywhere along it $(p \geq q)/2 = \frac{1}{2} \left(\frac{M}{\pi r^2} - \frac{3}{2} \frac{F}{\pi r^2} \right)$. Carefully inspected, fringe 12 can be counted at the point of load application. Therefore, $r = \frac{1}{12} \sqrt{\frac{M}{\pi (p - q/2)}}$ $65/0.282 = 2,770 \text{ lb/in}^2 (195 \text{ kgf/cm}^2)$.

THREE-DIMENSIONAL PHOTOELASTICITY

Stress "freezing" and slicing, wherein a plastic model is brought up to its critical temperature, loaded as desired, and while loaded, slowly brought back to room temperature, are techniques which freeze the fringe pattern into the model. The model can be cut into slices without disturbing the "frozen" strains. Two-dimensional models are usually machined from plate stock, and three-dimensional models are cast. The frozen stress model is sliced so that the desired information can be obtained by normal incidence using the previous formulations.

When normal incidence is not possible, oblique incidence becomes necessary. Oblique-incidence patterns are usable in two-dimensional as well as three-dimensional stress separation. The measurement of fractional fringes is often required when using oblique incidence. With a crossed, circular, monochromatic polariscope, oriented to the principal Fig. 5.2.74 stresses at a point, the analyzer is rotated through some angle θ until extinction occurs. The fringe value n is $n = n_0 \sin^2 \theta / 180$, where n_0 is the

The Stress-Optic Law In a transparent, isotropic plate subjected to order of the last visible fringe. Whether the fractional term is added or

a biaxial stress field within the elastic limit, the relative retardation R_t subtracted depends upon the direction in which the analyzer is rotated

between the two components produced by temporary double refraction (established by inspection).

is $R_t = Ct(p \geq q) \pm n$, where C is the stress-optic coefficient, t is the plate thickness, p and q are the principal stresses, n is the fringe order (the number of fringes which have passed the point during application of load), and λ is the wavelength of monochromatic light used. Thus, $(p \geq q)/2 \pm M \pm n/(2Ct) \pm nf/t$

If the material-fringe value f is determined with the same light source (generally a mercury-vapor lamp emitting light having a wavelength of 5,461 Å) as used in the model study, the maximum shearing stress, or one-half the difference between the principal stresses, is directly deter-

mined. The calibration is a matter of obtaining the material-fringe value

in lb/in² per fringe per inch (kgf/cm² per fringe per cm). Oblique-incidence calculations are based on the stress-optic law:

Isoclinics, or the direction of the principal planes, can be obtained with $n \pm R_1 \pm t(p \geq q)/f \pm tp/f \geq tg/f \pm np \geq nq$. Also, when polarized light

a plane polariscope. A new isoclinic parameter is observed each time is directed through the slice at an angle α to a principal plane, either by

the polarizer and analyzer are rotated simultaneously into a new posi- rotating the slice away from normal to the light ray or by cutting it at the

tion. A white-light source reveals a more distinct isoclinic, as the black angle α (see Fig. 5.2.75), the fringe order becomes

curve is more distinguishable against a colored background. $t \pm t$

Isostatics, or stress trajectories, are curves the tangents to which rep- $n \pm x \pm (p \geq q) \pm (p \geq q \cos^2 \alpha)$

resent the progressive change in principal-plane directions. They are $f \pm f \cos \alpha$

constructed graphically using the isoclinics. Since there are two princi- $(np \geq nq \cos^2 \alpha)/\cos \alpha$

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Solving algebraically, containing the “active” gage, the electric-resistance temperature effect is canceled out. Thus the active gage reports only that which is taking

$np \pm (n \pm x \cos \alpha \geq nn \cos^2 \alpha)/\sin^2 \alpha$

place in the stressed plate. The power supply can be either ac or dc.

and $nq \pm (n \pm x \cos \alpha \geq nn)/\sin^2 \alpha$

It is sometimes useful to make both gages active — e.g., mounted on

If orders nn and $n \pm x$ are thus measured at a point, np and nq can be opposite sides of a beam, with one gage subjected to tension and the

computed. The principal stresses are then determined from $p \pm fnp/t$ and other to compression. Temperature effects are still compensated, but the

q Δ fnq /t. bridge output is doubled. In other instances, it may be desirable to make The material-fringe value f in these equations is at the “freezing” all four bridge arms active gages. The experimenter must determine the temperature (critical temperature). The angle of incidence, as well as the most practical arrangement for the problem at hand and must bear in fringe orders, must be accurately measured if errors are to be mini- mind that the bridge unbalances in proportion to the difference in the mized. strains of gages located in adjacent legs and to the sum of strain in gages located in opposite legs.

Bonded Metallic Gages

Strain measurements down to one-millionth inch per inch (one-millionth cm/cm) are possible with electrical-resistance wire gages. Such gages can be used to measure surface strains (stress by Hooke’s law) on any shape or size of object. Figure 5.2.76 illustrates schematically the gage construction with a grid of fine alloy wire or thin foil, bonded to paper and covered for protection with a felt pad. In use, the gage is cemented rigidly to the surface of the member to be analyzed. The strain relation is $\Delta R/R = (1/Gf) \epsilon$ in/in (cm/cm). Thus, if the resistance R and gage factor Gf (given by the gage manufacturer) are known and the change in resistance ΔR is measured, the strain which caused the resistance change can be determined and Hooke’s law can be applied to determine the stress.

Fig. 5.2.77

Dynamic strains can be detected using circuits such as the ballast type shown in Fig. 5.2.77b. The capacitor coupling passes only rapidly varying or dynamic strains. The capacitor’s infinite impedance to a steady voltage filters out any static effects or strains. The circuit is dc powered.

Transverse Sensitivity Grid-type gages possess some strain sensitivity in the direction perpendicular to the gage axis. In a uniaxial stress field, this transverse sensitivity is of no concern because the gage factor was obtained in such a field. However, in a biaxial stress field, neglect of transverse sensitivity will give slightly erroneous strains. When accounted for, the true strains in the axial direction of gage, ϵ_1 , and at Fig. 5.2.76 right angles to it, ϵ_2 , are $\epsilon_1 = (1 \pm k)(\Delta R_1 \pm k\Delta R_2)/(1 \pm k^2)$ and $\epsilon_2 = (1 \pm k)(\Delta R_2 \pm k\Delta R_1)/(1 \pm k^2)$, where the apparent strains are ΔR_1

Gages must be properly selected in accordance with manufacturer’s $\Delta R_1/(RGf)$ and $\Delta R_2/(RGf)$, measured by cementing a gage in

recommendations. The surface to which the gage is applied must be each direction 1 and 2. The factor ν is Poisson’s ratio of the material to

clean, the proper cement must be used, and the gage assembly must be which gages are cemented, and k (usually provided by the gage manu-

coated for protection against environmental conditions (e.g., moisture). The factor $G_1 = R_1/(R_0 \epsilon)$ is the coefficient of transverse sensitivity of the gage. The gage

A gaging unit, usually a Wheatstone bridge or a ballast circuit (see Fig. 5.2.77 and Sec. 15), is needed to detect the signal resulting from the direction, and the uniaxial stress is applied in its axial

Fig. 5.2.77 and Sec. 15), is needed to detect the signal resulting from the direction, and the resistance change and strain are measured. The gage

change in resistance of the strain gage. The strain and, therefore, the factor $G_1 = R_1/(R_0 \epsilon)$ is computed. A uniaxial stress is next applied

signal are often too small for direct handling, so that amplification is transversely to the gage. Again the resistance change and strain are

needed, with a metering discriminator for magnitude evaluation. measured and G_2 computed. Then $k = (G_2 \leq G_1)/(G_1 \leq G_2)$.

The signal is read or recorded by a galvanometer, oscilloscope, or Strain Rosettes In a general biaxial stress field, the principal plane

other device. Equipment specifically constructed for strain measure- directions, as well as the stresses, are unknown. Thus, three gages

ment is available to indicate or record the signal directly in strain units. mounted in three differing directions are needed if the three unknowns

Static strains are best gaged on a Wheatstone bridge, with strain gages are to be determined. Three standard gage combinations, called strain

wired to it as indicated in Fig. 5.2.77a. With the bridge set so that the rosettes, are commercially available and are best for the purpose. These

only unbalance is the change of resistance in the active-strain gage, the are the rectangular strain rosette (Fig. 5.2.78a), which covers a mini-

potential difference between the output terminals becomes a measure- mum of area and is therefore best where the strain gradient is high; the

ment of strain. Since the gage is sensitive to temperature as well as equiangular strain rosette (Fig. 5.2.78b), where the gages do not over-

strain, it will measure the combined effect. However, if a "dummy" lap and which can be used where the strain gradient is low; the T-delta

gage, cemented to an unstressed piece of the same metal subjected to the strain rosette (Fig. 5.2.78c), which occupies no more area than the

same climatic conditions, is wired into the bridge leg adjacent to the one equiangular rosette and which provides an extra check, or "insurance"

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Fig. 5.2.78

gage. The wiring and instrumentation of gages in rosettes do not differ croinches per inch (400 microcentimeters per centimeter). If desired, the

from those of individual gages. approximate strain (probably within 10 percent) may be established

The true strains along the gage-length directions are found according using the calibration strip sprayed with the test part. The strip is placed

to the following equations, in which $R_n = \frac{R_n}{[RF1(1 \geq k/2)]}$ and b is in a loading device and bent as a cantilever beam by means of a cam at

1/k. the free end, causing the coating to crack on the tension surface. Crack

RECTANGULAR ROSETTE (SEE FIG. 5.2.78a) spacing varies with the strain, being close at the fixed end and diminish-

ing toward the free end down to threshold sensitivity values. The strip is

$$\sigma_1 \geq R_1 \geq R_3 / b$$

placed in a holder containing strain graduations. A visual comparison of

$$\sigma_2 \geq R_2(1 \leq 1/b) \geq (1/b)(R_1 \leq R_3)$$

cracks on the testpart surface with those on the strip reveals the strain

$$\sigma_3 \geq R_3 \geq R_1 / b$$

magnitude which caused the cracks.

EQUIANGULAR ROSETTE (SEE FIG. 5.2.78b)

Birefringent Coatings

$$\sigma_1 \geq R_1 \geq (1/b)(R_2 \leq R_3)$$

A birefringent coating is one which becomes double refractive when

$$\sigma_2 \geq R_2 \geq (1/b)(R_1 \leq R_3)$$

strained. The principle is quite old, but plastics, which adhere to all

$$\sigma_3 \geq R_3 \geq (1/b)(R_1 \leq R_3)$$

kinds of materials, which have stable optical-strain constants, and which

T-DELTA ROSETTE (SEE FIG. 5.2.78c) are sufficiently sensitive to be practical, are of recent development. The

trade name applied to this technique is Photostress. Photostress plastics

$$\sigma_1 \geq R_1(1 \leq 1/b) \geq (1/b)(R_3 \leq R_4)$$

can be obtained either as thin sheets (0.040, 0.080, and 0.20 in) or in

$$\sigma_2 \geq R_2(1 \leq 1/b) \geq (1/b)(R_3 \leq R_4)$$

liquid form. The sheet material can be bonded to a surface with a special

$$\sigma_3 \geq R_3 \geq (1/b)R_4$$

adhesive. The liquid can be brushed or sprayed on, or the part can be

$$\sigma_4 \geq R_4 \geq (1/b)R_3$$

dipped in the liquid. The layer should be at least 0.004 in (0.010 cm)

Foil Gages Foil gages are produced from thin foil by photoetching thick. It is often necessary to apply several successive coatings, with

techniques and are applied, instrumented, read, and evaluated just like heat curing of each layer in turn. Two sheet types and two liquids are

the wire-grid type. Foil gages, being much thinner, may be applied available; these differ in stretching ability and in magnitude of the

easily to curved surfaces, have lower transverse sensitivity, exhibit neg- strain-optical constant.

Each of the sheet materials is available metal-

ligible hysteresis under cycling loads, creep little under sustained loads, lized on one face, to reflect polarized light even when cemented to a dull

and can be stacked on top of each other. surface.

The principles involved are the same as those for conventional pho-

Brittle-Coating Analysis to elasticity. One frequent advantage is the fact that the plastic (sheet or liquid) can be applied directly to the part, which can then be subjected to Brittle coatings which adhere to the surface well can reveal the strain in actual operating loads. A special reflecting polariscope must be used. It the underlying material. Probably the first such coating used was mill contains only one polarizer and quarter-wave disk because the light scale, a thin iron oxide which forms on hot-rolled steel stock. Many passes back through the same pair after reflection by the stressed surface coatings such as whitewash, portland cement, and shellac have been face-plastic interface. The only limitation rests in the geometry of the tried.

structural component to be examined; not only must it be possible to The most popular of presently available strain-indicating brittle coatings apply the plastic to the surface, but the surface must be accessible to ings are the wood-rosin lacquers supplied by the Magnaflux Corporation light.

tion under the trade name Stresscoat. Several Stresscoat compositions The strain-optic law, since the light passes the plastic thickness twice, are available; the suitability of a particular lacquer depends upon the becomes

prevailing temperature and humidity. The lacquer is usually sprayed to a thickness of 0.004 to 0.008 in (0.01 to 0.02 cm) upon the surface, n E which must be clean and free of grease and loose particles. Calibration $p \approx q$ $2t K(1 \pm \nu)$

bars are sprayed at the same time. Both must be dried at an even temperature for up to 24 h. To facilitate observation of cracks, an under- where n is fringe order, E is modulus, ν is Poisson's ratio of workpiece

coating of bright aluminum is often applied. material, and K (supplied by the manufacturer) is the strain-optic coefficient

When the cured test piece is subjected to loads, the lacquer will first crient of the plastic. As in conventional photoelasticity, isoclinics are

begin to crack at its threshold sensitivity in the area of the largest principal stress, with the parallel cracks perpendicular to the principal stress.

This information is often sufficient, as it reveals the critical area and the Holography direction of normal stress. A more recently developed technique applicable to stress, or rather, The threshold sensitivity of Stresscoat lacquers is 600 to 800 microstrain analysis as well as to many other purposes is that of holography. It

microinches per inch (600 to 800 microcentimeters per centimeter) in a is made possible by the laser, an instrument which produces a highly

uniaxial stress field. Exact control of lacquer selection, thickness, curvature, concentrated, thin beam of light of single wavelength. The helium-neon

ing, and testing temperatures may reduce the threshold to 400 microstrain (He-Ne) laser, emitting at the red end of the visible spectrum at a wave-

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PIPELINE FLEXURE STRESSES 5-55

length of 633 nanometers, has found much favor. The output of a he- bench so that beam coherence is assured, required coherence depth

lium-cadmium (He-Cd) laser is at half the wavelength of the He-Ne satisfied, and the object/reference angle θ consistent with the fringe

laser; accordingly, the He-Ne laser is twice as sensitive to displace- spacing desired. The film must also possess adequate sensitivity in the

ments. spectral range of the laser beam used. It is important to recognize the

The laser beam is split into two components, one of which is directed inherent hazards of the high-intensity radiation in laser beams and to

upon the object (or specimen) and then onto the photographic plate. It is practice every precaution in the use of lasers.

identified as the object beam. The other component, referred to as the reference beam, propagates directly to the plate. Interference between the beams resulting from retardations caused by displacements or strains forms fringes which in turn provide a measure of the disturbance.

Spacing of such fringes depends upon Bragg's law:

$d \sin \theta$

$2 \sin (\theta/2)$

where d is the distance between fringes, λ is the wavelength of the light source, and θ is the angle between the object and reference ray at the plate.

A simple holographic setup consists of the laser source, beam splitter, reflecting surfaces, filters, and the recording plate. A possible arrangement is depicted in Fig. 5.2.79. Some arrangement for loading the spec- Fig. 5.2.79 Simple holographic setup. (1) Laser source; (2) beam splitter;

imen must also be provided. Additional auxiliary and refining hardware (3) reflecting surfaces; (4) circular polarizers; (5) loaded specimen (birefringent);

becomes necessary as the analysis assumes greater complexity. Thus (6) photographic plate.

the system layout is limited only by test requirements and the experi-

menter's imagination. However, only a thorough understanding of the Holography, using pulsed lasers, can be used to measure transient

laws of optics and interferometry will make possible a reliable investi- disturbances. Thus vibration studies are possible. Fatigue detection

gation and interpretation of results. using holographic techniques has also been undertaken.

Holography has

Stability of setup must be assured via a rigid optical bench and sup- been used in acoustical studies and in automatic gaging as well. It is a

porting brackets. Component instruments must be spaced upon the versatile engineering tool.

5.3 PIPELINE FLEXURE STRESSES

by Harold V. Hawkins

EDITOR'S NOTE: The almost universal availability and utilization of personal $\odot y$ is same as $\odot x$ but parallel to y direction, in. Note that computers in engineering practice has led to the development of many competing $\odot x$ and $\odot y$ are positive if under the change in and complementary forms of piping stress analysis software. Their use is wide- temperature the end opposite the origin tends to spread, and individual packaged software allows analysis and design to take into move in a positive x or y direction, respectively. account static and dynamic conditions, restraint conditions, aboveground and buried $\square$ wall thickness of pipe, in (m) ied configurations, etc. The reader is referred to the technical literature for the most suitable and current software available for use in solving the immediate r $\square$ mean radius of pipe cross section, in (m) problems at hand. $\square$ $\square$ constant $\square$ tR/r^2

The brief discussion in this section addresses the fundamental concepts entailed I $\square$ moment of inertia of pipe cross section about pipe and sets forth the solution of simple systems as an exercise in application of the centerline, in4 (m4) principles. E $\square$ modulus of elasticity of pipe at actual working temperature, lb/in² (N/m²)

REFERENCES: Shipman, Design of Steam Piping to Care for Expansion, Trans. ASME, 1929, Wahl, Stresses and Reactions in Expansion Pipe Bends, Trans. K $\square$ flexibility index of pipe. K $\square$ 1 for all straight pipe ASME, 1927. Hovgaard, The Elastic Deformation of Pipe Bends, Jour. Math. sections, K $\square$ $(10 \leq 12 \square^2) / (1 \leq 12 \square^2)$ for all Phys., Nov. 1926, Oct. 1928, and Dec. 1929. M. W. Kellogg Co., "The Design of curved pipe sections where $\square \rightarrow 0.335$ (see Fig. Piping Systems," Wiley. 5.3.3)

For details of pipe and pipe fittings see Sec. 8.7. $\square$, $\square$, $\square$ $\square$ ratios of actual max longitudinal flexure, trans- Nomenclature (see Figs. 5.3.1 and 5.3.2) verse flexure, and shearing stresses to Mr/I for curved sections of pipe (see Fig. 5.3.3)

M_0 $\square$ end moment at origin, in $\square$ lb (N $\square$ m)
 M $\square$ max moment, in $\square$ lb (N $\square$ m)
 F_x $\square$ end reaction at origin in x direction, lb (N)
 F_y $\square$ end reaction at origin in y direction, lb (N)
 S_l $\square$ (Mr/I) , $\square$ $\square$ max unit longitudinal flexure stress, lb/in² (N/m²)
 S_t $\square$ (Mr/I) , $\square$ $\square$ max unit transverse flexure stress, lb/in² (N/m²)
 S_s $\square$ (Mr/I) , $\square$ $\square$ max unit shearing stress, lb/in² (N/m²)
 $\odot x$ $\square$ relative deflection of ends of pipe parallel to x direction caused by either temperature change or

support movement, or both, in (m) Fig. 5.3.1 Fig. 5.3.2

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5-56 PIPELINE FLEXURE STRESSES

A, B, C, F, G, H constants given by Table 5.3.2 General Discussion

θ angle of intersection between tangents to direction Under the effect of changes in temperature of the pipeline, or of move-

of pipe at reactions ment of support reactions (either translation or rotation), or both, the

$\Delta\theta$ change in θ caused by movements of supports, or determination of stress distribution in a pipe becomes a statically inde-

by temperature change, or both, rad terminate problem. In general the problem may be solved by a slight

ds an infinitesimal element of length of pipe modification of the standard arch theory: $\Delta x \approx K \Delta My ds(EI)$, $\Delta y \approx$

s length of a particular curved section of pipe, in $K \Delta Mx ds/(EI)$, and $\Delta \theta \approx K \Delta M ds/(EI)$ where the constant K is intro-

(m) duced to correct for the increased flexibility of a curved pipe, and where

R radius of curvature of pipe centerline, in (m) the integration is over the entire length of pipe between supports. In

Table 5.3.1 are given equations derived by this method for moment and thrust at one reaction point for pipes in one plane that are fully fixed, hinged at both ends, hinged at one end and fixed at the other, or partly fixed. If the reactions at one end of the pipe are known, the moment distribution in the entire pipe then can be obtained by simple statics. Since an initially curved pipe is more flexible than indicated by its moment of inertia, the constant K is introduced. Its value may be taken from Fig. 5.3.3, or computed from the equation given below. $K \approx 1$ for all straight pipe sections, since they act according to the simple flexure theory.

In Fig. 5.3.3 are given the flexure constants K, Δ , θ , and ϕ for initially curved pipes as functions of the quantity $\phi \approx tR/r^2$. The flexure constants are derived from the equations.

$$K \approx (10 \leq 12\phi^2)/(1 \leq 12\phi^2) \text{ when } \phi \rightarrow 0.335$$

$$\Delta \approx 2/3K\sqrt{(5 \leq 6\phi^2)/18} \approx 1.472$$

$$\Delta \approx K(6\phi^2 \geq 1)/(6\phi^2 \leq 5) \approx 1.472$$

$$\theta \approx 18\Delta / (1 \leq 12\phi^2)$$

$$\phi \approx [8\phi \geq 36\phi^3 \leq (32\phi^2 \leq 20/3)]$$

$$\phi \sqrt{(4/3)\phi^2 \leq 5/18} \approx (1 \leq 12\phi^2) \text{ when } \phi \rightarrow 0.58$$

Fig. 5.3.3 Flexure constants of initially curved pipes. $\phi (12\phi^2 \leq 18\phi \geq 2) / (1 \leq 12\phi^2)$ when $\phi \rightarrow 0.58$

Table 5.3.1 General Equations for Pipelines in One Plane (See Figs. 5.3.1 and 5.3.2)

Symmetric

Type of supports Unsymmetric about y-axis

$$EI \otimes x(CF \geq AB) \leq EI \otimes y(BF \geq AG) \quad EI \otimes x F$$

Both ends fully fixed $M_0 \neq 0$

$$2ABF \leq CGH \geq B^2H \geq A^2G \geq CF^2 GH \geq F^2$$

$$EI \otimes x(CH \geq A^2) \leq EI \otimes y(BH \geq AF) \quad EI \otimes x H$$

$$F_x \neq F_x$$

$$2ABF \leq CGH \geq B^2H \geq A^2G \geq CF^2 GH \geq F^2$$

$$EI \otimes x(BH \geq AF) \leq EI \otimes y(GH \geq F^2) \quad F_y \neq 0$$

$$F_y \neq 0 \neq 0$$

$$2ABF \leq CGH \geq B^2H \geq A^2G \geq CF^2$$

$$0 \neq 0$$

Both ends hinged $M_0 \neq 0$ $M_0 \neq 0$

$$EI \otimes x C \leq EI \otimes y B \quad EI \otimes x$$

$$F_x \neq F_x$$

$$CG \geq B^2 G$$

$$EI \otimes x B \leq EI \otimes y G \quad F_y \neq 0$$

$$F_y \neq$$

$$CG \geq B^2 \geq 0 \quad xF$$

$$0 \neq$$

$$0 \otimes x(AB \geq CF) \leq 0 \otimes y(AG \geq BF) \quad G$$

$$0 \neq$$

$$CG \geq B^2$$

Origin end only hinged, other end fully $M_0 \neq 0$

$$\text{fixed } EI \otimes x C \leq EI \otimes y B$$

$$F_x \neq$$

$$CG \geq B^2$$

$$EI \otimes x B \leq EI \otimes y G$$

$$F_y \neq$$

$$CG \leq B^2$$

$$0 \otimes x(AB \geq CF) \leq 0 \otimes y(AG \geq BF)$$

$$0 \neq$$

$$CG \geq B^2$$

$$EI \otimes x(CF \geq AB) \leq EI \otimes y(BF \geq AG) \leq EI \otimes (CG \geq B^2)$$

In general for any specific rotation $0 \neq M_0 \neq$

$$\text{and movement } 0 \otimes x \text{ and } 0 \otimes y \dots 2ABF \leq CGH \geq A^3G \geq CF^2 \geq B^2H$$

$$EI \otimes x(CH \geq A^2) \leq EI \otimes y(BH \geq AF) \leq EI \otimes (CF \geq AB)$$

$$F_x \neq$$

$$2ABF \leq CGH \geq A^2G \geq CF^2 \geq B^2H$$

$$EI \otimes x(BH \geq AF) \leq EI \otimes y(GH \geq F^2) \leq EI \otimes (BF \geq AG)$$

$$F_y \neq$$

$$2ABF \leq CGH \geq A^2G \geq CF^2 \geq B^2H$$

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PIPELINE FLEXURE STRESSES 5-57

The increased flexibility of the curved pipe is brought about by the fact that there is no transverse flexure stress since all sections are straight. The maximum tendency of its cross section to flatten. This flattening causes a transverse shearing stress is either (1) one-half of the maximum longitudinal fiber stress, (2) one-half of the maximum longitudinal fiber stress whose maximum is S_t . Because the maximum longitudinal fiber stress is as given above, (2) one-half of the hoop-tension stress caused by an internal radial pressure. (3) one-half the difference of the maximum longitudinal and maximum transverse stresses do not occur at the same pressure that might exist in the pipe, or (3) one-half the difference of the maximum longitudinal fiber stress and hoop-tension stress, whichever of these three possibilities is numerically greatest. In the straight sections of the pipe, $\mu = 1$, the transverse stress disappears, and $\mu = 1/2$. This discussion of S_s does not include the uniform transverse or longitudinal tension stresses induced by the internal pressure in the pipe; their effects should be added if appreciable.

Table 5.3.2 gives values of the constants A, B, C, F, G, and H for use in equations listed in Table 5.3.1. The values may be used (1) for the solution of any pipeline or (2) for the derivation of equations for standard shapes composed of straight sections and arcs of circles as of Fig. 5.3.4.

5.3.5. Equations for shapes not given may be obtained by algebraic addition of those given. All measurements are from the left-hand end of the pipeline. Reactions and stresses are greatly influenced by end conditions. Formulas are given to cover the extreme conditions. The following suggestions and comments should be considered when laying out a right-angle pipeline (Fig. 5.3.5) may be determined from the following equations:

pipeline:

Avoid expansion bends, and design the entire pipeline to take care of its own expansion. $M_0 = C_1 EI \Delta x/R^2$
 $F_x = C_2 EI \Delta x/R^3$

The movement of the equipment to which the ends of the pipeline are attached must be included in the Δx and Δy of the equations.

Maximum flexibility is obtained by placing supports and anchors so that in these equations, Δx is the x component of the deflection between reaction points caused by temperature change only. The values of C_1 , C_2 , and C_3 are given that they will not interfere with the natural movement of the pipe. in Fig. 5.3.6 for $K = 1$ and $K = 2$. For other values of K , interpolation may be used.

That shape is most efficient in which the maximum length of pipe is employed working at the maximum safe stress. Excessive bending moment at joints is more likely to cause trouble

than excessive stresses in pipe walls. Hence, keep pipe joints away from points of high moment.

Reactions and stresses are greatly influenced by flattening of the cross section of the curved portions of the pipeline.

It is recommended that cold springing allowances be discounted in stress calculations.

Application to Two- and Three-Plane Pipelines Pipelines in more Fig. 5.3.5 Right angle pipeline.

than one plane may be solved by the successive application of the preceding data, dividing the pipeline into two or more one-plane lines. EXAMPLE 3. With $a/R = 20$ and $b/R = 3$, the value of C_1 is 0.185 for $K = 1$

EXAMPLE 1. The unsymmetric pipeline of Fig. 5.3.4 has fully fixed ends. and 0.165 for $K = 2$. If $K = 1.75$, the interpolated value of C_1 is 0.175.

From Table 5.3.2 use $K = 1$ for all sections, since only straight segments are involved. Elimination of Flexure Stresses Pipeline flexure stresses that nor-

Upon introduction of $a = 120$ in (3.05 m), $b = 60$ in (1.52 m), and $c = 180$ in mally would result from movement of supports or from the tendency of

(4.57 m), into the preceding relations (Table 5.3.3) for A, B, C, F, G, H, the

the pipes to expand under temperature change often may be avoided

equations for the reactions at 0 from Table 5.3.1 become

entirely through the use of expansion joints (Sec. 8.7). Their use may

$M_0 = EI \phi_x (\geq 7.1608 \times 10^5) \leq EI \phi_y (\geq 8.3681 \times 10^5)$ simplify both the design of the pipeline and the support structure. When

$F_x = EI \phi_x (\leq 1.0993 \times 10^5) \leq EI \phi_y (\leq 3.1488 \times 10^6)$ using expansion joints, the following suggestions should be considered:

$F_y = EI \phi_x (\leq 3.1488 \times 10^6) \leq EI \phi_y (\leq 1.33717 \times 10^6)$ (1) select expansion joint carefully for maximum temperature range

(and deflection) expected so as to prevent damage to expansion fitting;

Also it follows that

(2) provide guides to limit movement at expansion joint to direction

$M_1 = M_0 \leq F_y a = EI \phi_x (\leq 3.0625 \times 10^4) \leq EI \phi_y (\leq 7.6779 \times 10^5)$ permitted by joint; (3) provide adequate anchors at one end of each

$M_2 = M_1 \geq F_x b = EI \phi_x (\geq 3.5333 \times 10^4) \leq EI \phi_y (\geq 1.1215 \times 10^4)$ straight section or along their midlength, forcing movement to occur at

$M_3 = M_2 \leq F_y c = EI \phi_x (\leq 2.1345 \times 10^4) \leq EI \phi_y (\leq 1.2854 \times 10^4)$ expansion joint yet providing adequate support for pipeline; (4) mount

expansion joints adjacent to an anchor point to prevent sagging of the

Thus the maximum moment M occurs at 3. pipeline under its own weight and do not depend upon the expansion

The total maximum longitudinal fiber stress ($a = 1$ for straight pipe)

joint for stiffness — it is intended to be flexible; (5) give consideration

$F_x M_r$ to effects of corrosion, since corrugated character of expansion joints

$SI = 3$

2nd rt I makes cleaning difficult.

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Table 5.3.2 Values of A, B C, F, G, and H for Various Piping Elements

$A = K_x x$ ds $B = K_{xy} x y$ ds $C = K_{x^2} x^2$ ds $F = K_y y$ ds $G = K_{y^2} y^2$ ds $H = K$ ds

s

$(x \leq x^2)$

2 1

$A_y s = s^2$

3

$\leq x^1 x^2 \leq y F_y s$

sx

A

2

$(y^1 \leq y^2) A_x$

s

2

$(y^1 \leq y^2) s = s^2$

3

$\leq y^1 y^2 \leq s$

s A s 2 s s 2

$(x^1 \leq x^2) (y^1 \leq y^2) (x^1 \leq x^1 x^2 \leq x^2 (y^1 \leq y^2) (y^1 \leq y^1 y^2 \leq y^2) s$

2 3 3 2 3

s

$\leq (x^1 y^1 \leq x^2 y^2)$

6

$A = y \leq$

2R

$(y \leq 2R) K_R F_y \leq 2y \leq$

$R = K_R^2$

$K_R x$

$A = x \leq$

R2

2x

≤ 2

K_R

$A = y \geq$

2R

$(y \geq 2R) K_R F_y \geq 2y \geq$

2

$\leq$

$R K_R^2$

$\neg \neg x$

2

 $\supset R \neg KR Ay \leq \neg \neg x \supset$

R

2

 $KR^2 Ax \leq \neg \neg R$

4

 $\supset x \neg KR^2 \neg \neg y$

2

 $\leq R \neg KR Fy \leq \neg \neg R$

4

 $\leq y \neg KR^2$
 $\neg KR$

2

 $Ay \leq \neg \neg x \leq$

R

 $\neg \neg y$
 $\leq R \neg Fy \leq \neg \neg R$
 $\leq y \neg$
 $\neg \neg \neg \neg$
 $KR^2 KR KR^2$
 $\neg x^2 \neg R^2 4$
 $\leq R KR Ax \leq \leq x KR^2$
 $\neg \neg \neg \neg \neg \neg$

2 4

 $R \neg y \neg R$
 $Ay \supset x \leq KR^2 \supset R KR Fy \leq \supset y KR^2 \neg KR$

2 2 4

2

 $\neg \neg x$

2

 $\supset R \neg KR Ay \supset \neg \neg x \supset$

R

2

 $KR^2 Ax \leq \neg \neg R$

4

 $\supset x \neg KR^2 \neg \neg y$

2

 $\supset R \neg KR Fy \leq \neg \neg R$

4

 $\supset y \neg KR^2$

$Ay \leq 1$
 $\sqrt{2}$
2
 $x \leq y$
4
 ≤ 1
 $\sqrt{2}$
2
 $\leq$
 $R \leq 1$
 $\sqrt{2}$
2
 y
 R
 ≤ 2
 ≤ 1
 $\leq \leq$
1
 $R \leq 2$
 $\leq x$
 R
 $\leq R$
 $4 \leq Ax \leq$
8
4
 $R \leq 4$
 $\leq$

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$\leq \leq \leq \leq \leq$

$4 \sqrt{2}$

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$x \sqrt{2}$

$\sqrt{2} \geq 2 \leq y \sqrt{2} \sqrt{2}$

$Ay \geq 1 \geq x \geq 1 \geq R \leq y \geq 1 \geq y$

$2 \leq 2 \leq 2$

R

4

$\leq 2 \geq \leq \leq$

8

1

4
R KR 2
KR
Ay ≤ 1
√2
2
x y
4
≤ 1
√2
2
R KR Fy ≤ 1
√2
2
y
4
R
KR 2
1
≤
1
R KR 2
R
4 Ax ≤ R 8 4
x R 8 4
KR
4 √2
Ay ≥ 1
√2
2
x ≤
x √2
2
KR 2
y
4
≥ 1
√2
2

$$R_{KR} F_y \geq 1$$

$$\sqrt{2}$$

$$2$$

$$y$$

$$R$$

$$4$$

$$\frac{1}{2} R_{KR} \geq \frac{1}{2} \frac{1}{2} \frac{1}{2}$$

$$8$$

$$1$$

$$4$$

$$R_{KR} \geq 2$$

$$[r(\frac{1}{2} \geq \frac{1}{2})$$

$$\geq R(\sin \frac{1}{2} \geq \sin \frac{1}{2})]_{KR}$$

$$A_y \geq \frac{1}{2} x(\cos \frac{1}{2} \geq \cos \frac{1}{2}) A_x \geq \frac{1}{2} x(\sin \frac{1}{2} \geq \sin \frac{1}{2}) [y(\frac{1}{2} \geq \frac{1}{2})$$

$$\geq R(\cos \frac{1}{2} \geq \cos \frac{1}{2})]_{KR}$$

$$F_y \geq \frac{1}{2} y(\cos \frac{1}{2} \geq \cos \frac{1}{2}) (\frac{1}{2} \geq \frac{1}{2})_{KR}$$

$$R_{KR} R$$

$$\frac{1}{2}$$

$$\leq (\sin^2 \frac{1}{2} \geq (\sin^2 \frac{1}{2} \geq \sin^2 \frac{1}{2}) \leq (\sin^2 \frac{1}{2} \geq \sin^2 \frac{1}{2})$$

$$2 \frac{1}{2} \frac{1}{2}$$

$$\geq \sin^2 \frac{1}{2}$$

$$\frac{1}{2}$$

$$R_{KR} \geq 2$$

$$\frac{1}{2}$$

$$1$$

$$R_{KR}$$

$$\geq (\frac{1}{2} \geq \frac{1}{2})_{KR} \geq (\frac{1}{2} \geq \frac{1}{2})_{KR} \geq 2$$

$$2 \frac{1}{2} \frac{1}{2}$$

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Fig. 5.3.6 Reactions for right-angle pipelines.

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PENETRANT METHODS 5-61

Table 5.3.3 Example 1 Showing Determination of Integrals

Values of integrals

Part of

pipe A B C F G H

$a_2 a_3$
 $0-1 \ 0 \ 0 \ 0 \ a$
 $2 \ 3$
 $ab_2 \ b_2 \ b_3$
 $1-2 \ ab \ a \ 2b \ b$
 $2 \ 2 \ 3$
 $c \ bc \ c_3$
 $2-3 \ (2a \leq c) \ (2a \leq c) \leq ac(a \leq c) \ bc \ b \ 2c \ c$
 $2 \ 2 \ 3$
 $a_2 \ ab \ 2 \ bc \ a_3 \ c_3 \ b_2 \ b_3$
 $\text{Total } 0 - 3 \leq ab \leq (2a \leq c) \leq a_2b \leq bc \leq b \ 2c \ a \leq b \leq c$
 $2 \ 2 \ 2 \ 3 \ 3 \ 2 \ 3$
 $c \leq ac(a \leq c)$
 $\leq (2a \leq c)$
 2

5.4 NONDESTRUCTIVE TESTING

by Donald D. Dodge

REFERENCES: Various authors, "Nondestructive Testing Handbook," 8 vols., wave direct current may be used for the location of surface defects.

American Society for Nondestructive Testing. Boyer, "Metals Handbook," vol. Half-wave direct current is most effective for locating subsurface de-

11, American Society for Metals. Heuter and Bolt, "Sonics," Wiley. fects. Magnetic particles may be applied dry or as a wet suspension in a

Krautkramer, "Ultrasonic Testing of Materials," Springer-Verlag. Spanner, liquid such as kerosene or water. Colored dry powders are advantageous

"Acoustic Emission: Techniques and Applications," Intex, American Society for

when testing for subsurface defects and when testing objects that have

Nondestructive Testing. Crowther, "Handbook of Industrial Radiography," Ar-

nold. Wiltshire, "A Further Handbook of Industrial Radiography," Arnold. rough surfaces, such as castings, forgings, and weldments. Wet particles

"Standards," vol. 03.03, ASTM, Boiler and Pressure Vessel Code, Secs. III, V, are preferred for detection of very fine cracks, such as fatigue, stress

XI, ASME. ASME Handbook, "Metals Engineering — Design," McGraw-Hill. corrosion, or grinding cracks. Fluorescent wet particles are used to in-

SAE Handbook, Secs. J358, J359, J420, J425 - J428, J1242, J1267, SAE. Materi- spect objects with the aid of ultraviolet light. Fluorescent inspection is

als Evaluation, Jour. Am. Soc. Nondestructive Testing. widely used because of its greater sensitivity. Application of particles

while magnetizing current is on (continuous method) produces stronger

Nondestructive tests are those tests that determine the usefulness, ser- indications than those obtained if the particles are applied after the

viceability, or quality of a part or material without limiting its useful- current is shut off (residual method). Interpretation of subsurface-defect

ness. Nondestructive tests are used in machinery maintenance to avoid indications requires experience. Demagnetization of the test object after costly unscheduled loss of service due to fatigue or wear; they are used inspection is advisable. in manufacturing to ensure product quality and minimize costs. Consid- Magnetic flux leakage is a variation whereby leakage flux due to eration of test requirements early in the design of a product may facili- flaws is detected electronically via a Hall-effect sensor. Computerized tate testing and minimize testing cost. Nearly every form of energy is signal interpretation and data imaging techniques are employed. used in nondestructive tests, including all wavelengths of the electro- Electrified particle testing indicates minute cracks in nonconducting magnetic spectrum as well as vibrational mechanical energy. Physical materials. Particles of calcium carbonate are positively charged as they properties, composition, and structure are determined; flaws are de- are blown through a spray gun at the test object. If the object is metal- tected; and thickness is measured. These tests are here divided into the backed, such as porcelain enamel, no preparation other than cleaning is following basic methods: magnetic particle, penetrant, radiographic, ul- necessary. When it is not metal-backed, the object must be dipped in an trasonic, eddy current, acoustic emission, microwave, and infrared. Numer- aqueous penetrant solution and dried. The penetrant remaining in cracks ous techniques are utilized in the application of each test method. Table provides a mobile electron supply for the test. A readily visible powder 5.4.1 gives a summary of many nondestructive test methods. indication forms at a crack owing to the attraction of the positively charged particles.

MAGNETIC PARTICLE METHODS

PENETRANT METHODS

Magnetic particle testing is a nondestructive method for detecting dis- continuities at or near the surface in ferromagnetic materials. After the Liquid penetrant testing is used to locate flaws open to the surface of test object is properly magnetized, finely divided magnetic particles are nonporous materials. The test object must be thoroughly cleaned before applied to its surface. When the object is properly oriented to the in- testing. Penetrating liquid is applied to the surface of a test object by a duced magnetic field, a discontinuity creates a leakage flux which at- brush, spray, flow, or dip method. A time allowance (1 to 30 min) is tracts and holds the particles, forming a visible indication. Magnetic- required for liquid penetration of surface flaws. Excess penetrant is then field direction and character are dependent upon how the magnetizing carefully removed from the surface, and an absorptive coating, known force is applied and upon the type of current used. For best sensitivity, as developer, is applied to the object to draw penetrant out of flaws, thus the magnetizing current must flow in a direction parallel to the principal showing their location, shape, and approximate size. The developer is

direction of the expected defect. Circular fields, produced by passing typically a fine powder, such as talc usually in suspension in a liquid.

current through the object, are almost completely contained within the Penetrating-liquid types are (1) for test in visible light, and (2) for test

test object. Longitudinal fields, produced by coils or yokes, create ex- under ultraviolet light (3,650 Å). Sensitivity of penetrant testing is

ternal poles and a general-leakage field. Alternating, direct, or half- greatest when a fluorescent penetrant is used and the object is observed

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5-62 NONDESTRUCTIVE TESTING

Table 5.4.1 Nondestructive Test Methods*

Method Measures or detects Applications Advantages Limitations

Acoustic emission Crack initiation and growth Pressure vessels Remote and continuous Transducers must be

rate Stressed structures surveillance placed on part surface

Internal cracking in welds Turbine or gearboxes Permanent record Highly ductile

during cooling Fracture mechanics Dynamic (rather than materials yield low-

Boiling or cavitation research static) detection of amplitude emissions

Friction or wear Weldments cracks Part must be stressed or

Plastic deformation Sonic-signature analysis Portable operating

Phase transformations Triangulation techniques to Interfering noise needs to

locate flaws be filtered out

Acoustic-impact (tapping) Debonded areas or delam- Brazed or adhesive-bonded Portable Part geometry and mass

inations in metal or structures Easy to operate influences test results

nonmetal composites or Bolted or riveted assem- May be automated Impactor and probe must

laminates blies Permanent record or be repositioned to fit

Cracks under bolt or Turbine blades positive meter readout geometry of part

fastener heads Turbine wheels No couplant required Reference standards

Cracks in turbine wheels or Composite structures required

turbine blades Honeycomb assemblies Pulser impact rate is criti-

Loose rivets or fastener cal for repeatability

heads

Crushed core

D-Sight (DiffRACTo) Enhances visual inspection Detect impact damage to Portable Part surface must reflect

for surface abnormali- composites or honey- Fast, flexible light or be wetted with

ties such as dents pro- comb corrosion in Noncontact a fluid

trusions, or waviness' aircraft lap joints Easy to use

Crushed core Automotive bodies for Documentable

Lap joint corrosion waviness

Cold-worked holes

Cracks

Eddy current Surface and subsurface Tubing No special operator skills Conductive materials cracks and seams Wire required Shallow depth of penetra-

Alloy content Ball bearings High speed, low cost tion (thin walls only)

Heat-treatment variations "Spot checks" on all types Automation possible for Masked or false indications

Wall thickness, coating of surfaces symmetric parts caused by sensitivity

thickness Proximity gage Permanent-record capabil- to variations such as

Crack depth Metal detector ity for symmetric parts part geometry

Conductivity Metal sorting No couplant or probe con- Reference standards

Permeability Measure conductivity in % tact required required

IACS Permeability variations

Magneto-optic eddy- Cracks Aluminum aircraft Real-time imaging Frequency range of 1.6 to

current imager Corrosion thinning in Structure Approximately 4-in area 100 kHz

aluminum coverage Surface contour

Temperature range of 32 to

90°F

Directional sensitivity to

cracks

Eddy-sonic Debonded areas in metal- Metal-core honeycomb Portable Specimen or part must core or metal-faced Metal-faced honeycomb Simple to operate contain conductive

honeycomb structures Conductive laminates such No couplant required materials to establish

Delaminations in metal as boron or graphite- Locates far-side debonded eddy-current field

laminates or composites fiber composites areas Reference standards

Crushed core Bonded-metal panels Access to only one surface required

required Part geometry

May be automated

Electric current Cracks Metallic materials Access to only one surface Edge effect

Crack depth Electrically conductive required Surface contamination

Resistivity materials Battery or dc source Good surface contact

Wall thickness Train rails Portable required

Corrosion-induced wall Nuclear fuel elements Difficult to automate

thinning Bars, plates other shapes Electrode spacing

Reference standards

required

Electrified particle Surface flaws in noncon- Glass Portable Poor resolution on thin

ducting material Porcelain enamel Useful on materials not coatings

Through-to-metal pinholes Nonhomogeneous practical for penetrant False indications from

on metal-backed materials such as inspection moisture streaks or lint

material plastic or asphalt Atmospheric conditions

Tension, compression, coatings High-voltage discharge

cyclic cracks Glass-to-metal seals

Brittle-coating stress cracks

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PENETRANT METHODS 5-63

Table 5.4.1 Nondestructive Test Methods* (Continued)

Method	Measures or detects	Applications	Advantages	Limitations
Filtered particle	Cracks	Porous materials such as	Colored or fluorescent	Size and shape of particles
Porosity	clay, carbon, powdered particles must be selected before			
Differential absorption	metals, concrete	Leaves no residue after use		
Grinding wheels	baking part over 400°F	Penetrating power of		
High-tension insulators	Quickly and easily applied	suspension medium is		
Sanitary ware	Portable	critical		
Particle concentration must				
be controlled				
Skin irritation				
Infrared (radiometry)	Hot spots	Brazed joints	Sensitive to 0.1°F	Emissivity
(thermography)	Lack of bond	Adhesive-bonded joints	temperature variation	Liquid-nitrogen-cooled
Heat transfer	Metallic platings or	Permanent record or detector		
Isotherms	coatings; debonded thermal picture	Critical time-temperature		
Temperature ranges	areas or thickness	Quantitative relationship		
Electrical assemblies	Remote sensing; need not	Poor resolution for thick		
Temperature monitoring	contact part specimens			
Portable	Reference standards			
required				
Leak testing	Leaks: Joints: High sensitivity to	Accessibility to both		
Helium	Welded extremely small, light surfaces of part required			
Ammonia	Brazed separations not	Smeared metal or		
Smoke	Adhesive-bonded detectable by other contaminants may			
Water	Sealed assemblies NDT methods prevent detection			
Air bubbles	Pressure or vacuum	Sensitivity related to	Cost related to	sensitivity
Radioactive gas chambers	method selected			
Halogens	Fuel or gas tanks			
Magnetic particle	Surface and slightly	Ferromagnetic materials; Advantage over penetrant		
Alignment of magnetic				
subsurface flaws; bar, plate, forgings, is that it indicates field is critical				
cracks, seams, porosity, weldments, extrusions, subsurface flaws, Demagnetization of parts				
inclusions etc. particularly inclusions required after tests				
Permeability variations	Relatively fast and low-cost	Parts must be cleaned be-		
Extremely sensitive for	May be portable fore and after inspection			
locating small tight	Masking by surface			

cracks coatings

Magnetic field (also Cracks Ferromagnetic materials Measurement of magnetic Permeability

magnetic flux leakage) Wall thickness Ship degaussing material properties Reference standards

Hardness Liquid-level control May be automated required

Coercive force Treasure hunting Easily detects magnetic Edge effect

Magnetic anisotropy Wall thickness of objects in nonmagnetic Probe lift-off

Magnetic field nonmetallic materials material

Nonmagnetic coating Material sorting Portable

thickness on steel

Microwave (300 MHz – Cracks, holes, debonded Reinforced plastics Between radio waves and Will not penetrate metals

300 GHz) areas, etc., in Chemical products infrared in Reference standards

nonmetallic parts Ceramics electromagnetic required

Changes in composition, Resins spectrum Horn-to-part spacing

degree of cure, Rubber Portable critical

moisture content Wood Contact with part surface Part geometry

Thickness measurement Liquids not normally required Wave interference

Dielectric constant Polyurethane foam Can be automated Vibration

Loss tangent Radomes

Liquid penetrants (dye or Flaws open to surface of All parts with Low cost Surface films such as

fluorescent) parts; cracks, porosity, nonabsorbing surfaces Portable coatings, scale, and

seams, laps, etc. (forgings, weldments, Indications may be further smeared metal may

Through-wall leaks castings, etc.). Note: examined visually prevent detection of

Bleed-out from porous Results easily interpreted flaws

surfaces can mask Parts must be cleaned

indications of flaws before and after

inspection

Flaws must be open to

surface

Fluoroscopy Level of fill in containers Flow of liquids High-brightness images Costly equipment

(cinefluorography) Foreign objects Presence of cavitation Real-time viewing Geometric unsharpness

(kinefluorography) Internal components Operation of valves and Image magnification Thick specimens

Density variations switches Permanent record Speed of event to be studied

Voids, thickness Burning in small solid-pro- Moving subject can be Viewing area

Spacing or position pellant rocket motors observed Radiation hazard

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5-64 NONDESTRUCTIVE TESTING

Table 5.4.1 Nondestructive Test Methods* (Continued)

Method Measures or detects Applications Advantages Limitations

Neutron radiology (thermal Hydrogen contamination of Pyrotechnic devices High neutron absorption by Very costly equipment
neutrons from reactor, titanium or zirconium Metallic, nonmetallic hydrogen, boron, Nuclear reactor or
accelerator, or alloys assemblies lithium, cadmium, accelerator required
Californium 252) Defective or improperly Biological specimens uranium, plutonium Trained physicists required
loaded pyrotechnic Nuclear reactor fuel Low neutron absorption by Radiation hazard
devices elements and control most metals Nonportable
Improper assembly of rods Complement to X-ray or Indium or gadolinium
metal, nonmetal parts Adhesive-bonded structures gamma-ray radiography screens required
Corrosion products
Gamma radiology (cobalt Internal flaws and Usually where X-ray Low initial cost One energy level per source
60, iridium 192) variations, porosity, machines are not Permanent records; film Source decay
inclusions, cracks, lack suitable because source Small sources can be Radiation hazard
of fusion, geometry cannot be placed in placed in parts with Trained operators needed
variations, corrosion part with small small openings Lower image resolution
thinning openings and/or power Portable Cost related to source size
Density variations source not available Low contrast
Thickness, gap, and Panoramic imaging
position
X-ray radiology Internal flaws and Castings Permanent records; film High initial costs
variations; porosity, Electrical assemblies Adjustable energy levels Orientation of linear flaws
inclusions, cracks, lack Weldments (5 kV – 25 meV) in part may not be
of fusion, geometry Small, thin, complex High sensitivity to density favorable
variations, corrosion wrought products changes Radiation hazard
Density variations Nonmetallics No couplant required Depth of flaw not indicated
Thickness, gap, and Solid-propellant rocket Geometry variations do not Sensitivity decreases with
position motors affect direction of increase in scattered
Misassembly Composites X-ray beam radiation
Misalignment Container contents
Radiometry X-ray, gamma Wall thickness Sheet, plate, strip, tubing Fully automatic Radiation
hazard
ray, beta ray) Plating thickness Nuclear reactor fuel rods Fast Beta ray useful for
(transmission or Variations in density or Cans or containers Extremely accurate ultrathin coatings
only
backscatter) composition Plated parts In-line process control Source decay
Fill level in cans or Composites Portable Reference standards
containers required
Inclusions or voids
Reverse-geometry digital Cracks Aircraft structure High-resolution 106 pixel Access to both sides of

X-ray Corrosion image with high object
 Water in honeycomb contrast Radiation hazard
 Carbon epoxy honeycomb
 Foreign objects
 X-ray computed Small density changes Solid-propellant rocket Measures X-ray opacity of Very expensive
 tomography (CT) Cracks motors object along many paths Trained operator
 Voids Rocket nozzles Radiation hazard
 Foreign objects Jet-engine parts
 Turbine blades
 Shearography electronic Lack of bond Composite-metal Large area coverage Requires vacuum thermal,
 Delaminations honeycomb Rapid setup and operation ultrasonic, or
 Plastic deformation Bonded structures Noncontacting microwave stressing of
 Strain Composite structures Video image easy to store structure to cause
 Crushed core surface strain
 Impact damage
 Corrosion in Al honeycomb
 Thermal (thermochromic Lack of bond Braze joints Very low initial cost Thin-walled surfaces only paint, liquid crystals) Hot spots Adhesive-bonded joints Can be readily applied to Critical time-temperature
 Heat transfer Metallic platings or surfaces which may relationship
 Isotherms coatings be difficult to inspect Image retentivity affected
 Temperature ranges Electrical assemblies by other methods by humidity
 Blockage in coolant Temperature monitoring No special operator skills Reference standards passages required
 Sonic (less than 0.1 MHz) Debonded areas or Metal or nonmetal Portable Surface geometry delaminations in metal composite or laminates Easy to operate influences test results or nonmetal composites brazed or adhesive- Locates far-side debonded Reference standards or laminates bonded areas required
 Cohesive bond strength Plywood May be automated Adhesive or core-thickness under controlled Rocket-motor nozzles Access to only one surface variations influence conditions Honeycomb required results
 Crushed or fractured core
 Bond integrity of metal
 insert fasteners

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RADIOGRAPHIC METHODS 5-65

Table 5.4.1 Nondestructive Test Methods* (Continued)

Method Measures or detects Applications Advantages Limitations

Ultrasonic (0.1 – 25 MHz) Internal flaws and Metals Most sensitive to cracks Couplant required variations; cracks, lack Welds Test results known Small, thin, or complex of fusion, porosity, Braze joints immediately parts may be difficult inclusions, Adhesive-bonded joints Automating and to inspect delaminations, lack of Nonmetallics permanent-record Reference standards bond, texturing In-service parts capability required Thickness or velocity Portable Trained operators for Poisson's ratio, elastic High penetration capability manual inspection modulus Special probes Thermoelectric probe Thermoelectric potential Metal sorting Portable Hot probe Coating thickness Ceramic coating thickness Simple to operate Difficult to automate Physical properties on metals Access to only one surface Reference standards Thompson effect Semiconductors required required P-N junctions in Surface contaminants semiconductors Conductive coatings

* From Donald J. Hagemeyer, "Metal Progress Databook," Douglas Aircraft Co., McDonnell-Douglas Corp., Long Beach, CA.

in a semidarkened location. After testing, the penetrant and developer A radiograph is a shadow picture, since X-rays and gamma rays are removed by washing with water, sometimes aided by an emulsifier, follow the laws of light in shadow formation. Four factors determine or with a solvent. the best geometric sharpness of a picture: (1) The effective focal-spot In filtered particle testing, cracks in porous objects (100 mesh or size of the radiation source should be as small as possible. (2) The smaller) are indicated by the difference in absorption between a cracked source-to-object distance should be adequate for proper definition of and a flaw-free surface. A liquid containing suspended particles is the area of the object farthest from the film. (3) The film should be as sprayed on a test object. If a crack exists, particles are filtered out and close as possible to the object. (4) The area of interest should be in concentrate at the surface as liquid flows into the additional absorbent the center of and perpendicular to the X-ray beams and parallel to the area created by the crack. Fluorescent or colored particles are used to X-ray film. locate flaws in unfired dried clay, certain fired ceramics, concrete, some Radiographic films vary in speed, contrast, and grain size. Slow films powdered metals, carbon, and partially sintered tungsten and titanium generally have smaller grain size and produce more contrast. Slow films carbides. are used where optimum sharpness and maximum contrast are desired. Fast films are used where objects with large differences in thickness are to be radiographed or where sharpness and contrast can be sacrificed to

RADIOGRAPHIC METHODS

shorten exposure time. Exposure of a radiographic film comes from direct radiation and scattered radiation. Direct radiation is desirable,

Radiographic test methods employ X-rays, gamma rays, or similar penetrating radiation; scattered radiation, which occurs in the object being X-rayed or in neighboring objects, produces undesirable images of objects. Electromagnetic energy wavelengths in the range of on the film and loss of contrast. Intensifying screens made of 0.005- or 0.01 to 10 Å ($1 \text{ Å} = 10^{-8} \text{ cm}$) are used to examine the interior of opaque 0.010-in- (0.13-mm or 0.25-mm) thick lead are often used for radiographing materials. Penetrating radiation proceeds from its source in straight paths at voltages above 100 kV. The lead filters out much of the low-energy scatter radiation. Under action of X-rays or gamma rays above 88 kV, a lead screen also emits electrons which, when in intimate contact with the film, produce additional coherent darkening of the film. X-rays of a variety of wavelengths result when high-speed electrons Exposure time can be materially reduced by use of intensifying screens in a vacuum tube are suddenly stopped. An X-ray tube contains a heated filament (cathode) and a target (anode); radiation intensity is almost directly proportional to filament current (mA); tube voltage (kV) determines the penetration capability of the rays. As tube voltage increases, small rectangular plate of the same material as the object being X-rayed. Penetrameters are used to indicate the contrast and definition which exist in a radiograph. The type generally used in the United States is a rectangular plate of the same material as the object being X-rayed. It is uniform in thickness (usually 2 percent of the object thickness) and has holes drilled through it. ASTM specifies hole diameters 1, 2, and 4 times the thickness of the penetrometer. Step, wire, and bead penetrameters are also used. (See ASTM Materials Specification E94.) Because of the variety of factors that affect the production and measurements of an X-ray image, operating factors are generally selected from reference tables or graphs which have been prepared from test data obtained for a range of operating conditions.

Gamma rays of a specific wavelength are emitted from the disintegrating nuclei of natural radioactive elements, such as radium, and from all materials may be inspected by radiographic means, but there are limitations to the configurations of materials. With optimum techniques, wires 0.0001 in (0.003 mm) in diameter can be resolved in small

phy. The half-life of an isotope is the time required for half of the electrical components. At the other extreme, welded steel pressure vessels with 20-in (500-mm) wall thickness can be routinely inspected by many years. use of high-energy accelerators as a source of radiation. Neutron radiography. Radiographs are photographic records produced by the passage of ionizing radiation through materials such as lead more readily than penetrating radiation onto a film. A void or reduced mass appears as a darker image on the film because of the lesser absorption of energy and materials such as plastics, usually because of their hydrogen content. the resulting additional exposure of the film. The quantity of X-rays Radiographic standards are published by ASTM, ASME, AWS, and absorbed by a material generally increases as the atomic number increases. API, primarily for detecting lack of penetration or lack of fusion in creases. welded objects. Cast-metal objects are radiographed to detect conditions such as shrink, porosity, hot tears, cold shuts, inclusions, coarse grain structure, and cracks. employed for flaw detection. Low frequencies (40 kHz to 1.0 MHz) are The usual method of utilizing penetrating radiation employs film. used on materials of low elastic modulus or large grain size. High frequencies (2.25 to 25 MHz) provide better resolution of smaller defects photoconductors (xeroradiography), scintillation crystals, and vidicon and are used on fine-grain materials and thin sections. Frequencies tubes (image intensifiers) are also used. Computerized digital radiography above 25 MHz are employed for investigation and measurement of physical properties related to acoustic attenuation. The dangers connected with exposure of the human body to X-rays Wave-vibrational modes other than longitudinal are effective in detecting flaws that do not present a reflecting surface to the ultrasonic beam, for the use of radiation equipment. NIST is a prime source of information or other characteristics not detectable by the longitudinal mode. They tion concerning radiation safety. NRC specifies maximum permissible exposure are useful also when large areas of plates must be examined. Wedges of exposure to be a 1.25 R/ 1/4 year. plastic, water, or other material are inserted between the transducer face and the test object to convert, by refraction, to shear, transverse, surface, or Lamb vibrational modes. As in optics, Snell's law expresses the

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5-66 NONDESTRUCTIVE TESTING

tions such as shrink, porosity, hot tears, cold shuts, inclusions, coarse grain structure, and cracks. employed for flaw detection. Low frequencies (40 kHz to 1.0 MHz) are The usual method of utilizing penetrating radiation employs film. used on materials of low elastic modulus or large grain size. High frequencies (2.25 to 25 MHz) provide better resolution of smaller defects photoconductors (xeroradiography), scintillation crystals, and vidicon and are used on fine-grain materials and thin sections. Frequencies tubes (image intensifiers) are also used. Computerized digital radiography above 25 MHz are employed for investigation and measurement of physical properties related to acoustic attenuation. The dangers connected with exposure of the human body to X-rays Wave-vibrational modes other than longitudinal are effective in detecting flaws that do not present a reflecting surface to the ultrasonic beam, for the use of radiation equipment. NIST is a prime source of information or other characteristics not detectable by the longitudinal mode. They tion concerning radiation safety. NRC specifies maximum permissible exposure are useful also when large areas of plates must be examined. Wedges of exposure to be a 1.25 R/ 1/4 year. plastic, water, or other material are inserted between the transducer face and the test object to convert, by refraction, to shear, transverse, surface, or Lamb vibrational modes. As in optics, Snell's law expresses the

relationship between incident and refracted beam angles; i.e., the ratio of the sines of the angle from the normal, of the incident and refracted beams in two mediums, is equal to the ratio of the mode acoustic velocities in the two mediums. Limiting conditions for ultrasonic testing may be the test-object shape, surface roughness, grain size, material structure, flaw orientation, selectivity of discontinuities, and the skill of the operator. Test sensitivity is liquid into the test object, where the ultrasonic energy is attenuated, less for cast metals than for wrought metals because of grain size and surface differences. Reflected, transmitted, or resonant sound energy is reconverted to electrical energy by a transducer and returned to the test instrument, where society, and company specifications (see references above). Evaluation is made by comparing received signals with signals obtained from reference blocks containing conditions of the test-object material. flat bottom holes between 1/64 and 8/64 in (0.40 and 0.325 cm) in diameter. Materials capable of being tested by ultrasonic energy are those which are free of, or from parts containing known flaws, drilled holes, or machined surfaces. Metals are tested in dimensions of up to notches. 30 ft (9.14 m). Noncellular plastics, ceramics, glass, new concrete, or organic materials, and rubber can be tested. Each material has a characteristic sound velocity, which is a function of its density and modulus (elastic or shear). Eddy current nondestructive tests are based upon correlation between material characteristics determinable through ultrasonics include electromagnetic properties and physical or structural properties of a test object. Eddy currents are induced in metals whenever they are brought into an ac magnetic field. These eddy currents create a secondary magnetic field, which opposes the inducing magnetic field. The presence of inclusions, porosity, lack of penetration, and lack of fusion. Other applications include location of unbond in laminated materials, location of the apparent impedance of the inducing coil or of a detection coil. Coil

fatigue cracks in machinery, and medical applications. Automatic test- impedance indicates the magnitude and phase relationship of the eddy

ing is frequently performed in manufacturing applications. currents to their inducing magnetic-field current. This relationship is

Ultrasonic systems are classified as either pulse-echo, in which a sin- dependent upon the mass, conductivity, permeability, and structure of

gle transducer is used, or through-transmission, in which separate send- the metal and upon the frequency, intensity, and distribution of the

ing and receiving transducers are used. Pulse-echo systems are more alternating magnetic field. Conditions such as heat treatment, composi-

common. In either system, ultrasonic energy must be transmitted into, tion, hardness, phase transformation, case depth, cold working,

and received from, the test object through a coupling medium, since air strength, size, thickness, cracks, seams, and inhomogeneities are indi-

will not efficiently transmit ultrasound of these frequencies. Water, oil, cated by eddy current tests. Correlation data must usually be obtained to

grease, and glycerin are commonly used couplants. Two types of testing determine whether test conditions for desired characteristics of a partic-

are used: contact and immersion. In contact testing, the transducer is ular test object can be established. Because of the many factors which

placed directly on the test object. In immersion testing, the transducer cause variation in electromagnetic properties of metals, care must be

and test object are separated from one another in a tank filled with water taken that the instrument response to the condition of interest is not

or by a column of water or by a liquid-filled wheel. Immersion testing nullified or duplicated by variations due to other conditions.

eliminates transducer wear and facilitates scanning of the test object. Alternating-current frequencies between 1 and 5,000,000 Hz are used

Scanning systems have paper-printing or computerized video equip- for eddy current testing. Test frequency determines the depth of current

ment for readout of test information. penetration into the test object, owing to the ac phenomenon of "skin

Ultrasonic transducers are piezoelectric units which convert electric effect." One "standard depth of penetration" is the depth at which the

energy into acoustic energy and convert acoustic energy into electric eddy currents are equal to 37 percent of their value at the surface. In a

energy of the same frequency. Quartz, barium titanate, lithium sulfate, plane conductor, depth of penetration varies inversely as the square root

lead metaniobate, and lead zirconate titanate are commonly used trans- of the product of conductivity, permeability, and frequency. High-fre-

ducer crystals, which are generally mounted with a damping backing in quency eddy currents are more sensitive to surface flaws or conditions

a housing. Transducers range in size from 1/16 to 5 in (0.15 to 12.7 cm) while low-frequency eddy currents are sensitive also to deeper internal

and are circular or rectangular. Ultrasonic beams can be focused to flaws or conditions.

improve resolution and definition. Transducer characteristics and beam Test coils are of three general types: the circular coil, which surrounds patterns are dependent upon frequency, size, crystal material, and con- an object; the bobbin coil, which is inserted within an object; and the struction. probe coil, which is placed on the surface of an object. Coils are further Test frequencies used range from 40 kHz to 200 MHz. Flaw-detection classified as absolute, when testing is conducted without direct compari- and thickness-measurement applications use frequencies between son with a reference object in another coil; or differential, when compar-

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ACOUSTIC SIGNATURE ANALYSIS 5-67

ison is made through use of two coils connected in series opposition. emit radiation at varying intensities, depending upon their temperature Many variations of these coil types are utilized. Axial length of a circu- and surface characteristics. A passive infrared system detects the natural lar test coil should not be more than 4 in (10.2 cm), and its shape should radiation of an unheated test object, while an active system employs a correspond closely to the shape of the test object for best results. Coil source to heat the test object, which then radiates infrared energy to a diameter should be only slightly larger than the test-object diameter for detector. Sensitive indication of temperature or temperature distribution consistent and useful results. Coils may be of the air-core or magnetic- through infrared detection is useful in locating irregularities in materi- core type. als, in processing, or in the functioning of parts. Emission in the infrared Instrumentation for the analysis and presentation of electric signals range of 0.8 to 15 μm is collected optically, filtered, detected, and resulting from eddy current testing includes a variety of means, ranging amplified by a test instrument which is designed around the characteris- from meters to oscilloscopes to computers. Instrument meter or alarm tics of the detector material. Temperature variations on the order of circuits are adjusted to be sensitive only to signals of a certain electrical 0.1°F can be indicated by meter or graphic means. Infrared theory and phase or amplitude, so that selected conditions are indicated while instrumentation are based upon radiation from a blackbody; therefore, others are ignored. Automatic and automated testing is one of the prin- emissivity correction must be made electrically in the test instrument or cipal advantages of the method. arithmetically from instrument readings. Thickness measurement of metallic and nonmetallic coatings on metals is performed using eddy current principles. Coating thicknesses ACOUSTIC SIGNATURE ANALYSIS measured typically range from 0.0001 to 0.100 in (0.00025 to 0.25 cm). For measurement to be possible, coating conductivity must differ from Acoustic signature analysis involves the analysis of sound energy emitted

that of the base metal. from an object to determine characteristics of the object. The object may be a simple casting or a complex manufacturing system. A passive MICROWAVE METHODS test is one in which sonic energy is transmitted into the object. In this case, a mode of resonance is usually detected to correlate with cracks or

Microwave test methods utilize electromagnetic energy to determine structure variations, which cause a change in effective modulus of the characteristics of nonmetallic substances, either solid or liquid. Fre-object, such as a nodular iron casting. An active test is one in which the quencies used range from 1 to 3,000 GHz. Microwaves generated in a object emits sound as a result of being struck or as a result of being in test instrument are transmitted by a waveguide through air to the test operation. In this case, characteristics of the object may be correlated to object. Analysis of reflected or transmitted energy indicates certain damping time of the sound energy or to the presence or absence of a material characteristics, such as moisture content, composition, struc-certain frequency of sound energy. Bearing wear in rotating machinery ture, density, degree of cure, aging, and presence of flaws. Other appli-can often be detected prior to actual failure, for example. More complex cations include thickness and displacement measurement in the range of analytical systems can monitor and control manufacturing processes, 0.001 in (0.0025 cm) to more than 12 in (30.4 cm). Materials that can be based upon analysis of emitted sound energy.

tested include most solid and liquid nonmetals, such as chemicals, min-Acoustic emission is a technology distinctly separate from acoustic erals, plastics, wood, ceramics, glass, and rubber.

signature analysis and is one in which strain produces bursts of energy INFRARED METHODS

in an object. These are detected by ultrasonic transducers coupled to the object. Growth of microcracks, and other flaws, as well as incipient

Infrared nondestructive tests involve the detection of infrared electro- failure, is monitored by counting the pulses of energy from the object or

magnetic energy emitted by a test object. Infrared radiation is produced recording the time rate of the pulses of energy in the ultrasonic range

naturally by all matter at all temperatures above absolute zero. Materials (usually a discrete frequency between 1 kHz and 1 MHz).